The Asset Market Channel of Market Power

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Abstract

We characterize the macroeconomic effects of product market power through asset markets. We propose a model and a sufficient statistics framework to quantify how aggregate markups, through their effects on the distribution of factor income and profits across heterogeneous households, alter the long-run equilibrium interest rate and cost of capital. Firm ownership is the key ingredient that determines how the profits generated by market power affect asset markets. Undiversified profits from private businesses support precautionary savings and therefore asset demand, while profits that back the market value of diversified public equity contribute to asset supply. Disciplined with U.S. data, we demonstrate that the estimated rise in aggregate markups from 1980 to 2018 can explain several important macroeconomic trends, including the sustained decline in interest rates, the divergence between labor and profit shares, and rising wealth concentration. The counterfactual contribution of the rise in aggregate markups to the decline in long-run rates is 1 percentage point, in line with empirical estimates of the trend decline in r^* prior to 2008. This decline in rates lowers the cost of capital and mitigates half of the 11 percent aggregate output loss from the product market distortions of markups relative to a representative household benchmark. Despite the smaller quantity distortions, the population consumption-equivalent welfare loss is five times as large, reflecting the income redistribution away from the large worker population toward the small group of business owners.

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1 Introduction

Market power shapes aggregate production and the division of national income between productive factors and profits. The conventional analysis of the macroeconomic consequences of market power focuses on its distortions in product markets. Firms are able to charge prices above marginal cost, which reduces their demand for factor inputs and lowers their production below efficient scale. We develop a complementary perspective in which the effects of market power on the distribution of factor income and profits shapes the balance of supply and demand in asset markets, which feeds back into product markets and the broader economy.

Our paper makes three main contributions. First, we develop a sufficient statistics framework linking aggregate markups to the equilibrium rate of return on assets and demonstrate the implications for the cost of capital and the long-run level of aggregate output. Second, we show that accounting for the distributional effects of market power through asset markets lowers the aggregate output losses from the estimated rise of aggregate markups since 1980 by 50% but *raises* the population welfare loss by a multiple of five, primarily because of the incidence of market power across the household income distribution. Third, we quantify how rising aggregate markups can explain a number of important macroeconomic trends, including the long-run decline in interest rates, the divergence in the labor and profit shares, and the secular rise of top wealth concentration.

The core idea of the "asset market channel" is that market power affects both the income sources that households save out of and the assets they save into. When household savings are imperfectly elastic to the long-run rate r, the distributional effects of market power on household income and savings changes r. Because asset returns determine the cost of capital in equilibrium, these asset market effects can mitigate or even amplify the original product market distortions on aggregate quantities. This setting contrasts with the standard representative household benchmark in which the steady state rate of return is fixed and determined by the discount factor $\beta(1+r) = 1$.

To formalize this idea, we propose an equilibrium model where product and asset markets are connected through the endogenous determination of the rate of return r. While production and imperfect competition between firms are standard, the central ingredients of our framework are two dimensions of household heterogeneity, which jointly affect their means and motives for saving. Households have ex-ante heterogeneity in their claims to the profits generated by firms: workers solely earn labor income and firm owners earn profit income. Within types, households are also ex-post heterogeneous due to the idiosyncratic income risk that they face in the presence of incomplete markets. As in the standard Aiyagari (1994) model, all households trade claims on physical capital to self-insure, and workers face labor income risk. Firm owners transition between private and public states, changing their exposure to undiversified business risk and their contribution to aggregate asset demand as public equity sales convert private business income into liquid financial wealth. These sales make profits from previously private businesses available to public shareholders, expanding the supply of total assets available for households to hold.

Our main theoretical contribution is a sufficient statistics representation of steady state asset market clearing, which is characterized by factor income shares (labor s_W, capital s_K, and profit s_Π), and household balance sheet and firm ownership statistics.

$$\bar{D}(r)\big[\underbrace{s_W + \lambda(r)s_\Pi}_{\text{Redistribution}}\big] = \underbrace{\frac{s_K}{r+\delta_K}}_{\text{Capital Wedge}} + \underbrace{Q(r)\omega s_\Pi}_{\text{Asset Creation}} \tag{1}$$

We group the effects of market power on asset market clearing into three forces: *Redistribution*, the *Capital Wedge*, and *Asset Creation*. Factor income shares determine the fraction of national income devoted to private savings, either by workers or firm owners, and hence to asset demand. The relative wealth-to-income ratio of firm owners to workers $\lambda(r)$ determines the change in aggregate savings when a unit of national income is redistributed from workers to firm owners with differing savings propensities, where $\bar{D}(r)$ is the worker's baseline asset demand per dollar of permanent labor income. An increase in markups induces a reallocation in national income from the labor share to the profit share, raising asset demand when $\lambda(r) > 1$ (*Redistribution*).

While all capital with depreciation rate δ_K is traded as in the baseline Aiyagari (1994) model, the fraction of aggregate profits accruing to publicly traded equity, ω, is capitalized into its market value. While an increase in markups depresses factor demand for capital, decreasing the capital share and the available quantity of capital to hold as an asset (*Capital Wedge*), it increases the quantity of tradable equity available to the public by increasing the aggregate profit share (*Asset Creation*). The market value of public equity also depends on the endogenous valuation multiple $Q(r)$, where a lower r raises $Q(r)$ because households discount the future profits of public firms by less. The net effect of these three forces determines whether a given increase in market power raises or lowers the equilibrium rate r.

While the sufficient statistic representation displayed above stems from our model, it also nests a number of other standard household consumption-savings models. These include the standard Aiyagari (1994) one-asset model, two-account heterogeneous-agent models as in Auclert, Rognlie and Straub (2024), and overlapping generations models as in Diamond (1965); Blanchard (1985); Yaari (1965). The representation can also be extended to accommodate a wide range of model generalizations. Our baseline assumption is that all firms operate a common Cobb-Douglas production function, which implies that factor shares are independent of the level of aggregate TFP. Therefore, the aggregate markup is the sufficient statistic for the effects of primitive changes in market power, such as in market structure or conduct, on stationary asset market equilibrium. However, in an extension with more general CES production functions and/or long-run TFP growth, the effects of market power on misallocation and TFP will matter for determining r. With government borrowing, the debt-to-GDP ratio appears on the right-hand side of our asset market clearing condition, and factor income shares reflect post-tax adjustments. Appendix G develops these generalizations further and examines their quantitative implications.

We proceed by quantifying the impact of the estimated rise of aggregate markups in the U.S. since 1980 on asset markets, inequality, output, and welfare. Beyond the rise of market power, there have been a number of other secular trends in recent history, such as the demographic transition (Eggertsson, Mehrotra and Robbins, 2019; Auclert et al., 2021; Hopenhayn, Neira and Singhania, 2022) and the increasing indebtedness of the U.S. government. To isolate the role of market power in explaining the evolution of macroeconomic variables over the past four decades, we start by calibrating our model to a baseline U.S. economy in 1980. We estimate the path of aggregate cost-weighted markups, following Ren and Zhang (2025), to compute the markup-implied path of national income shares annually between 1980 and 2018. Our model-implied labor share falls by 5.6 and the profit share rises by 8.4 percentage points, largely in line with literature estimates. Feeding our path of national income shares into the sufficient statistics representation yields the counterfactual model-implied path of rates, holding model primitives and other economic conditions fixed to the 1980 baseline.

Our first main quantitative finding is that the roughly 10 percentage point increase in aggregate markups over the past four decades has contributed to a 1 percentage point decline in the long-run rate of return r. Our model-implied path of the decline in r closely aligns with the pre-2008 trend decline in the long-run neutral rate r^* estimated in Holston, Laubach and Williams (2017), prior to the precipitous drop in r^* following the 2008 Global Financial Crisis and the onset of quantitative easing by the Federal Reserve. Our estimated decline in r coincides with an untargeted increase in top wealth concentration with the top 1 percent total wealth share in our model rising from about 23 in 1980 to 34 percent by 2018, compared with the World Inequality Database estimated increase from 23 to 35 percent over the same period. In our model, one of the primary drivers of the decline in r is the increasing share of income earned by high-saving business owners due to the redistributive effects of market power, providing a microfoundation for the recently observed savings glut of the rich Mian, Straub and Sufi (2021*b*).

While rising market power lowers long-run returns r and increases wealth concentration in our model, this is *not* a foregone conclusion of commonly used heterogeneous-agent models of consumption-savings (Aiyagari, 1994; Kaplan, Moll and Violante, 2018; Auclert, Rognlie and Straub, 2024; Auclert et al., 2025). We can interpret the reason for this through the *Redistribution* mechanism in our asset market channel. In models with one permanent household type, if profits, like wage income, are distributed in proportion to idiosyncratic productivity then swapping aggregate income shares between profits to labor does not change aggregate asset demand and is largely neutral for wealth inequality. This can be interpreted through our framework as setting $\lambda(r) = 1$, where pure redistribution between the labor and profit share is neutral for aggregate asset demand when households have identical claims and savings responses to both income sources. If instead aggregate profits are distributed lump-sum in environments with only idiosyncratic labor income risk, then they act as a form of insurance for households *reducing* asset demand and raising r [1] and

[1] A related insurance mechanism appears in heterogeneous-agent models with nominal price rigidities. As empha-

lowering wealth inequality.

Using data from the Survey of Consumer Finances 1983, we calibrate the initial $\lambda_0 = 3.31$ to match the relative financial wealth to income ratio of large-scale business owners ($\geq$\$250K annual business sales) to households that solely earn labor income. As in our model, this measure of wealth subtracts the value of privately held business equity because it is undiversified, risky, and relatively illiquid. These features of private business wealth provide reasons for private business owners to save into other forms of liquid financial wealth as a means of self-insurance, given that they tend to face substantially higher volatility in their business income than comparable workers as documented by DeBacker, Panousi and Ramnath (2023). Calibrated to λ_0 and estimates of idiosyncratic income risk, the stationary savings policy functions of our worker and owner types then supply the components of aggregate asset demand $\bar{D}(r), \lambda(r)$ to use in our sufficient statistics representation.

A common alternative assumption to the direct distribution of profits is that all profits are capitalized and paid out as dividends from a publicly traded asset (Eggertsson, Robbins and Wold, 2021; Moll, Rachel and Restrepo, 2022; Del Negro, Giannoni and Patterson, 2023; Auclert et al., 2025). In this case, rising markups disproportionately increase total asset supply as highlighted by the *Asset Creation* component of the asset market channel. We measure the publicly traded profit share parameter $\omega = 0.34$ in our model using IRS, BEA, and Compustat data along with literature estimates, finding that a majority share of aggregate profits are generated by private firms or closely held by corporate insiders of public companies, such as former founders and upper-level executives (Helwege, Pirinsky and Stulz, 2007; Eisfeldt and Shi, 2025; Peter, 2026). The consequence of the higher savings propensities of firm owners and the larger share of profits directly accruing to them is that rising markups increase asset demand relatively more than asset supply and lower the long-run equilibrium r.

Our second main finding is that the decline of r in our model mitigates 50 percent of the aggregate output distortion from the direct effect of markups on product markets because of the endogenous decline in the cost of capital. The rental cost of capital net of depreciation, equivalently the return on capital, equals the return on other assets in the economy through no-arbitrage. In the representative household benchmark, this return r is fixed in steady state $\beta(1 + r) = 1$, which shuts down any opportunity for changes in asset markets to feed back into product markets. However, in our model where r is variable, rising markups increase the effective capital that households are willing to supply for production, dampening the initial markup distortion on firms' capital demand.

In spite of this, the welfare loss across our model's population is significantly larger at -1.52 percent consumption-equivalent variation (CEV) units compared to the representative household, whose CEV declines by -0.32 percent as a consequence of the estimated rise of markups. The reason

sized by Werning (2015), lump-sum profit rebates along with the countercyclical markups implied by sticky prices result in procyclical household income risk, which attenuates the effects of forward guidance.

that welfare losses are larger in our model is because of the incidence of market power across the household income distribution, and in particular between workers and owners. The substantial capital savings of owners in response to markups supports the aggregate size of the economy, in terms of its long-run level of output. However, the shift in national income away from labor more than offsets the smaller decline in aggregate output, leaving workers relatively worse off. While the median CEV decline for the worker population is -1.7 percent, business owners gained substantially from the windfalls of rising markups with a median CEV increase of 19.5 percent relative to their baseline consumption allocation.

Our baseline model focuses on product market power in the form of markups, where the profits derived from market power increase the demand and supply for assets in a single asset market. We consider two extensions of our model framework. In the first extension, we consider an economy with a safe and a risky asset class and analyze how market power alters the composition of assets supplied. Given our estimated increase in markups, the supply of risky public equity expands, whereas the supply of the safe government debt in our counterfactual analysis, does not. The differential increase in the supply of risky assets relative to safe assets, raises the relative returns on risky assets required to clear asset markets, which drives up the equity risk premium by 43 basis points by 2018. However, because markups produce a concentrated increase in profits earned by high-saving business owners, aggregate demand for all asset classes increases, pushing the level of both the risky and safe returns downward with the long-run safe return declining by 123 basis points, a roughly 20 percent larger decline relative to our one asset benchmark r. To evaluate the resulting output distortion from market power in this setting, we allow the weighted average cost of capital to be determined by both the risky and safe rates in proportion to their historical financing shares as of 1980. Given the larger decline in safe rates in this multi-asset model relative to our one-asset baseline, the output loss is mitigated by 0.5 percentage point more. In the second extension, we study the asset market effects of an increase in labor market power through a rise in the aggregate wage markdown. The primary difference relative to the product market power case, is that markdowns do not directly distort the factor demand for capital, and hence there is no *Capital Wedge* effect on asset markets.

Related work. Our paper connects work on the macro finance consequences of market power with work on asset demand, asset supply, and the long run interest rate. Eggertsson, Robbins and Wold (2021) show that rising monopoly power, together with a declining real interest rate, can account for the rise in financial wealth and Tobin's Q. Farhi and Gourio (2018) explain related macro finance trends through market power, intangibles, and risk premia, while Ball and Mankiw (2023) characterize the wedge that market power creates between the interest rate earned by savers and the marginal product of capital. On the asset market side, Auclert et al. (2025) study a race between asset demand and asset supply, with inequality among the forces raising asset demand and capitalized profits among the components of asset supply. Rachel (2025) develops a related

capital market equilibrium framework for household saving, business capital, and government debt. In Eggertsson, Robbins and Wold (2021); Moll, Rachel and Restrepo (2022); Auclert et al. (2025), profits accrue as income on assets held by households. We distinguish private business income from tradable claims to that income and formalize the demand and supply effects of market power on asset markets through our *Redistribution* and *Asset Creation* mechanisms. The demand side is related to the saving glut of the rich in Mian, Straub and Sufi (2021*b*) and, more broadly, to their theory of how heterogeneous propensities to save shape equilibrium demand and the natural rate (Mian, Straub and Sufi, 2021*a*). Impullitti and Rendahl (2025) study a related Asset Creation mechanism in which higher market power raises profits, firm valuations, and asset returns.

We also build on quantitative work on the production costs of market power. Edmond, Midrigan and Xu (2023) decompose the cost of product markups into the aggregate markup, misallocation, and inefficient entry. Berger, Herkenhoff and Mongey (2022) estimate a general equilibrium model of oligopsony and quantify its output and welfare costs. These papers discipline the production side benchmark. Our departure is in the asset market. With imperfectly elastic asset demand, the asset market channel determines the equilibrium interest rate response, which changes the user cost of physical capital and mitigates the direct production loss. Gomez and Gouin-Bonenfant (2026) study an economy in which investment requires specialized labor in finite supply. In their model, the long run supply of physical capital is upward sloping, so shifts in desired saving move capital goods prices as well as quantities.

Our welfare analysis relates to a growing literature on how the distribution of income and firm ownership shapes the incidence of economic distortions across households. Boar and Midrigan (2025) place concentrated firm ownership at the center of product market policy. Atkin et al. (2025) trace the incidence of distortions through consumption, factor income, firm ownership, taxes, and production networks. Their implementation distributes firm profits to owners as income, while noting that profits retained inside firms instead raise ownership values until dividends are paid or firms are sold. In our framework, this distinction matters for asset market clearing: profit income received by owners supports asset demand, while tradable equity enters asset supply. Incorporation links the two sides of the asset market through the liquid proceeds received by the founders who sell their businesses. Our measurement of *Redistribution* builds on evidence that business income and wealth are concentrated among active owners (Smith et al., 2019; Smith, Zidar and Zwick, 2023) and that income received by owners of pass through businesses is risky and persistent (DeBacker, Panousi and Ramnath, 2023). Models of owner saving explicitly account for uninsurable business risk (Quadrini, 2000; Cagetti and De Nardi, 2006; Boar, Gorea and Midrigan, 2026), while Di Tella, Malgieri and Tonetti (2025) develop a related theory in which markups compensate owners for business risk. Bhandari and McGrattan (2021) document the large value and limited transferability of private business sweat equity, supporting the distinction between rents retained in private ownership and profits backing tradable equity.

Outline. Section 2 connects firm level market power to aggregate production distortions, shows when misallocation leaves the interest rate unchanged, and derives how the interest rate response changes output. Section 3 develops the incorporation economy and the sufficient-statistic characterization of asset market clearing. Section 4 measures its inputs and quantifies the markup counterfactual, including the nonlinear transition. Section 5 studies household welfare and its sources. Section 6 examines multiple assets and wage markdowns, and Section 7 concludes.

2 Market Power and the Asset Market Channel

We begin by characterizing how firm level market power affects aggregate production and the division of national income. Changes in technology, firm conduct, or market structure affect these outcomes through the aggregate markup μ and effective productivity A. While both affect output, we show that under our baseline assumptions only the aggregate markup changes the stationary equilibrium interest rate. We then characterize how the resulting change in the cost of capital mitigates or amplifies the production distortions of market power. Section 3 develops the household and ownership structure that determines this interest rate response.

2.1 From firm-level market power to aggregate wedges

Following Edmond, Midrigan and Xu (2023), we take firm level markups as given. Firms indexed by i produce differentiated goods using $y_i = z_i F(k_i, \ell_i)$, where F is a common constant returns technology. Final output is a homothetic aggregate of firm outputs, with its price normalized to one. Capital and labor are competitively supplied at rental rate r^k and wage w. The product markup $\mu_i \geq 1$ is the ratio of price to marginal cost, so the firm optimality conditions are

$$r^k = \frac{p_i z_i}{\mu_i} F_K(k_i, \ell_i), \qquad w = \frac{p_i z_i}{\mu_i} F_L(k_i, \ell_i). \tag{2}$$

Taking their ratio eliminates the markup, price, and productivity. The common relative factor price therefore determines a common capital labor ratio across firms.

Let $s_i = p_i y_i / Y$ denote firm i's revenue share. The aggregate markup is

$$\frac{1}{\mu} = \int s_i \frac{1}{\mu_i} \, di, \tag{3}$$

the sales weighted harmonic mean of firm level markups, equivalently their cost weighted arithmetic mean (Edmond, Midrigan and Xu, 2023). The common input mix allows the firm conditions to aggregate to

$$Y = AF(K, L), \qquad r^k = \frac{A}{\mu} F_K(K, L), \qquad w = \frac{A}{\mu} F_L(K, L), \tag{4}$$

where A is effective productivity. Define the income shares $s_K = r^k K / Y$, $s_W = W / Y$, and $s_\Pi = \Pi / Y$,

with aggregate labor income $W = wL$ and aggregate profit income $\Pi = Y - W - r^k K$. Writing the production elasticities as $\varepsilon_K = F_K K/F$ and $\varepsilon_L = F_L L/F$, evaluated at (K, L), gives

$$s_K = \frac{\varepsilon_K}{\mu}, \qquad s_W = \frac{\varepsilon_L}{\mu}, \qquad s_\Pi = 1 - \frac{1}{\mu}, \tag{5}$$

where constant returns implies $\varepsilon_K + \varepsilon_L = 1$. Write effective productivity as $A = \Omega \mathcal{Z}^*$, where $\mathcal{Z}^*$ is frontier productivity and $0 < \Omega \leq 1$ measures the productivity consequences of misallocation across firms (Hsieh and Klenow, 2009; Restuccia and Rogerson, 2013; Edmond, Midrigan and Xu, 2023). Within the maintained production and demand environment, changes in technology, conduct, or market structure can move μ, Ω, and $\mathcal{Z}^*$. The aggregate production wedge μ distorts factor payments relative to marginal products, while Ω and $\mathcal{Z}^*$ jointly determine effective productivity A. Given μ and A, (4) summarizes the production conditions. For example, among the mechanisms studied by De Loecker, Eeckhout and Mongey (2026) is a decline in the number of potential competitors. This change weakens competition and alters both aggregate markups and the allocation of production across heterogeneous firms.

In the homothetic product market environments we consider[2], a change in aggregate factor prices scales the common component of firms' costs while leaving relative costs unchanged. The interest rate therefore changes aggregate inputs without changing the relative allocation of production. Section 6.2 considers labor market power, which adds a wedge to the labor condition in (4).

2.2 Productivity, misallocation, and long run interest rate

To characterize the asset market channel through μ, we must account for accompanying changes in effective productivity $A = \Omega \mathcal{Z}^*$. Such changes leave the stationary interest rate unchanged when asset demand and asset supply scale together. At each candidate interest rate r, the production conditions (4), capital rental rate $r^k = r + \delta_K$ with capital depreciation δ_K, and labor supply determines productive capital K and the associated income flows W and Π. Given these income flows, the household and ownership blocks determine desired asset holdings and the value of tradable profit claims. Let $D(r; \mu, A)$ and $S(r; \mu, A)$ denote the resulting stationary asset demand and asset supply, with supply including physical capital and tradable equity. Next, we show that, under our baseline environment, the asset market condition is invariant to changes in effective TFP.

Proposition 1 (Effective productivity and the interest rate)**.** *Hold μ, the production function F, the labor supply schedule, and household and ownership primitives fixed. If $D(r; \mu, A)$ and $S(r; \mu, A)$ are homogeneous of the same degree in A, changes in effective productivity leave the set of stationary equilibrium*

[2]These environments encompass many commonly used models of product demand and competition. Examples include the HSA and HDIA classes characterized by Matsuyama and Ushchev (2017), which contain familiar demand systems such as translog and generalized Kimball, as well as standard oligopolistic environments such as Atkeson and Burstein (2008). Entry, heterogeneous financing costs, or other scale-dependent forces can break this separation.

interest rates unchanged. If the equilibrium interest rate is unique, it is independent of A.

Our baseline environment satisfies this condition. With Cobb–Douglas production, and standard preference specifications governing labor supply, including separable CRRA and GHH preferences, aggregate labor income W, aggregate profit income Π, and productive capital K all scale with A. Section 3 establishes that stationary household asset demand is homogeneous in income, including proceeds from incorporation, building on Straub (2019). Tradable profit claims inherit the same scaling. Asset demand and asset supply are therefore homogeneous of the same degree in A.

Therefore, changes in misallocation that operate through Ω, or changes in frontier technology that operate through Hicks–neutral productivity $\mathcal{Z}^*$, affect output without generating a stationary interest-rate response. More generally, for primitive changes whose relevant aggregate effects are summarized by (μ, A), only the induced change in μ affects the stationary interest rate. This separation holds conditional on normalized income risk and the ownership structure, as shown in Section 3. If changes in effective TFP directly affect factor shares, however, this homogeneity property need not hold exactly. Appendix A.2 illustrates one such departure under CES production and shows that the resulting deviation from the homogeneous benchmark is quantitatively very small.

2.3 Output and the asset market channel

The previous subsection showed that, under our baseline environment, changes in effective productivity A affect output but do not generate a long run interest rate response. We now show how A, the aggregate production wedge μ, and the equilibrium interest rate jointly determine output. This makes clear why the interest rate response is central to the long run aggregate consequences of market power. For the output mapping, we specialize the labor supply to the GHH preferences used in our framework and specified formally in Section 3. Let φ denote the associated Frisch elasticity, so that

$$\varphi \equiv \frac{d \log L(w)}{d \log w}.$$

Let $\xi \equiv -\frac{\partial \log Y}{\partial \log r^k} > 0$ be the local output response to the capital user cost. The production conditions then imply

$$d \log Y = \underbrace{d \log A}_{\text{direct productivity}} + \underbrace{\varphi \left(d \log A - d \log \mu \right)}_{\text{labor adjustment}} + \underbrace{\xi \left(d \log A - d \log \mu - d \log r^k \right)}_{\text{capital adjustment}}, \tag{6}$$

where user cost of capital is $r^k = r + \delta_K$[3]. Equation (6) separates three margins. Higher effective productivity raises output directly and induces greater use of labor and capital. At a fixed interest rate, a higher markup works in the opposite direction: it lowers the wage and desired capital, reducing labor input and capital intensity. The asset market matters through the last term. If the equilibrium interest rate falls, the capital rental rate falls and firms use more capital than they would at a fixed interest rate, mitigating the output loss from the production distortion. If the interest rate rises, the same feedback amplifies the loss. Since $r^k = r + \delta_K$,

$$d \log r^k = \frac{r}{r + \delta_K} d \log r.$$

Thus, once (A, μ) are given, the remaining equilibrium object needed to determine the output response is the response of the interest rate.

The two roles of the markup. The aggregate markup enters the economy in two conceptually distinct ways. First, it distorts firms' factor demand: from (4), a higher μ lowers factor payments relative to marginal products and therefore depresses labor and capital use at a given interest rate. Second, the same markup determines how much output accrues as profits. Under constant returns,

$$s_K + s_W = \frac{1}{\mu}, \qquad s_\Pi = 1 - \frac{1}{\mu}.$$

A higher markup therefore not only creates a production distortion, but also shifts income away from factor payments and toward profits.

In the representative-household benchmark, these two roles largely separate. Because factor payments and profits ultimately accrue to the same household, only the production-wedge role matters for aggregate output. With heterogeneous ownership and imperfectly elastic asset demand, however, this separation breaks down. Who receives the profits affects household asset demand, while their capitalization into tradable claims affects asset supply. The incidence and capitalization of profits therefore determine the equilibrium interest rate that enters (6), and hence how strongly the production distortion translates into lower capital and output. Section 3 turns to this asset-market determination of the interest rate.

3 Households, Firm Ownership, and Asset Market Equilibrium

Section 2 characterizes the production distortions of market power and its effects on the division of national income. With inelastic asset demand, the distribution of income affects the equilibrium interest rate through both the savings of households receiving profits and the capitalization of those

[3]Here φ is the Frisch elasticity of labor supply. $\xi \equiv -\partial \log Y / \partial \log r^k = (\sigma_{KL} + \varphi) s_K / s_W > 0$ is the elasticity of output with respect to the capital user cost, where σ_{KL} is the local elasticity of substitution between capital and labor. It captures both the direct adjustment in capital intensity and the induced labor response through wages.

profits into tradable equity. Models of business owner saving emphasize profits received directly as income (Quadrini, 2000; Cagetti and De Nardi, 2006; Mian, Straub and Sufi, 2021*b*), while other models assume that future profits are capitalized into publicly traded equity (Eggertsson, Robbins and Wold, 2021; Auclert et al., 2025). Our model connects these mechanisms by allowing private businesses to go public. Workers earn labor income, while business owners initially receive profits directly from their private businesses. When a business goes public, its owner sells a share of future profits as tradable equity and retains the remaining share. The sale increases asset supply while providing the owner with liquid financial wealth that supports asset demand. Our framework accounts for both effects when characterizing how the distribution of factor income and profits affects the equilibrium interest rate.

Household problem Aggregate output is divided into labor income W, capital payments $r^k K$, and profits $\Pi = Y - W - r^k K$, where $r^k = r + \delta_K$ is the rental cost of capital. There are two household types, workers and business owners, who take aggregate labor income W, aggregate profit income Π, and the interest rate r as given. We use the subscript O to denote business owners.

Workers consume, save, and supply labor. They have GHH preferences over the composite $c - v(\ell)$, with isoelastic labor disutility

$$v(\ell) = \frac{\psi_\ell}{1 + 1/\varphi} \ell^{1+1/\varphi}.$$

With wage per efficiency unit w, idiosyncratic efficiency e, and $R = 1 + r$, a worker solves

$$\begin{aligned} V^W(a, e) = &\max_{c,\, a',\, \ell} \left\{ u(c - v(\ell)) + \beta_W \, \mathbb{E}\left[V^W(a', e') \mid e \right] \right\}, \\ c + a' = &Ra + we\ell, \qquad a' \geq 0. \end{aligned}$$

The GHH structure makes the intratemporal labor choice independent of wealth: $v'(\ell^*(w, e)) = we$. Define the maximized labor surplus

$$y^W(w, e) := \max_{\ell} \{ we\ell - v(\ell) \}.$$

With the isoelastic v, both $y^W(w, e)$ and aggregate labor compensation W scale with $w^{1+\varphi}$. Hence $y^W(w, e) = W\widetilde{e}$ for a normalized Markov state $\widetilde{e}$ independent of the aggregate scale. This scaling property allows us to express worker asset demand per dollar of aggregate labor income, as we show below.

Owners initially hold private businesses. A private business generates profit income Πz, where z is the owner's share of aggregate profits. An owner with share z receives Πz, and the distribution

of z determines the distribution of profits across owners without changing aggregate profits:

$$\pi = \Pi z, \qquad \int z \, di = 1.$$

The state z is stochastic with mean normalized to one. Appendix A.3 derives this process from relative demand shocks that change the owner's share of aggregate profits. A private business goes public at an iid rate κ, and publicly incorporated businesses exit at an iid rate δ_C. Upon public incorporation, the owner sells the fraction $1 - \theta_F$ of future profits at an equilibrium price $q(z; \Pi, r)$ and retains a nontradable claim to the remaining fraction θ_F. While the business remains private, the owner's problem is

$$\begin{aligned} V^P(a, z) = &\max_{c,\, a'} \left\{ u(c) + \beta_O \Big[(1-\kappa) \mathbb{E}\left[V^P(a', z') \mid z \right] + \kappa V^L\big(q(z; \Pi, r), a', z\big) \Big] \right\}, \\ c + a' = &Ra + \Pi z, \qquad a' \geq 0 \end{aligned}$$

Private business ownership corresponds to the closely held equity studied by Bhandari and McGrattan (2021). The owner retains nontradable equity after public incorporation, consistent with the insider equity holdings studied by **?**. We use $V^L(\mathcal{T}, a, z)$ for the state after public incorporation, where $\mathcal{T}$ records the one-period sale proceeds. In the period of public incorporation $\mathcal{T} = q(z; \Pi, r)$; thereafter $\mathcal{T} = 0$. While the publicly incorporated business survives, the owner receives retained profits $\theta_F \Pi z$ and the business continues to face idiosyncratic profit risk:

$$\begin{aligned} V^L(\mathcal{T}, a, z) = &\max_{c,\, a' \geq 0} \left\{ u(c) + \beta_O \left[\delta_C \, \mathbb{E}_\pi V^P(a', z') + (1 - \delta_C) \mathbb{E}\left[V^L(0, a', z') \mid z \right] \right] \right\}, \\ c + a' = &Ra + \mathcal{T} + \theta_F \Pi z, \qquad a' \geq 0. \end{aligned}$$

Upon exit, the retained claim disappears and the owner draws a new private business opportunity. Setting $\theta_F = 0$ eliminates retained equity and makes z irrelevant in the state after public incorporation. In our baseline model, business owner status, the profit share process, and public incorporation are exogenous. These assumptions allow us to isolate how profits contribute to asset demand and asset supply. Allowing for endogenous entry, financing, or public incorporation would affect the distribution of profits and the household saving responses that enter our sufficient statistics representation.

3.1 From private business income to publicly traded equity

Profits from private businesses and retained equity are received directly as income by business owners. When a firm goes public, the owner also receives the proceeds from the sale of equity, while the equity purchased by other households enters asset supply. The market value of this tradable equity determines the contribution of Asset Creation to aggregate asset supply.

Let $q(z; \Pi, r)$ denote the ex dividend value of tradable equity in a publicly incorporated business

with business profit share z. It satisfies

$$(1+r)\,q(z;\Pi,r) = \mathbb{E}\left[(1-\theta_F)\Pi z' + (1-\delta_C)\,q(z';\Pi,r) \mid z\right]. \tag{7}$$

The equity value reflects the next dividend payment and the expected continuation value, accounting for the probability of firm exit. Averaging over the stationary distribution of z, which has mean one, gives the market value of tradable equity per dollar of publicly traded profit income (for $1-\theta_F > 0$):

$$Q(r) \equiv \frac{\mathbb{E}_\pi\, q(z;\Pi,r)}{(1-\theta_F)\Pi} = \frac{1}{r+\delta_C}. \tag{8}$$

Thus $Q(r)$ values each dollar of publicly traded profit income at the capitalization rate $r+\delta_C$. We use $Q(r) = 1/(r+\delta_C)$ also when no equity is marketed, in which case the publicly traded profit share is zero. A constant risk premium or expected cash flow growth would enter through the corresponding $Q(r)$; the baseline sets those terms to zero (Farhi and Gourio, 2018). We allow for a risk premium in the two asset extension below, which is developed further in Appendix H.

Let ω^{inc} denote the share of aggregate profits generated by publicly incorporated businesses. We assume that a firm's probability of public incorporation is independent of its business profit share z. Under this assumption, the publicly incorporated profit share equals the stationary share of publicly incorporated businesses derived in Appendix A.4. The publicly traded equity share of aggregate profits is

$$\omega = (1-\theta_F)\omega^{\text{inc}}, \qquad \omega^{\text{inc}} = \frac{\kappa}{\kappa+\delta_C}. \tag{9}$$

Without independence, ω^{inc} is the profit-weighted share of publicly incorporated businesses. Aggregate profits divide into private, retained, and publicly traded flows:

$$1 = (1-\omega^{\text{inc}}) + \theta_F\omega^{\text{inc}} + \omega.$$

Tradable claims add $Q(r)\omega\Pi$ to asset supply. Retained profits are received directly as income by the owner and do not create an additional tradable asset.

3.2 Aggregate asset demand and sufficient statistics

We next characterize aggregate asset demand for workers and business owners. Despite idiosyncratic income risk, borrowing constraints, and changes in firm ownership, household asset demand remains linear in each group's aggregate income. We can therefore express asset demand per dollar of aggregate labor income for workers and aggregate profit income for business owners. This property also extends to environments with aggregate risk, multiple assets, and balanced growth, which we discuss below.

Proposition 2 (Linearity of household asset demand)**.** *Suppose household preferences are homothetic and the normalized worker and business owner problems admit stationary distributions with finite mean*

asset holdings. Then stationary group asset demands are linear in their respective income scales:

$$D_W(r, W) = \bar{D}(r)W, \qquad D_O(r, \Pi) = \bar{D}_O(r)\Pi \equiv \lambda(r)\bar{D}(r)\Pi.$$

Here $\bar{D}(r)$ and $\bar{D}_O(r)$ are asset demand per dollar of worker labor compensation and business owner profit income, respectively, and

$$\lambda(r) \equiv \frac{\bar{D}_O(r)}{\bar{D}(r)}$$

is the relative wealth-to-income ratio of owners to workers.

Idiosyncratic income risk, borrowing constraints, retained profits, IPO proceeds, entry, and exit affect household saving without changing the linearity of asset demand in aggregate income. Worker asset demand per dollar of income is summarized by $\bar{D}(r)$, while $\lambda(r)$ is the relative wealth-to-income ratio of owners to workers. This result builds on the homothetic linearity results used in the wealth distribution literature (Constantinides and Duffie, 1996; Angeletos, 2007; Straub, 2019; Gomez, 2023). Appendix C states the broader homogeneity principle and shows how the same structure arises in Aiyagari, perpetual youth OLG, and asset in utility economies. Our sufficient statistics representation therefore applies beyond the particular household and ownership structure of our baseline model.

Aggregating asset demand. Aggregating worker and business owner asset demand gives:

$$D(r, W, \Pi) = \bar{D}(r)\big[W + \lambda(r)\Pi\big].$$

Asset supply consists of physical capital and tradable equity:

$$S(r) = K + \underbrace{Q(r)\,\omega\,\Pi}_{\text{tradable equity}}. \tag{10}$$

Private and retained profits, together with the proceeds from going public, provide resources that business owners can save and therefore contribute to asset demand. Profits capitalized into tradable equity instead contribute to asset supply. Combining these expressions yields our sufficient statistics representation of asset market clearing.

Theorem 1 (Asset Market Clearing)**.** *In a stationary equilibrium, asset market clearing per unit of output is*

$$\underbrace{\bar{D}(r)\big[s_W + \lambda(r)s_\Pi\big]}_{\textit{asset demand}} = \underbrace{\frac{1}{r + \delta_K}s_K + Q(r)\omega s_\Pi}_{\textit{asset supply}}, \tag{11}$$

$$Q(r) = \frac{1}{r + \delta_C}.$$

Here (s_K, s_W, s_Π) are the capital, labor, and profit shares.

Capital Wedge, Redistribution, and Asset Creation Equation (11) groups the asset market effects of market power into three forces. The Capital Wedge operates through the physical capital supply term $s_K/(r+\delta_K)$: a lower capital share reduces the supply of physical capital and lowers the equilibrium interest rate. Redistribution operates through aggregate asset demand $\bar{D}(r)[s_W + \lambda(r)s_\Pi]$, so the lower labor share and higher profit share enter jointly on the demand side. Asset Creation operates through $Q(r)\omega s_\Pi$: higher profits raise the value of tradable equity and shift asset supply outward. The equilibrium interest rate therefore depends on the balance of these three forces.

Once the asset demand schedule $\bar{D}(r)$, the factor shares (s_W, s_K, s_Π), and the relative wealth-to-income ratio, ownership share, and valuation multiple (λ, ω, Q) are known, equation (11) determines the long run equilibrium interest rate. Section 3.3 uses this characterization to explain how these forces determine the direction of the interest rate response.

3.3 Asset Market Equilibrium: Redistribution and Asset Creation

Market power affects asset demand through the distribution of factor income and profits and asset supply through physical capital and tradable equity. Combining the factor shares from Section 2 with asset market clearing from Section 3 allows us to characterize the resulting interest rate response.

We consider two special cases that isolate the Redistribution and Asset Creation mechanisms. Under pure Redistribution, profits are received directly as income by business owners and support asset demand. Under pure Asset Creation, owners sell tradable equity and draw down the proceeds, while the equity remains in asset supply. With inelastic asset demand, these cases imply opposite interest rate responses to the same increase in market power.

Assumption 1 (A transparent benchmark)**.** *(i) There is no depreciation: $\delta_K = 0$. (ii) Claims are priced at $Q(r) = 1/r$, with $r > 0$. (iii) Technology is Cobb–Douglas with capital elasticity $\alpha \in (0,1)$. (iv) Worker asset demand satisfies $d[r\bar{D}(r)]/dr > 0$ over the relevant range of returns.*

Under Assumption 1, asset market clearing per unit of output is

$$\underbrace{\bar{D}(r)\,(s_W + \lambda\, s_\Pi)}_{\text{asset demand per unit of output}} \quad = \quad \underbrace{\frac{s_K + \omega\, s_\Pi}{r}}_{\text{asset supply per unit of output}} .$$

Workers receive the labor share s_W. Profits support asset demand through λs_Π and add tradable equity to asset supply through $\omega s_\Pi/r$. Both contributions are present in the general model, while the special cases isolate each mechanism. Cobb–Douglas technology makes the share response to a product markup explicit:

$$s_K = \frac{\alpha}{\mu}, \qquad s_W = \frac{1-\alpha}{\mu}, \tag{12}$$

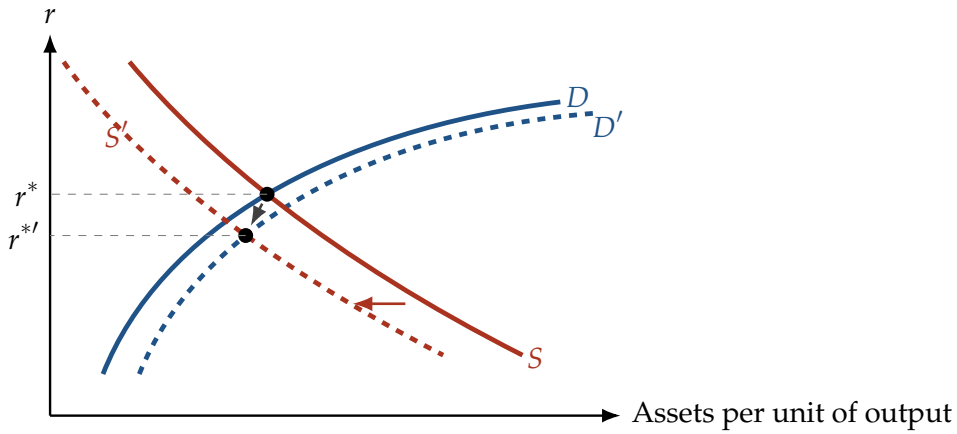


Figure 1: Redistribution. A higher markup lowers the capital share s_K and raises the noncapital income share $s_W + s_\Pi$. Redistribution shifts asset demand outward, the lower capital share reduces asset supply from productive capital, and the equilibrium interest rate falls from r^* to $r^{*\prime}$.

with profits accounting for the remaining share of output. A higher markup reduces both factor shares and raises the profit share. The interest rate response depends on whether the additional profits accrue directly to owners or are capitalized into tradable equity.

Redistribution ($\omega = 0$, $\lambda = 1$). Profits remain private income, and owners and workers demand the same assets per dollar of income. Asset market clearing becomes

$$\bar{D}(r)(s_W + s_\Pi) = \frac{s_K}{r} \quad \Rightarrow \quad r\bar{D}(r) = \frac{s_K}{1 - s_K} = \frac{\alpha}{\mu - \alpha}. \tag{13}$$

Physical capital is the only asset supplied, while the combined labor and profit share, $s_W + s_\Pi = 1 - s_K$, supports asset demand. A higher markup lowers s_K. Redistribution shifts asset demand outward, and the Capital Wedge reduces asset supply from physical capital. At the initial interest rate, asset demand therefore exceeds asset supply, so the equilibrium interest rate falls. Figure 1 shows the decline from r^* to $r^{*\prime}$.

Asset Creation ($\omega = 1$, $\lambda = 0$). Set $\theta_F = \delta_C = 0$ and maintain $\kappa > 0$. Every business eventually incorporates into a publicly traded business, its owner sells the entire claim once, and the claim remains permanently tradable. With $\beta_O(1 + r) < 1$, owners draw down their assets, yielding $\lambda = 0$ at the endpoint (Lemma 1). Asset market clearing becomes

$$\bar{D}(r)\, s_W = \frac{s_K + s_\Pi}{r} \quad \Rightarrow \quad r\bar{D}(r) = \frac{s_K + s_\Pi}{s_W} = \frac{1 - s_W}{s_W} = \frac{\mu - 1 + \alpha}{1 - \alpha}. \tag{14}$$

Profits now contribute to asset supply through tradable equity. A higher markup lowers the labor share supporting asset demand and raises total asset supply, $s_K + s_\Pi = 1 - s_W$. At the initial interest rate, asset supply therefore exceeds asset demand, so the equilibrium interest rate rises for every $\alpha \in (0, 1)$. The two special cases coincide at $\mu = 1$, where both give $r\bar{D}(r) = \alpha/(1 - \alpha)$, and

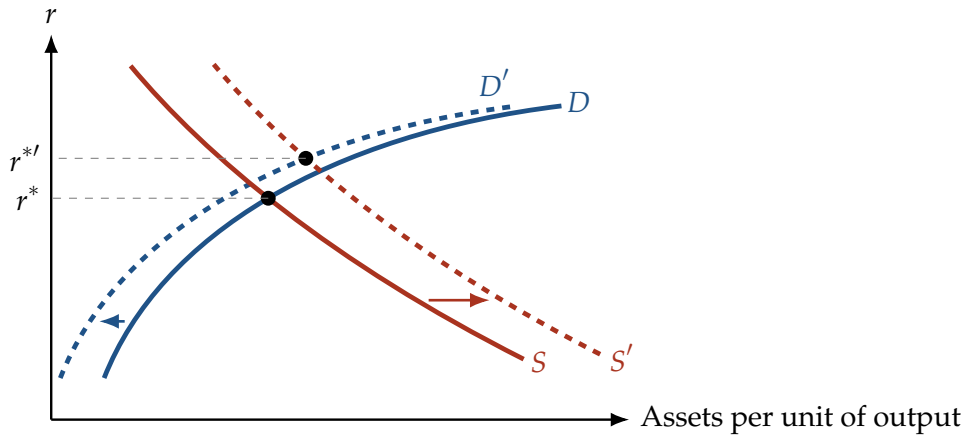


Figure 2: Asset Creation. A higher markup lowers the labor share that supports asset demand and raises asset supply from tradable equity. Asset demand shifts inward, asset supply shifts outward, and the equilibrium interest rate rises from r^* to $r^{*\prime}$.

imply different interest rate responses when profits are positive. Figure 2 shows the increase from r^* to $r^{*\prime}$.

Between the special cases. The economy with public incorporation in Theorem 1 nests both special cases, with their relative importance determined by the model primitives. Without public incorporation, $\kappa = 0$, no profits are capitalized into tradable equity: $\omega = 0$, profits are received directly by owners, yielding pure Redistribution (the illustration above additionally assumes $\lambda = 1$). Pure Asset Creation requires two additional conditions. The claim must be complete and permanent, $\theta_F = 0$ and $\delta_C \downarrow 0$ at fixed $\kappa > 0$, so that $\omega \to 1$ and $Q(r) \to 1/r$. Retained equity continues to generate owner income for the life of the firm and supports asset demand in the same way as private profit income. A positive θ_F therefore allows Redistribution to operate after public incorporation. In addition, owners must draw down the proceeds from the equity sale. At the time of an IPO, the purchaser acquires the equity and the owner receives the same value in sale proceeds. The increase in asset supply is therefore initially matched by the resources received by the owner. With $\beta_O R < 1$, which precautionary saving delivers in a stationary incomplete-market economy, owners consume the proceeds over time while the equity remains in asset supply; Proposition 3 shows that owner asset demand, and with it λ, is of order δ_C. Were $\beta_O R = 1$, the owner would hold the proceeds permanently. The resulting asset demand would exactly offset the additional asset supply, eliminating the net effect of Asset Creation. Retained equity, firm exit, and owner time preferences jointly determine how profits contribute to asset demand and asset supply. Our calibrated model includes private profit income, retained equity, and firm exit, allowing both mechanisms to operate.

Labor market power. A wage markdown allows us to isolate the role of the relative wealth-to-income ratio of owners to workers. It leaves the capital share unchanged and redistributes

income from labor to profits, dollar for dollar. In the Redistribution benchmark with $\lambda(r) = 1$ this redistribution has no effect on asset demand: workers and owners save the same amount per dollar, so aggregate asset demand and the interest rate do not move. With $\lambda(r) > 1$ the same transfer raises asset demand and lowers the interest rate, and the Capital Wedge does not contribute to the interest rate response. Section 6.2 quantifies this mechanism using the estimated U.S. markdown path.

The relative wealth-to-income ratio of owners to workers. The two special cases set λ at one and at zero. In the data, wealthy business owners hold substantially more assets per dollar of income than workers (Straub, 2019; Mian, Straub and Sufi, 2021*b*). We capture this difference with the relative wealth-to-income ratio of owners to workers, $\lambda(r)$: when $\lambda(r) > 1$, a dollar of business owner profit income generates more asset demand than a dollar of worker labor income. A larger $\lambda(r)$ therefore strengthens Redistribution. On the supply side, the publicly traded equity share of profits ω and the valuation multiple $Q(r)$ govern Asset Creation.

A higher markup lowers the labor and capital shares and raises the profit share. The additional profit income enters asset demand with weight $\lambda(r)$, while tradable equity enters asset supply with weight $Q(r)\omega$. The balance between these two terms determines the sign of the equilibrium interest rate response.[4] The asset demand semi elasticity then governs the size of the interest rate response. It combines the return responses of $\bar{D}(r)$ and $\lambda(r)$, holding factor shares fixed.

3.4 Beyond the baseline

Our sufficient statistics representation extends to environments with fiscal policy, more general production technologies, growth, and multiple assets. In each case, market power affects the distribution of income supporting asset demand and the assets available for households to hold. The extensions modify the income shares, household saving responses, or asset valuations that enter the representation. Section 4 examines their quantitative implications, while the appendices provide the derivations.

Fiscal policy. With taxes on labor, capital, and profits, together with government debt, asset market clearing becomes

$$\underbrace{\bar{D}(r)\left[\widetilde{s}_W + \lambda(r)\widetilde{s}_\Pi\right]}_{\text{asset demand}} = \underbrace{\frac{\widetilde{s}_K}{r} + Q(r)\omega\widetilde{s}_\Pi + b}_{\text{asset supply}}.$$

Here $\widetilde{s}_W$ and $\widetilde{s}_\Pi$ are after tax labor and profit income shares, $\widetilde{s}_K = rK/Y$ is net capital income paid to savers, and $b = B/Y$ is government debt relative to output. Taxes change the income supporting

[4] With a product markup and the local stability condition in Appendix A.8, the interest rate falls when $\lambda > (1-\alpha)Q(r)\omega(r+\delta_K)/\alpha$. In our standard calibration, the threshold is 0.79, well below the measured value $\lambda = 3.31$.

asset demand, while government debt adds asset supply. Debt service links the two sides through the government budget. Appendix G.2 derives this fiscal sufficient statistic.

Balanced growth. On a balanced growth path, asset market clearing is

$$\underbrace{\bar{D}(r,g)\left[s_W + \lambda(r,g)s_\Pi\right]}_{\text{asset demand}} = \underbrace{\frac{s_K}{r+\delta_K} + \omega Q^g(r,g)s_\Pi}_{\text{asset supply}}.$$

Here g is trend growth and $Q^g(r,g)$ is the valuation multiple for growing publicly traded profit income. Growth changes household asset demand and the duration and value of tradable equity. Appendix G.3 derives $\bar{D}(r,g)$ and $Q^g(r,g)$.

CES production. For the markup experiment with CES production, the corresponding equation at date t is

$$\underbrace{\bar{D}(r)\left[s_{W,t}(r) + \lambda(r)s_{\Pi,t}\right]}_{\text{asset demand}} = \underbrace{\frac{s_{K,t}(r)}{r+\delta_K} + Q(r)\omega s_{\Pi,t}}_{\text{asset supply}}.$$

The labor and capital shares now depend on the equilibrium interest rate. Appendix G.4 gives the exact share functions.

Misallocation. When misallocation affects factor shares, asset market clearing becomes

$$\underbrace{\bar{D}(r)\left[s_W(\Omega,r) + \lambda(r)s_\Pi(\Omega,r)\right]}_{\text{asset demand}} = \underbrace{\frac{s_K(\Omega,r)}{r+\delta_K} + Q(r)\omega s_\Pi(\Omega,r)}_{\text{asset supply}}.$$

In our baseline model, misallocation changes effective TFP without changing factor shares. When factor shares also respond to misallocation, these changes can shift asset demand and asset supply. Appendix A.2 derives this interaction.

Multiple assets. With several assets, the scalar sufficient statistic becomes a vector system:

$$\underbrace{\mathbf{D}(\mathbf{r})}_{\text{asset demand}} = \underbrace{\mathbf{S}(\mathbf{r})}_{\text{asset supply}}.$$

The distribution of income continues to determine total asset demand, while household portfolio choices determine demand for each asset. The discount rate for each asset and firms' financing structures determine the corresponding asset supplies. The two asset extension in Section 6.1 separates the safe return, the equity required return, and the cost of capital relevant for firms' production decisions. Appendix H derives household portfolio choices and the resulting market clearing system.

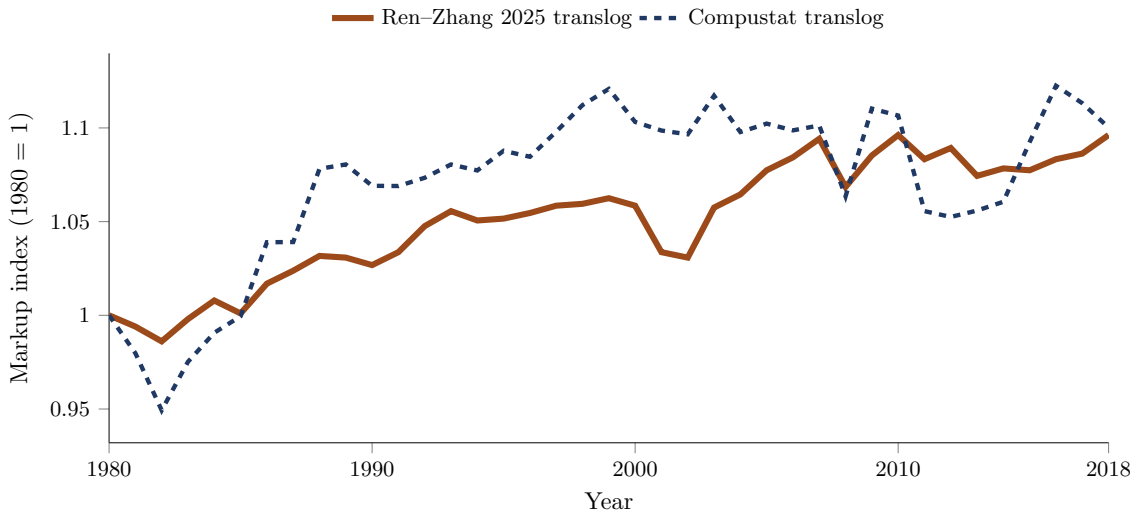


Figure 3: The path of market power. The solid line is the baseline: the index of the measured wedge estimated by Ren and Zhang (2025) on Census matched firms with COGS as the flexible input, built from sales weighted harmonic means of firm level estimates within two digit industries. The dashed line is a translog estimate on the full Compustat sample, the sales weighted harmonic mean across all firms. Both series are normalized to one in 1980.

4 Quantification

We now use our sufficient-statistic characterization of asset market clearing in Theorem 1 to quantify the effects of the secular increase in market power on long-run output and interest rates. Using U.S. data on production costs, inputs, and sales, we estimate the path of annual, aggregate cost-weighted markups μ using the production function approach. We measure the relative wealth-to-income ratio of business owners to workers λ using household balance sheet data and ω using the allocation of aggregate profits earned by U.S. private businesses relative to publicly traded corporations. We then estimate the counterfactual contributions[5] of the estimated path of markups μ starting from a 1980 baseline calibration of the U.S. economy on steady state r and Y year-by-year, as in Auclert et al. (2025); De Loecker, Eeckhout and Mongey (2026), and compare this sequence of steady state allocations to those obtained from the actual transition dynamics given various assumptions on households' foresight with respect to the estimated path of markups $\{\hat{\mu}_t\}_{t\geq 1980}$.

4.1 Estimating the path of market power

To quantify the consequences of aggregate market power, we require a time series for the aggregate markup μ. With heterogeneous firm level markups, the relevant aggregate object for factor payments is the cost weighted markup in (3), equivalently the sales weighted harmonic

[5]That is, holding fixed demographics, fiscal policy, and other important determinants of long-run interest rates (Rachel and Summers, 2019; Auclert et al., 2021; Platzer and Peruffo, 2022; Kopecky and Taylor, 2022).

mean of firm level markups:

$$\mu_t = \left(\sum_i s_{it} \mu_{it}^{-1} \right)^{-1} = \frac{\sum_i p_{it} y_{it}}{\sum_i p_{it} y_{it} / \mu_{it}}, \qquad s_{it} \equiv \frac{p_{it} y_{it}}{\sum_j p_{jt} y_{jt}}. \tag{15}$$

We therefore use this measure throughout the quantification rather than the sales-weighted arithmetic markup commonly reported in the markup literature. We use the estimated series from Ren and Zhang (2025), which estimates markups and markdowns using publicly traded firms matched to the confidential Census Longitudinal Business Database from 1977 to 2019. The Census data provide payroll information that is largely unavailable in Compustat. This matters because COGS includes labor expenditures, so conventional markup measures can reflect both price markups and wage markdowns. After separating the two sources of market power, the secular rise in market power is concentrated in wage markdowns, while price markups remain relatively stable. Section 6 revisits our results under this interpretation, but for now we treat the estimated composite market power series as a markup.

The implied factor shares, through which market power enters the sufficient statistic, are also closer to those observed than under conventional Compustat estimates. There are two reasons. First, the timing lines up with the evolution of factor shares. Conventional Compustat measures place most of the increase in the 1980s, when the U.S. labor share changed relatively little. The Census matched series instead rises gradually through 2010. Second, the increase is modest: the cost weighted markup rises by 9.6 percent between 1980 and 2018, at the bottom of the 10 to 20 percent range of conventional estimates. The estimation follows the production function approach of Hall (1988) and De Loecker and Warzynski (2012) and, as in De Loecker, Eeckhout and Unger (2020), treats cost of goods sold as the flexible input, giving the conventional firm level measure

$$\widehat{\mu}_{it}^{\text{meas}} = \widehat{\theta}_{it}^{v} \frac{\text{Sales}_{it}}{\text{COGS}_{it}}, \tag{16}$$

where $\widehat{\theta}_{it}^{v}$ is the estimated output elasticity of the flexible input. The firm level estimates are aggregated with the harmonic formula (15) within two digit industries and combined into an index I_t of the growth of the measured wedge within industries. We follow the conventional interpretation here and attribute the measured wedge to the product markup μ. Appendix D.1 describes the construction and Appendix D.2 the interpretation.

The production function estimates discipline changes in market power rather than its absolute level (De Ridder, Grassi and Morzenti, 2025). We therefore rebase the index to the model's initial year, 1980, and calibrate the initial level separately:

$$\mu_t^{\text{model}} = \mu_{1980} \frac{I_t}{I_{1980}}, \tag{17}$$

where $\mu_{1980} = 1/(1 - s_{\Pi,1980}) = 1.0417$ is the level implied by the 1980 profit share of 4 percent

targeted in Table 2.

Figure 3 plots the resulting index together with a conventional Compustat estimate. The measured wedge rises by 9.6 percent between 1980 and 2018. The increase is modest during the 1980s, continues through the 1990s and 2000s, and is essentially complete by 2010. Through (5), with $s_{W,t} + s_{K,t} = 1/\mu_t$ and $s_{\Pi,t} = 1 - 1/\mu_t$, this path determines the factor shares that enter the Capital Wedge, Redistribution, and Asset Creation terms in the sufficient statistic. The sufficient statistic calculation and the nonlinear transition use the same path.

4.2 Measurement and calibration of the model parameters

Publicly incorporated profit share ω^{inc} The profit share of public firms ω^{inc} measures the fraction of aggregate profits generated by publicly incorporated businesses, before accounting for retained equity. Given the difficulty of measuring this concept, we consider three different methods for measuring ω^{inc} and choose a baseline value of $\omega^{\text{inc}} = 0.4$, the midpoint of the three methods.

The most direct measurement of ω^{inc} comes from estimates obtained from IRS data used in Bhandari et al. (2020), which contains reported taxable net income by legal form: C-corporations, S-corporations, partnerships and sole proprietorships. In addition to the aggregate IRS Statistics of Income (SOI) net income measures for each legal form, the authors additionally use the confidential IRS Schedule M-3 microdata to separately compute the net income of public and private C-corporations, using data from 2004 through 2014. The implied ω^{inc} from the aggregated net income measures is 0.39 and with underreporting adjustments for private businesses falls to 0.345. While these data are high quality and allow for the separation of public and private business income using IRS data, there is suggestive evidence that ω^{inc} may have shifted over the past couple of decades: The share of firms that are publicly listed has declined Gao, Ritter and Zhu (2013), listed firms seem to account for a smaller share of economic activity Schlingemann and Stulz (2022), and the relative profitability of public versus private firms also seems to have fallen Davis, Sollaci and Traina (2021).

We use two other measurement approaches for ω^{inc} using U.S. data from 1980. The first combines IRS SOI data and Compustat data on federal tax expense `TXFED` or `TXC - TXFO - TXS`[6] to "gross up" (Manzon and Plesko, 2002) the taxable net income using the top statutory marginal federal corporate income tax at the time of 46%. This calculation yields an $\omega^{\text{inc}} = 0.41$. There are a number of other tax adjustments that may occur in a given year that do not directly affect taxable net income, which might make the gross up method an inaccurate proxy for income. Two adjustments we consider are 1) adding back investment tax credits `ITCI`, which reduce tax liability but not income, raising ω^{inc} to 0.5 from the baseline and separately 2) subtracting the annual increase in the stock of tax-loss carryforwards $-\Delta$`TLCF` lowering ω^{inc} to 0.4. An alternative direct measurement approach uses direct measures of earnings from the BEA, Compustat, and CRSP,

[6]Total current tax expense minus foreign and state taxes, used as a backup measure if (`TXFED`) is missing.

computing ω^{inc} as the ratio of total Compustat/CRSP EBIT of publicly traded firms to the total BEA private-enterprise net operating surplus as used in Hartman-Glaser, Lustig and Xiaolan (2019), yielding an $\omega^{\text{inc}} = 0.39$ for nonfinancial firms and winsorized at the 1% level, or $\omega^{\text{inc}} = 0.47$ without these filters. The primary drawback to this measure is the misalignment in geographic coverage in the numerator and denominator. While the EBIT measure in the numerator can include earnings from foreign operations, the BEA operating surplus measure is restricted to domestic earnings.

Retained public equity share θ_F, private IPO rate κ, and public exit rate δ_C The publicly incorporated profit share ω^{inc} and the retained equity share θ_F jointly determine the publicly traded equity share of profits ω. In many countries with public equity markets, a substantial share of public equity remains closely held by insiders, such as the founders that took the companies public and their families or corporate executives (Peter, 2026). We therefore allow for a positive fraction, θ_F, for the retained share of equity of liquid business owners in our model. Drawing on empirical evidence from U.S. data, Using data from the Securities Data Corporation spanning 1970 through 2001, Helwege, Pirinsky and Stulz (2007) document that the average share of insider ownership following an IPO reaches 15% after year 30. Eisfeldt and Shi (2025) construct a new dataset spanning 1996 to 2021 on founder and venture capital ownership and find that by years 10 and 15 the mean retained equity shares are 16.6% and 17.3%. While long-horizon retained equity shares do not necessarily imply literal non-tradability of equity holdings, an assumption that holds in our model, we treat these empirical estimates as suggestive that there may be other preferences or frictions that might induce these concentrated holdings by past founders. We use the Helwege, Pirinsky and Stulz (2007) estimate as our baseline $\theta_F = 0.15$ value. Together with $\omega^{\text{inc}} = 0.40$, this implies $\omega = (1 - 0.15) \times 0.40 = 0.34$.

Because of the simplicity of our stationary model, the public firm exit rate δ_C is overdetermined by a number of independent moments in the data. The key stationary flow condition in our model connects the masses of private N_P and public N_C businesses with the iid transition rate κ from private $P \to L$ and δ_C back from public $L \to P$ to private. By independence of IPO probability across firms, this implies

$$\omega^{\text{inc}} = \frac{N_C}{N_P + N_C} = \frac{\kappa}{\kappa + \delta_C}$$

That is, the profit share of public firms is equivalent to the population share of public firms in stationary equilibrium, which in turn is pinned down by these transition rates. For our model, it is important to align ω^{inc} with the profit share of public firms, which restricts one degree of freedom in determining κ, δ_C.

We consider two separate ways of motivating our baseline estimate of $\delta_C = 0.05$, which in turn pins down $\kappa = 0.03$. First, δ_C determines the average duration of a public claim and consequently the average spell length that a business owner remains liquid. Using CRSP data on firms first

listed from 1980-1984, we can calculate the total number of non-financial spells and potentially destructive exits, which includes liquidation, bankruptcy, or other adverse delistings, which yields $\delta_C = 0.049$. A separate way in which δ_C matters is through the aggregate valuation multiple $Q(r)$, where period flow profits that are publicly traded contribute a multiple of $(r + \delta_C)^{-1}$ to total asset supply. Using data from Damodaran (2026) on the aggregate valuation multiple for the S&P 500 in 1980, and given our baseline composite asset return $r = 0.06$, this implies $\delta_C = 0.05$. While we do not technically have an equity premium in our baseline model, in an extension in Section 6.1 we consider a multi-asset framework and align this measured equity valuation multiple with the risky equity asset there.

Relative wealth-to-income ratio of owners to workers λ Our model allows for differences in total household savings of business owners relative to workers per unit of permanent income earned. The relative wealth-to-income ratio, λ, consequently summarizes the aggregate asset demand shift for a unit of permanent income redistribution between workers and owners.[7]

We use household balance sheet data from the Survey of Consumer Finances (SCF) 1983 wave to separately measure the wealth and income of business owners and workers, focusing on the household population between ages 25-64. In the data, there is a large degree of heterogeneity in the observable characteristics of business owners, which reflects the widely varying motives for becoming an entrepreneur, see Sterk, Sedláček and Pugsley (2021). In lieu of choosing the entire SCF sample of households earning business income to map to our business owner type, we focus on a smaller sub-sample of this population that better reflects the owner type in our model.

The first sub-samples we consider from the SCF are households who are active business owners, operating larger scale businesses as measured by those with annual gross receipts or sales of $\geq$ \$250K and $\geq$ \$500K, which constitute 2.8 and 2.1 percent of the total SCF 1983 household population and 19.8 and 15.2 percent of the total active business owner population respectively. The defining feature of private business owners in our model is that the aggregate component of their flow income is determined by aggregate profits derived from market power. Given recent evidence that a substantial driver of the rise of aggregate markups seems to be a reallocation of economic activity toward larger firms (Autor et al., 2020; De Loecker, Eeckhout and Unger, 2020), this operating scale filter seems to be a reasonable and transparent way to focus on likely business owners of these types of firms. We consider an alternative sub-sample that isolates the subset of business income earners with high incomes, in particular those that fall within the top 5% and 1% of the income distribution in the SCF 1983. The benefit of this screen is that we can also consider the contribution of passive business income earners, tied to the idea in our model that some households may be limited partners, family shareholders, or former owner-operators who

[7] While in our model, $\lambda \neq 1$ follows from workers and business owners differing in terms of their time preferences β and idiosyncratic income processes, an alternative mechanism that yields aggregate asset demand consequences of permanent income redistribution is allowing for non-homothetic preferences for consumption-savings as in Straub (2019); Morrison (2026).

Table 1: Alternative SCF measures of relative wealth to income

Screen	Owner definition	λ_0	Overall population share	Business owner population share
Business sales $\geq$ \$250K (baseline)	Active	3.31	2.83%	19.77%
Business sales $\geq$ \$500K	Active	3.41	2.17%	15.15%
Top 5% of household income	Active or passive	3.13	3.04%	18.50%
Top 1% of household income	Active or passive	3.67	0.80%	4.89%

Notes: The sample is the 1983 SCF population ages 25–64. Workers report no business income and hold no active or passive private-business interest. The income screens rank households in the overall SCF income distribution and include owners with either an active or a passive private-business interest. The business owner share uses all active owners as the denominator for the sales screens and all active or passive owners for the income screens.

have retained equity after sale. In terms of population shares and balance sheet characteristics, households from this income filter are largely similar to those from our operating scale filter. We choose the $\geq$ \$250K subset of households as our baseline specification. We will map our model worker type to households that earn no business income and do not actively or passively own a business. Table 1 summarizes the resulting estimates.

We measure household wealth using the SCF's net-worth measure (B3323), subtracting the value of all privately held business equity.[8] This produces the SCF's nonbusiness net-worth measure, B3322. A key distinction required by our model is that the relevant measure of financial wealth to define the numerator of λ does not include the capitalized value of ownership claims to flow labor and private business income. For income, we use total pretax household income (B3201) for business owners, rather than solely relying on reported business and wage income. Using total income reduces the sensitivity of our results to different classifications of business owners' income across sources, consistent with the idea that Smith et al. (2019) find that switching from C-corporation legal status to pass-through reduces reported wages and increases reported profits. However, because total income includes interest, dividends, and capital gains, which our model would classify as asset income, we adjust for this income concept misalignment by adding back interest income in the denominator when aligning the model-implied λ with the SCF λ.

Previously, we described how to construct a "micro-based" measurement of the λ parameter based purely on SCF data. Alternatively, one could rely on the homogeneity properties of aggregate asset demand implied by our model to derive a "macro-based" measurement equation for λ. This calculation measures the relative wealth-to-income ratio of owners to workers, λ, using aggregate owner financial wealth relative to the profit share and aggregate worker wealth relative to the labor share.

$$\lambda = \frac{D_O / s_\Pi}{D_W / s_W} \approx 5.8$$

where the relative wealth ratio $D_O / D_W = 0.36$ is computed from the SCF 1983 and the relative income shares $s_W / s_\Pi = 16$ under our baseline calibration. This calculation for λ is substantially

[8] This is given by the sum of equity in businesses in which the household has a management role (B3502) and equity in businesses in which it has no management role (B3501).

higher than our micro-based approach for two main reasons. First, while the business owner income concept we used in the micro-based approach was purposefully broad to account for potential misclassification, this macro-based approach narrowly defines business income as profit income. While the stark separation of income sources between workers and business owners is useful as a modeling device, it likely understates business income and hence overestimates λ. Second, there is substantial uncertainty around the measurement of the aggregate profit share s_Π itself. We use a more conservative accounting-based measure of the profit share from the BEA NIPA Table 1.11: after-tax domestic corporate profits, including inventory valuation and capital consumption adjustments, as a share of gross domestic income, which yields a profit share of about 4% in 1980. Alternative s_Π estimates for 1980 from the literature, such as Karabarbounis and Neiman (2018); Hasenzagl and Pérez (2026), yield 11.3% and 7.7% respectively, implying macro-based λ estimates of 2.0 and 3.0.

Asset demand elasticities for owners and workers The remaining objects in Theorem 1 for us to characterize are the unit-income asset demand curves for workers and owners $\bar{D}(r)$, $\bar{D}_O(r) \equiv \lambda(r)\bar{D}(r)$. In principle, our sufficient-statistics approach could be used without fully specifying and solving the household problem. For example, one could assume that both asset demand curves have constant semi-elasticities, implying

$$\bar{D}(r) = \bar{D}(r_0)\exp[\varepsilon_W(r - r_0)]\,, \qquad \lambda(r) = \lambda(r_0)\exp[(\varepsilon_O - \varepsilon_W)(r - r_0)]$$

Hence, as r changes from its baseline value r_0, the relative wealth-to-income ratio of owners to workers $\lambda(r)$ changes because the savings of one type, say workers, are more sensitive to changes in equilibrium returns. However, existing empirical evidence primarily identifies the aggregate responses of wealth or capital supply to net returns, as opposed to separate estimates based on subsets of the household population, see Moll, Rachel and Restrepo (2022). Using our calibrated model to recover these two group-specific semi-elasticities yields the following estimates

$$\hat{\varepsilon}_W = 16.46, \quad \hat{\varepsilon}_O = 15.49$$

Aggregate asset demand per dollar of income $D(r)$ is the income share weighted average of worker and business owner asset demand per dollar of income at the baseline factor shares.[9] The semi-elasticity of this aggregate asset demand is 16.2963, which lies roughly at the center of the 1 to 35 range from the quasi-experimental estimates described in Moll, Rachel and Restrepo (2022). For simplicity and consistency with the sufficient statistics approach, we adopt the constant

[9] The weighted average is

$$D(r) \equiv \frac{s_{W,0}\bar{D}(r) + s_{\Pi,0}\lambda(r)\bar{D}(r)}{s_{W,0} + s_{\Pi,0}}.$$

Multiplying by $s_{W,0} + s_{\Pi,0}$ gives aggregate desired assets per dollar of output. This constant normalization leaves the asset demand semi elasticity unchanged.

Table 2: Calibration

Panel A. Parameters and discipline

Parameter	Value	Source / target
Frisch elasticity; EIS	0.50; 0.50	Standard
α	1/3	Standard
δ_K	0.0685	$K/Y = 2.49$; BEA
$\delta_C = \rho_L$	0.05	Exit proxy; Damodaran 1980 P/E
ω^{inc}	0.40	Tax and earnings calculations
ω	0.34	$(1-\theta_F)\omega^{\text{inc}}$
θ_F	0.15	Helwege, Pirinsky and Stulz (2007); Eisfeldt and Shi (2025)
κ	0.0333	Implied by δ_C and ω^{inc} in steady state
Owner fraction	2.831%	1983 SCF active owners; sales at least \$250,000
ρ_e, σ_e	0.91, 0.613	$\text{Var}[\log(\ell_i e_i)] = 0.92^2$; Auclert et al. (2021)
ρ_z, σ_z	0.799, 1.432	DeBacker, Panousi and Ramnath (2023), Tables 2 and 5; log AR(1)

Panel B. Solved parameters

Parameter	Value	Notes
β_W, β_O	0.8872, 0.9032	Targeting $r_0 = 6\%$ and $\lambda_0 = 3.315$
μ_0	1.0417	Targeting $s_{\Pi,0} = 4\%$, implying $s_{W,0} = 64\%$

Notes: The income processes are annual, and σ_e and σ_z are stationary log standard deviations. The baseline profit share is the BEA NIPA accounting measure described above. Given $\alpha = 1/3$, the calibrated markup implies $s_{K,0} = 32\%$ and $s_{W,0} = 64\%$. The SCF target is the private-owner-to-worker wealth-to-total-income ratio, including interest income. The reported λ_0 is its profit-normalized model counterpart.

semi-elasticity form for $\bar{D}(r), \lambda(r)$, using $\hat{\varepsilon}_W, \hat{\varepsilon}_O$ from our calibrated model. This approximation is quantitatively accurate, resulting in only a 5 basis point discrepancy in r between the constant semi-elasticity approximation and the full model stationary equilibrium evaluated at the terminal 2018 markup level.

Household model parameters. Workers have mass $N_W = 1 - 0.0283$, and business owners have mass $1 - N_W = 0.0283$, matching the 1983 SCF share of active owners with business sales of at least \$250,000. Both types have an elasticity of intertemporal substitution (EIS) of 0.5, and workers have GHH preferences with a Frisch elasticity of 0.5. Worker log productivity follows an AR(1) process with persistence 0.91 and stationary standard deviation 0.613. Under GHH preferences, earnings are proportional to productivity raised to 1.5, so this productivity dispersion produces a log earnings standard deviation of 0.92, following Auclert et al. (2021).

Business owners tend to have significantly more volatile income than labor income earners and often earn zero income or incur losses in a given year. Because the richer business income process estimated by DeBacker, Panousi and Ramnath (2023) from IRS microdata includes these zeros and losses, we cannot import it directly as the positive log income process required for our model business owners. In lieu of adding model complications, such as bankruptcy and occupational switching back to being a labor income earner, we instead approximate a positive business income process from the results in DeBacker, Panousi and Ramnath (2023). To do so, we use Table 2 and the

interval boundaries in the authors' Footnote 26 to assign deciles to their midpoints. For example, the fifth decile of business income is the range [5000; 9000], which we assign to a realization of 7000. The exterior deciles are unbounded, with $< -6,000$ for the bottom and $> 147,000$. To infer point estimates for these deciles, we choose values to ensure that collectively all deciles produce first and second moments matching their full-sample means and standard deviations reported in their Table 2. We retain the seven deciles with income above \$1,000, condition the Table 5 transition matrix on remaining within these states, and estimate the best-fitting log AR(1) process on the process implied by the tables. This yields persistence $\rho_z = 0.799$ and stationary log standard deviation $\sigma_z = 1.432$. The resulting dispersion is close to the log standard deviation of 1.466 among SCF active owners ages 25–64 with positive business income.[10] We choose the household discount factors jointly to clear the baseline asset market at $r_0 = 6\%$ and match the SCF relative wealth-to-total-income ratio. This implies $\beta_W = 0.8872$, $\beta_O = 0.9032$, and $\lambda_0 = 3.31$.

4.3 Quantifying the asset market channel

We proceed to use Theorem 1 to map the model-implied factor shares $(s_{K,t}, s_{W,t}, s_{\Pi,t})$ generated by our estimated markup path into our model's implied path of interest rates and output r_t, Y_t. Our sufficient-statistic calculations solve for a sequence of stationary equilibria indexed by the estimated markup $\hat{\mu}_t$ in each year, as in Auclert et al. (2025); De Loecker, Eeckhout and Mongey (2026). For each equilibrium, aggregate prices and quantities, e.g. r_t, Y_t, are constant, while distributions of households and ownership statuses are stationary even as individual households and owners may transition between idiosyncratic states.

These calculations omit the transition dynamics associated with the gradual evolution of states and the anticipation of future aggregate changes. Therefore, we also compute the full transition dynamics of our structural equilibrium model under two sets of assumptions on the degree of foresight of forward-looking agents in our model. In the first case, agents have perfect foresight with respect to aggregate variables upon the unanticipated announcement of the full $t = 1980, ..., 2018$ path of markups $\{\hat{\mu}_t\}$ in period $t = 1980$. We then solve for the full non-linear transition dynamics given this "MIT shock". In the second case, agents are completely myopic and assume that the newly observed markup in the current period $\hat{\mu}_t$ will last indefinitely. At time-t, they still anticipate the gradual transition of endogenous variables back to a steady state indexed by $\hat{\mu}_t$ after the initial surprise of learning $\hat{\mu}_t$, shifting from their prior time-$t-1$ belief that $\hat{\mu}_{t-1}$ would last indefinitely. Hence, we solve for this non-linear transition given this "sequence of MIT shocks", where in each period-t, aggregate state variables may have still been in transition from past-period surprises of learning $\hat{\mu}_{t-1}$ at time-$t-1$, $\hat{\mu}_{t-2}$ at time-$t-2$ and so on.

[10] DeBacker, Panousi and Ramnath (2023) also report an analogous transition matrix for labor earnings in Table 6. Applying the same procedure, with the open top decile chosen to match average labor earnings in Table 3, implies persistence 0.880 and a stationary log earnings standard deviation of 0.866. Under GHH labor supply, this corresponds to a standard deviation of the productivity process of 0.577, close to the persistence of 0.91 and productivity standard deviation of 0.613 in our Auclert et al. (2021) calibration.

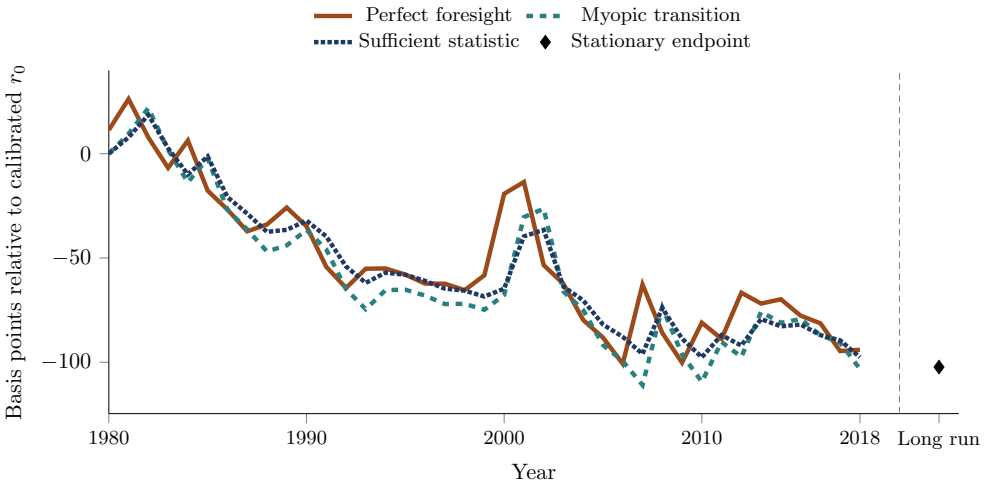


Figure 4: Interest rate r along the perfect foresight transition, the myopic transition, and the two-group sufficient statistic. All three use the same cost-weighted markup path and report changes from the calibrated 1980 economy in basis points. The diamond reports the stationary equilibrium associated with the 2018 markup.

Figure 4 displays the path of rates r_t implied by our sufficient-statistic calculation, and the perfect foresight and myopic transition dynamics, and their common stationary endpoint after 2018 once markups $\hat{\mu}_t$ cease to change. The perfect-foresight path responds when the future markup path is announced: r initially rises by 11.4 basis points, while the myopic and sufficient-statistic paths begin at the 1980 baseline. While the perfect-foresight path tends to move in advance of the other paths, although all three remain close. By 2018, the interest rate responses are -94.0, -103.2, and -97.6 basis points under perfect foresight, myopic expectations, and the sufficient statistic, respectively, with a long-run response of -102.3 basis points.

Comparisons to benchmark assumptions on asset markets In our model, profits directly accrue to private business owners as income or to liquid owners as wealth after taking their companies public, where the profits associated to the businesses sold by liquid owners are capitalized and publicly traded as equity. The existing literature has often made simplifying assumptions, either that all profit income is directly earned by the business sector as in the entrepreneurship literature (Quadrini, 2000; Cagetti and De Nardi, 2006), or that all profit income is capitalized and traded in equity markets (Farhi and Gourio, 2018; Eggertsson, Robbins and Wold, 2021; Auclert et al., 2025).

Figure 5 shows how these different assumptions on the ownership of profits changes the response of rates r through asset markets to our estimated markup path, and consequently of output Y to the endogenous response of the user cost of capital induced by r. Additionally, we plot the response of output Y under the representative household benchmark, where the rate is fixed $r = \beta^{-1} - 1$ by the discount rate. The calculations displayed in Figure 5 are sequences of steady states implied by our sufficient-statistic formula.

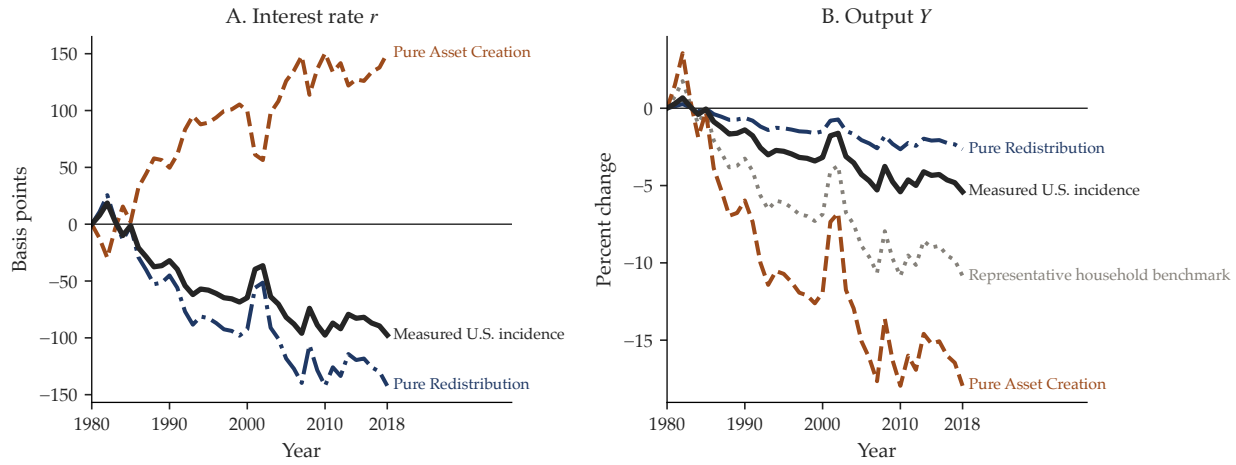


Figure 5: Ownership and the markup counterfactual. Interest rate r and output Y responses under alternative incidence assumptions, together with the representative household benchmark. The two-group sufficient statistic uses the same cost weighted markup path in all cases. Changes are relative to the calibrated 1980 economy.

To construct the "Pure Redistribution" path, we set $\kappa = \omega = 0$, so all profits directly accrue to private business owners who never IPO and enter into the liquid state. Since private business owners face idiosyncratic risk and now have more income, they increase their demand for diversified assets, which pushes r downward. To construct the "Pure Asset Creation" path, we set $\lambda_0 = \theta_F = \delta_C = 0$ and consequently $\omega = 1$. This case can be interpreted as all private business owners eventually selling their claims in an IPO and perpetually remaining as liquid owners without any retained equity. They draw down their assets over time until they make a negligible contribution to aggregate asset demand resulting in $\lambda_0 = 0$. One can essentially reinterpret existing models that assume all profit income is capitalized and traded as following from this sequence of assumptions in our model. Now because all profit income is steadily increasing and fully capitalized, the total market value of publicly traded equity is increasing. To clear asset markets, the return on this asset r must rise to induce households to be willing to save more to hold this increasing stock of assets in equilibrium.

Panel A of Figure 5 shows that the implications of these different assumptions for the path of rates r are quantitatively significant. Under "Pure Redistribution", rates r fall by 142.2 basis points, whereas under "Pure Asset Creation" rates r *rise* by 150.5 basis points. Given our calibration on U.S. data, where a significant portion of profits are earned by private businesses and a non-trivial share of publicly traded equity is retained by insiders, our calibrated model finds that rates r fall by 97.6 basis points by 2018. Panel B of Figure 5 shows that these interest rate responses substantially change the output effect of heightened markups. Output falls by 2.65 percent under Pure Redistribution and by 17.95 percent under Pure Asset Creation. Under measured U.S. incidence, the output response is -5.40 percent, compared with -10.85 percent in the representative household benchmark, where the rate r is fixed.

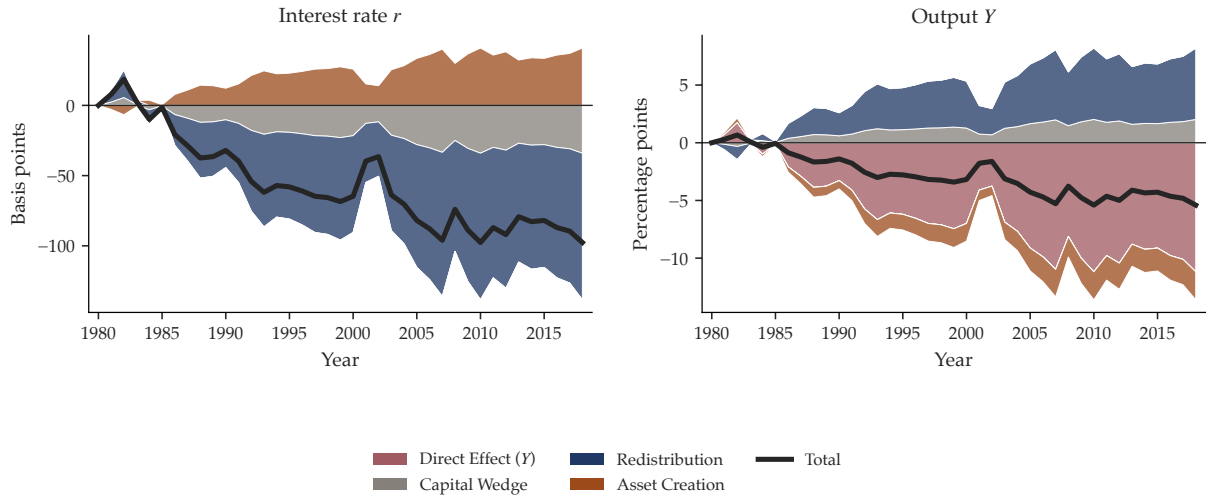


Figure 6: Decomposition of the markup counterfactual. The left panel reports contributions to the interest rate r response in basis points. The right panel reports percentage-point contributions to output Y, separating the Direct Effect in the output equation from the Capital Wedge's effect through the endogenous interest rate. In both panels, the black line is the total two-group sufficient-statistic response relative to the calibrated 1980 economy.

Figure 6 decomposes the sufficient-statistic response into the Capital Wedge, Redistribution, and Asset Creation. For output, we separate the Direct Effect of the markup on capital and labor demand from the Capital Wedge's effect through the endogenous interest rate. Following Shapley (1953), we average each force's marginal contribution across all orderings of the three interest rate forces or four output forces. By 2018, Redistribution contributes -104.4 basis points to the interest rate response, more than offsetting the 40.8 basis point increase from Asset Creation. The right panel shows that Redistribution and the Capital Wedge's interest rate effect mitigate the direct output loss from the markup's distortions of capital and labor demand.

4.4 Model extensions

The interest rate decline remains large across the model extensions in Table 3. Each exercise adds one feature to the baseline price-markup counterfactual and uses the corresponding sufficient statistic from Section 3.4. Appendix G re-solves the household problem when an extension changes the calibrated 1980 economy.

Balanced growth produces the largest attenuation, reducing the interest rate decline from 97.6 to 87.2 basis points. Fiscal policy and CES production imply responses of -96.1 and -95.6 basis points, respectively. The local misallocation exercise moves the baseline interest rate by at most 8.3 basis points under the separate productivity perturbation. Redistribution continues to dominate Asset Creation in each exercise.

These exercises retain a common return on physical capital and tradable equity and treat market power as a price markup. Section 6 studies the wage-markdown and multiple-asset extensions.

Table 3: Interest rate response across model extensions

Specification	Added feature	2018 response (bp)	Difference from baseline (bp)
Baseline	None	−97.6	0.0
Fiscal policy	Taxes on labor, profits, capital, and government debt	−96.1	+1.5
Balanced growth	Trend growth of 2 percent	−87.2	+10.4
CES production	Substitution elasticity $\sigma_{KL} = 0.7$ in place of Cobb–Douglas	−95.6	+2.0
Misallocation	A 3 percent productivity loss with aggregate wedges fixed, $\sigma_{KL} \in [0.6, 1.2]$	n.a.	$\lvert\Delta r\rvert \leq 8.3$

Notes: Responses use the baseline calibration in Table 2, with a 6 percent 1980 interest rate and a 4 percent initial profit share, and are changes from that 1980 interest rate. A positive difference from the baseline denotes a smaller interest rate decline. The household block is re-solved where the 1980 problem changes. The misallocation row bounds a local change from the 1980 baseline under a separate productivity perturbation, rather than a 2018 markup-path response. Appendices G.3 and G.4 report the exact stationary heterogeneous agent solutions as checks; the misallocation exercise provides only a local bound.

4.5 Robustness

Appendix E.9 varies the incidence of profits across private business owners and public equity. Appendix E.7 reports how asset demand sensitivities change between the baseline and long run stationary equilibria.

5 Welfare

The asset market channel dampens output loss from the production distortions of market power. In spite of this, we show that the welfare effects of market power in our model are significantly amplified due to its negative distributional consequences.

We proceed by quantifying the welfare effects of our estimated, increasing path of markups from Section 4.1. Welfare is measured in consumption equivalent variation (CEV) units, the permanent proportional change in the initial consumption path prior to the markup transition required to match discounted, lifetime utility along the post-markup transition path. We evaluate this measure separately under perfect foresight and myopia as in Section 4.3. Note that in this section we do not rely on the sufficient-statistics calculation. All results follow from actual dynamic transitions.

Table 4 summarizes our main welfare findings. The first two rows report our earlier results, which show that under our baseline heterogeneous-agent (HA) model with workers and owners, the long-run output decline from the increase in market power from 1980 to 2018 is mitigated by about 50% relative to the representative household (RA) benchmark.

Are aggregate quantities, such as a smaller decline of aggregate output or consumption, informative about the magnitude or direction of the average welfare consequences of markups? As a benchmark, we compute a CEV welfare loss of -0.32% for a representative household given the

Table 4: Long run output and welfare consequences of market power

Outcome	Perfect foresight	Myopic
Panel A. Model aggregates and representative-agent welfare		
Long run output change, RA	-10.85%	
Long run output change, HA	-5.12%	
RA CEV with RA allocations	-0.32%	-0.25%
RA CEV with HA allocations	-0.18%	-0.15%
Panel B. Welfare heterogeneity in our model		
Population-level CEV	-1.52%	-1.43%
Worker type-level CEV	-1.65%	-1.53%
Business owner type-level CEV	+20.04%	+14.71%

Notes: Long run output compares stationary equilibria at the 2018 and initial markups, so each value spans both expectations columns. All CEVs use a permanent proportional change in baseline actual consumption. For myopic expectations, the final markup observation supplies the infinite welfare horizon. The population-level CEV uses the model's cardinal normalization across types. The welfare figure below uses perfect foresight.

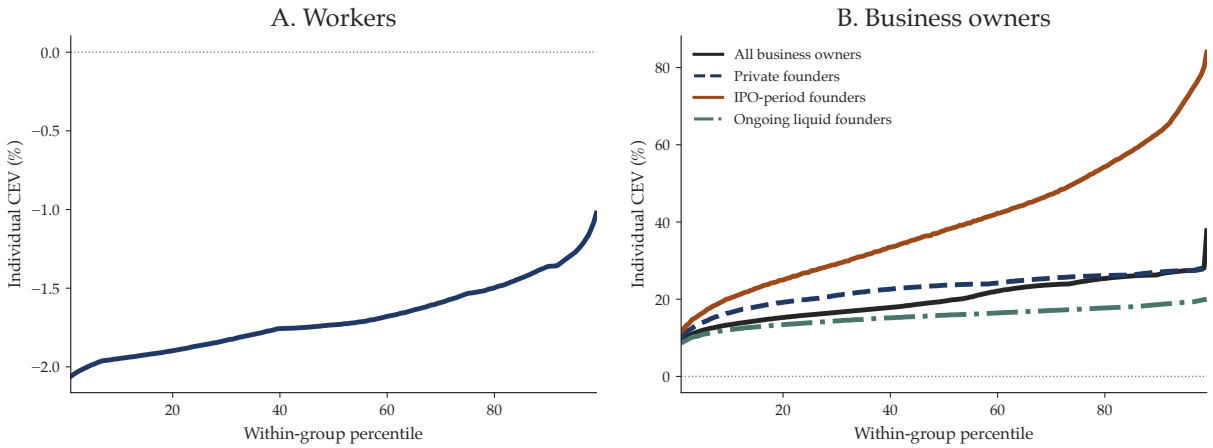


Figure 7: Distribution of individual welfare effects under perfect foresight. Curves show weighted quantiles of individual CEVs from the 1st through 99th percentile within each household type, using the 1980 stationary distribution. Business owner curves are also shown conditional on initial founder status. Positive CEVs denote welfare gains.

estimated markup transition path. As an illustrative experiment, we apply the same representative household's preferences (equivalently, the preferences of our worker type) to the equilibrium paths of aggregate consumption and labor generated by our baseline HA model (row four of Table 4). The welfare loss to a representative household given these allocations is -0.18% CEV units, a nearly 50% welfare loss mitigation. Nonetheless, we show in Panel B that the actual welfare effects of market power in our model are significantly larger due to its heterogeneous incidence.

The population-level CEV in our model, which is the common proportional change in baseline consumption that equates population share-weighted lifetime utility to that under the markup transition, is -1.52 percent. This substantially larger welfare loss compared to the RA benchmark is driven by the welfare losses of the large worker population, at the benefit of the smaller business owner population.

The welfare effects of market power are more unequal under perfect foresight than under myopia. One of the primary driving factors is the larger revaluation effect on public equity claims that occurs under perfect foresight. Wealthier business owners, who hold a substantial share of aggregate wealth, gain disproportionately more from the wealth effects due to this markup-induced asset revaluation than workers who hold less per capita wealth. In addition, because of the large valuation increase in aggregate equity under perfect foresight, suddenly a larger quantity of assets has to clear asset markets in the initial period. This induces an increase in the equilibrium r relative to the myopic case, where the valuation effect is smaller given only a single, one-time markup increase is expected to persist compared to the steady markup increase over time under foresight. This relatively higher r then crowds out capital K, further lowering wages for workers beyond the direct distortions induced by the realized markup increase.

Figure 7 further decomposes the welfare effects to the individual-level CEVs and orders them across the CEV distribution to show the heterogeneity in welfare loss within types. Unsurprisingly, workers lose and business owners gain across their full within-type distribution, with business owners entering their IPO gaining the most. Appendix Sections E.2 and E.3 provide the representative-household benchmark, formal CEV definitions, and supplementary welfare source decomposition and figures for further reference.

6 Generalizations: multiple assets and labor market power

Our baseline model considers product market power, where the profits derived from market power increase the demand and supply for assets in a single asset market. In this section we propose two key extensions of our model framework. We first consider the implications of the asset market channel, when there are multiple asset classes whose asset demand and supply are differentially affected by an increase in market power. This results in differential implications of market power on returns across asset classes. We apply this generalized framework to study the effects of market power on the increasing market value of U.S. public equity and the evolution of

the equity premium. We next consider the implications of the asset market channel when firms have market power over factor input markets and concretely consider the case of labor market power. With Cobb-Douglas production and labor market power, income shifts toward profits only from the labor share, rather than proportionally from both the labor and capital share under product market power.

6.1 Multiple assets

In the one asset economy, the interest rate r determines both the financing cost of physical capital K and the value of tradable equity. We separate these roles by introducing a safe asset and equity into the fiscal economy of Appendix G.2. Household saving determines total asset demand D, while portfolio choice allocates it across the two markets. Appendix H gives the household problems and derives the market clearing system.

Let r^b denote the safe return and r^e the equity required return, and write

$$\mathbf{r} = (r^b, r^e).$$

Physical capital is financed by a share θ of bond-like claims and a share $1-\theta$ of equity. Its cost of capital is therefore

$$r^c(\mathbf{r}) = \theta r^b + (1-\theta) r^e.$$

The same tax-adjusted user cost as in the fiscal extension maps r^c into firms' demand for physical capital. Tradable equity backed by aggregate profit income Π is instead discounted at the equity required return, with valuation multiple per dollar of publicly traded profit income

$$Q^{\Pi}(r^e) = \frac{1}{r^e + \delta_C}.$$

The cost of capital r^c therefore takes the role of r in transmitting the asset market response to output Y through the user cost of physical capital K. Government debt belongs to safe asset supply and pays r^b. The two returns consequently play different roles even before household portfolio choice is introduced: r^c governs the user cost of physical capital, r^e governs the value of tradable equity, and r^b governs government debt service.

Asset market clearing. For group $g \in \{W, O\}$, let D_g denote total asset demand and ϕ_g its equity share. Homogeneity in each group's income scale gives the following extension of Theorem 1:

$$\underbrace{\bar{D}(\mathbf{r}) \begin{pmatrix} [1-\phi_W(\mathbf{r})]\widetilde{s}_W + \lambda(\mathbf{r})[1-\phi_O(\mathbf{r})]\widetilde{s}_\Pi \\ \phi_W(\mathbf{r})\widetilde{s}_W + \lambda(\mathbf{r})\phi_O(\mathbf{r})\widetilde{s}_\Pi \end{pmatrix}}_{\text{asset demand}} = \underbrace{\begin{pmatrix} \theta \dfrac{s_K}{u_K(\mathbf{r})} + b \\ (1-\theta)\dfrac{s_K}{u_K(\mathbf{r})} + Q^{\Pi}(r^e)\omega\widetilde{s}_\Pi \end{pmatrix}}_{\text{asset supply}}. \tag{18}$$

Here $\widetilde{s}_W$ and $\widetilde{s}_\Pi$ are after tax labor and profit income shares, $\bar{D}(\mathbf{r})$ is worker asset demand per dollar of after tax aggregate labor income, and $\lambda(\mathbf{r})$ is the relative wealth-to-income ratio of owners to workers. The tax-adjusted user cost $u_K(\mathbf{r})$ determines physical capital relative to output, $K/Y = s_K/u_K(\mathbf{r})$, and $b = B/Y$ is government debt relative to output. The first row clears the safe asset market. Its supply consists of the bond-like claims financing physical capital and government debt. The second row clears the equity market. Its supply consists of the equity financing physical capital and the tradable equity created by Asset Creation.

Adding the two rows preserves the aggregate accounting of the one asset model: Redistribution still determines how much total asset demand is generated by the shift of income toward business owners, and Asset Creation still adds the value of tradable equity to total asset supply. What changes is that these forces no longer enter the same market at the same price. Portfolio choice determines where household demand is directed, while the safe return, equity required return, and cost of capital can move separately.

Household asset demand and portfolio allocation. We discipline two household responses: how total asset demand D_g changes with returns, and how the equity share ϕ_g changes with the equity premium. The household problems in Appendix H allow for aggregate and idiosyncratic risk while preserving homogeneity in income. We combine asset demand responses from the calibrated household problems with an empirical portfolio response rather than estimate a particular portfolio friction.[11]

Around the baseline portfolio, we parameterize these local responses as

$$\begin{pmatrix} d\log D_g \\ d\phi_g \end{pmatrix} = \begin{pmatrix} \epsilon_g(1-\phi_{g,0}) & \epsilon_g\phi_{g,0} \\ -\eta_g & \eta_g \end{pmatrix} \begin{pmatrix} dr^b \\ dr^e \end{pmatrix}.$$

The first row says that total asset demand responds to the return on the household's baseline portfolio,

$$dr_g^p = (1-\phi_{g,0})dr^b + \phi_{g,0}dr^e,$$

with semi elasticity ϵ_g. A change in the equity premium that leaves this portfolio return unchanged has no additional first order effect on total desired assets. The second row says that the equity share responds to the equity premium,

$$d\phi_g = \eta_g\, d(r^e - r^b).$$

This separates the amount of saving from the allocation of that saving across assets.

The distinction is also consistent with the available empirical evidence. Ring (2025) studies household saving responses to wealth taxation, which changes after-tax asset returns broadly, and finds a substantial response of financial saving with relatively little movement in portfolio

[11] The appendix allows borrowing and portfolio restrictions and nonpecuniary portfolio frictions consistent with this homogeneity property.

shares. We therefore take the total asset demand responses from the one asset fiscal household calibration. For portfolio composition, we use the long run response of equity holdings to the equity premium estimated by Fagereng, Guiso and Ring (2026). Their design exploits the 1998 removal of preferential wealth-tax treatment of equity in Norway. Because our exercise concerns long run risky steady states, we use their long run estimate rather than the shorter horizon response.

We compare *risky steady states* as the markup changes: the allocation at the mean aggregate state once the wealth distribution has adjusted, with households still facing the same aggregate and idiosyncratic risk. In our approximation, the asset demand semi elasticities and portfolio slopes summarize household responses around the baseline, including precautionary and hedging motives. For each markup, the pair (r^b, r^e) clears the safe asset and equity markets jointly, while the government budget determines the balancing labor tax and the cost of capital and equity values adjust with the two returns.[12] These are comparisons of equilibria at the dated markup values, not the nonlinear transition studied in Section 4.3.

Calibration. The asset demand responses carry over directly from the one asset fiscal household block of Appendix G.2, whose discount factors are re-solved at the fiscal 1980 baseline. At that baseline, the asset demand semi elasticities are

$$\epsilon_W = 18.13, \qquad \epsilon_O = 16.89.$$

These determine how much total desired wealth workers and business owners demand in response to the return on their respective reference portfolios. Differences in wealth holdings per dollar of income between the two groups continue to be summarized by the relative wealth-to-income ratio of owners to workers. Because this extension includes the fiscal closure, the relevant relative wealth-to-income ratio of owners to workers, measured using after-tax income, is $\lambda = 3.05$.

The baseline portfolio shares come from the SCF. Using gross financial wealth and the same age, wealth, and business owner definitions used to measure λ, the equal-weighted average over the 12 SCF waves from 1989 to 2022 gives

$$\phi_{W,0} = 48.88\%, \qquad \phi_{O,0} = 52.23\%.$$

For portfolio allocation, the long run estimate of Fagereng, Guiso and Ring (2026) implies

$$\eta_W = \eta_O = 6.9.$$

Thus a 100 basis point increase in the equity premium raises the equity share by 6.9 percentage points locally. We apply the same slope to workers and business owners.

We retain the 6.00 percent baseline cost of capital used throughout the paper and set the

[12] Away from the baseline, a logit schedule extends the equity share response while reproducing the local estimate η_g.

Table 5: Markup counterfactual in the one asset and two asset fiscal economies

Outcome	One asset fiscal	Two asset fiscal
Safe return r^b	–	−128.0
Equity required return r^e	–	−80.2
Cost of capital	−96.1	−102.2
Equity premium $r^e - r^b$	–	47.8
Labor tax rate (percentage points)	−0.78	−0.92
Output (percent)	−5.00	−4.54

Notes: Counterfactual responses to the estimated cost weighted markup increase from 1980 to 2018, relative to the calibrated 1980 economy. Returns and the equity premium are in basis points, the labor tax is in percentage points, and output is in percent. The one asset column is the full fiscal closure of Appendix G.2; its single return is reported on the cost of capital row. The two asset column reports the corresponding risky steady state comparison. Both economies are recentered to the same 1980 labor tax.

initial equity premium to 4.0 percentage points, a conventional long run calibration rather than an independently targeted moment. We do not separately calibrate the financing share θ. Given the initial premium and the 6.00 percent cost of capital, safe asset market clearing pins it down. The resulting baseline is

$$\theta = 45.91\%, \qquad r_0^b = 3.84\%, \qquad r_0^e = 7.84\%, \qquad r_0^c = 6.00\%.$$

The equity required return therefore exceeds the cost of capital by 1.84 percentage points at the baseline. This gap will matter directly for Asset Creation because future tradable equity is discounted at r^e, rather than at r^c.

The remaining fiscal objects are inherited from Appendix G.2: government debt held by the public is 26.2 percent of GDP, the initial effective labor tax is 26.4 percent, the marginal corporate-tax user-cost factor is 1.331, and the other fiscal ratios are held at their matched 1980 values. [13]

Quantitative results. Table 5 compares the markup counterfactual across the two fiscal economies. At the 2018 markup, the cost of capital r^c falls by 102.2 basis points in the two asset economy, compared with 96.1 basis points in the matched one asset fiscal economy. Output, which falls by 5.00 percent in the one asset fiscal economy, falls by 4.54 percent: through the user cost, the additional decline in the cost of capital is worth roughly 0.46 percentage points of output.

The two required returns move quite differently. The safe return falls by 128.0 basis points, while the equity required return falls by 80.2 basis points. The equity premium therefore widens by 47.8 basis points. The fact that r^e falls less than the one asset return does not prevent the financing cost of physical capital from falling more. Physical capital is financed by both bond-like claims and equity, and the much larger decline in r^b pulls r^c below its one asset counterpart. The stronger

[13] Government debt now pays r^b rather than the common one asset return. We therefore adjust the residual fiscal term so that the two asset economy reproduces the same 26.4 percent labor tax in 1980. This keeps the comparison focused on the change in asset structure rather than a different initial fiscal position.

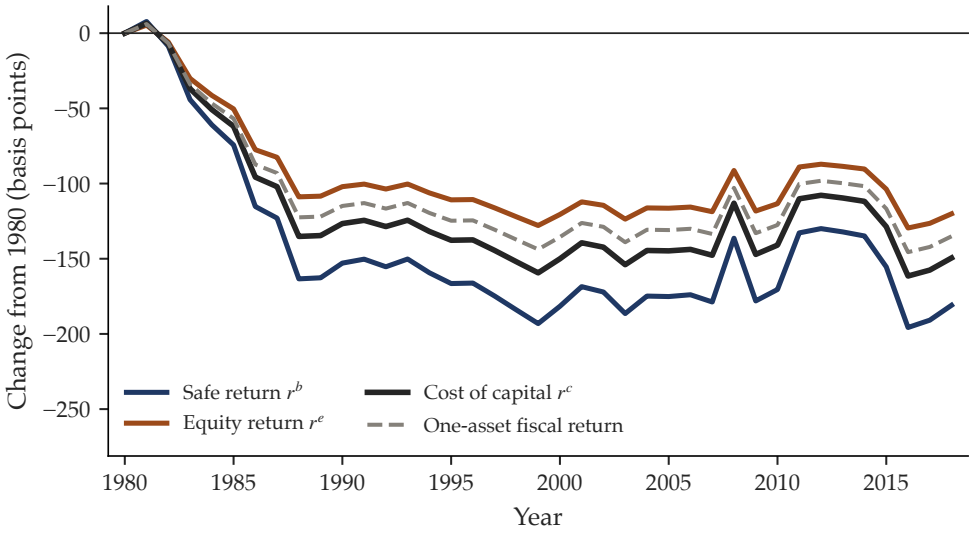


Figure 8: Returns in the two asset markup counterfactual. Changes in the safe return, equity required return, cost of capital, and matched one asset fiscal return, in basis points relative to the calibrated 1980 economy. The curves are stationary sufficient statistic comparisons, not nonlinear transition paths.

decline in the safe return also reduces government debt service further: the labor tax falls by 0.92 percentage points, compared with 0.78 percentage points in the one asset fiscal economy.

Figure 8 reports these returns across the dated markup values. The cost of capital r^c lies between the safe return r^b and equity required return r^e and falls below the matched one asset fiscal return at markup values from 1984 onward.

Equity valuation and portfolio allocation. The larger decline in the cost of capital comes mainly from the valuation of tradable equity, rather than from portfolio substitution. Because the system is nonlinear, the numerical contributions depend on the order in which the margins are introduced. We therefore use the following sequence only as a diagnostic.

Starting from the one asset fiscal economy, replacing the aggregate asset demand schedule with separate worker and business owner demand schedules leaves the cost of capital response essentially unchanged at -96.1 basis points. This step retains a single asset market and isolates the effect of using the separate household demand schedules.

We then introduce a distinct equity required return while holding the equity premium fixed at its baseline value. Tradable equity is now discounted at r^e, while physical capital continues to depend on r^c. The cost of capital response moves to -102.2 basis points, an additional -6.1 basis points. This is the largest step. At the baseline, discounting profit income at $r_0^e = 7.84\%$ rather than at the 6.00 percent cost of capital reduces the equity market value supported by a dollar of underlying profit income by about 14 percent. Asset Creation therefore offsets less of Redistribution.

Finally, allowing household portfolios to respond endogenously moves the cost of capital to -102.2 basis points, a further 0.1 basis points. Portfolio substitution has a much smaller effect on the cost of capital than equity valuation. Its main role is instead to determine the equity premium required for households to absorb the additional equity supply. With the calibrated long run portfolio response, households shift toward equity as the premium rises, and the equilibrium premium widens by 47.8 basis points.

The accounting behind Redistribution and Asset Creation is thus preserved, but equity valuation and portfolio allocation determine how the adjustment is divided across returns. In this calibration, weaker Asset Creation produces a larger decline in the cost of capital r^c and greater mitigation of the output distortion.

6.2 Price Markups or Wage Markdowns?

The production block in Section 2 distinguishes the price markup μ, which distorts both factor conditions, from the wage markdown ν, which adds a wedge only to labor. This distinction matters for the asset market channel because the two wedges change factor income shares differently. It also matters for measurement: using matched Census–Compustat data to separate labor expenses from other variable inputs, Ren and Zhang (2025) find that the increase in measured market power is concentrated in wage markdowns rather than price markups. We use their estimates to study a wage markdown counterfactual from the same 1980 baseline.

The source of the ambiguity is particularly transparent with linear production. Suppose firm i produces according to $y_{it} = A_{it}\ell_{it}$. Let μ_{it} denote its price markup and ν_{it} its wage markdown. The firm's optimal price and wage choices imply

$$p_{it} = \mu_{it}\nu_{it}\frac{w_{it}}{A_{it}}. \tag{19}$$

A researcher who assumes that labor is supplied competitively would therefore infer from price over marginal cost that

$$\widetilde{\mu}_{it} = \frac{p_{it}A_{it}}{w_{it}} = \mu_{it}\nu_{it}. \tag{20}$$

The same measured wedge can therefore reflect product market power, labor market power, or both.[14]

Wage markdowns and asset market clearing. Under the Cobb–Douglas production assumptions of Section 2, the factor income shares are

$$s_K = \frac{\alpha}{\mu}, \qquad s_W = \frac{1-\alpha}{\mu\nu}, \qquad s_\Pi = 1 - s_K - s_W. \tag{21}$$

[14]The argument does not depend on linear production. With a general differentiable production function, the labor ratio estimator satisfies $\theta^\ell_{it}/\alpha^\ell_{it} = \mu_{it}\nu_{it}$, where θ^ℓ_{it} is the output elasticity of labor and α^ℓ_{it} is labor's revenue share.

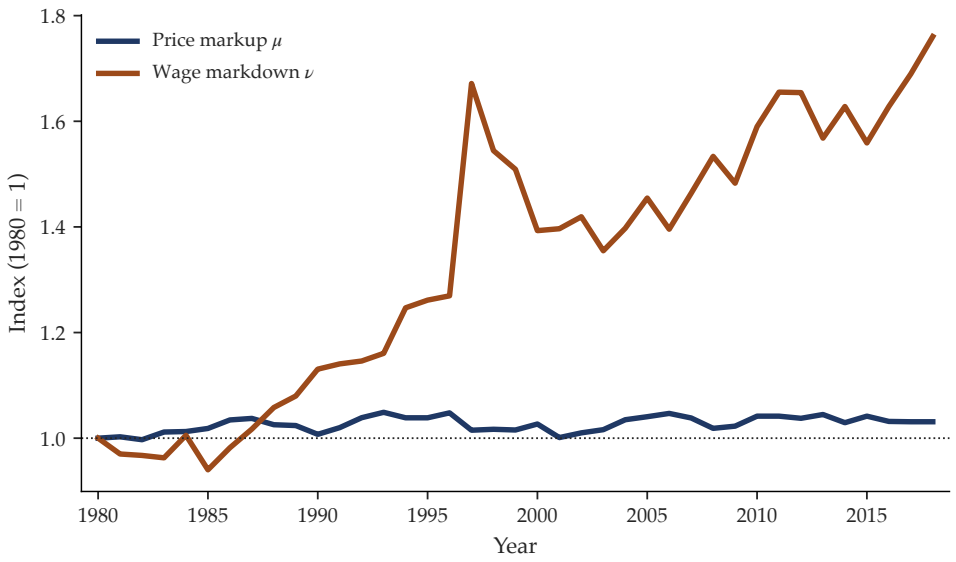


Figure 9: Price markups and wage markdowns in Ren and Zhang (2025). Both series are normalized to one in 1980.

A higher price markup creates rents by reducing both labor and capital payments. A higher wage markdown is different. Holding the price markup fixed,

$$ds_K = 0, \qquad ds_W = -ds_\Pi < 0. \tag{22}$$

The capital income share s_K is unchanged: the rise in the aggregate profit income share s_Π comes entirely from the aggregate labor income share s_W. With endogenous labor supply, the lower wage also directly reduces labor L and output Y.

The role of the asset market channel is clearest with fixed aggregate labor supply, $L = \bar{L}$. The capital first-order condition is

$$r + \delta_K = \frac{\alpha}{\mu} \frac{Y}{K}. \tag{23}$$

Holding μ, r, and $\bar{L}$ fixed, a higher wage markdown leaves physical capital K and output Y unchanged. It only transfers income from workers to profit recipients. Redistribution shifts asset demand and Asset Creation shifts asset supply. With inelastic asset demand, these shifts move the equilibrium interest rate and hence physical capital and output. The Capital Wedge contributes nothing to the interest rate response in this markdown-only experiment because s_K is unchanged.

Joint changes in product markups and wage markdowns. For the quantitative exercise, we use both paths from Ren and Zhang (2025) shown in Figure 9. We multiply the calibrated 1980 product markup and wage markdown by their respective indices, each normalized to one in 1980. Product markups rise by 3.11 percent and wage markdowns by 76.04 percent through 2018. All other parameters remain at their baseline values, and we use the sufficient statistic with the calibrated

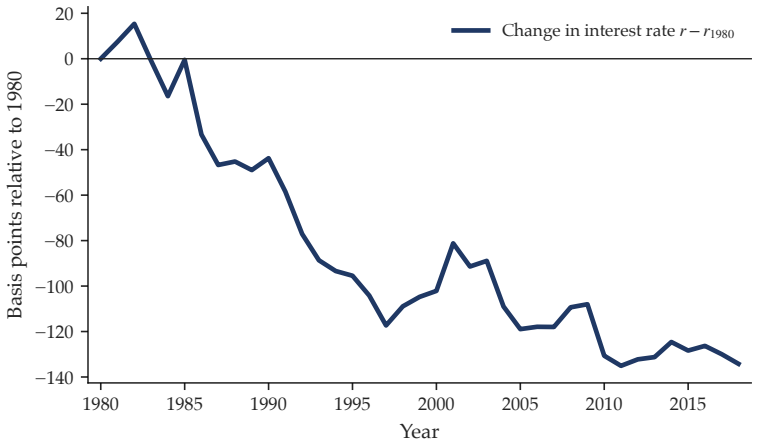


Figure 10: Product markups, wage markdowns, and the asset market channel. Change in the interest rate r relative to 1980, computed with the sufficient statistic using both paths in Figure 9.

worker and owner asset demand semi elasticities. Unlike the markdown-only benchmark, this joint experiment includes both the asset market incidence of rents and the Capital Wedge associated with a higher product markup.

Figure 10 shows that the interest rate r falls by 134.2 basis points between the calibrated 1980 economy and the 2018 joint wedges. Redistribution dominates this response. Using the decomposition of Section 4.3, Redistribution contributes -275.7 basis points, while Asset Creation partially offsets it with a contribution of 155.8 basis points. The Capital Wedge contributes only -14.3 basis points. Thus, even with both product markups and wage markdowns changing, the asset market incidence of additional rents accounts for most of the interest rate decline.

In the representative household benchmark, the joint increase in market power reduces output Y by 27.22 percent. The equilibrium interest rate decline raises physical capital and mitigates this loss: output falls by 20.16 percent instead. The asset market channel therefore mitigates 26.0 percent of the representative household output loss, approximately one quarter.

7 Conclusion

How market power redistributes factor income and profits determines its effects on asset markets and the broader economy. We characterize how rising markups contribute to an endogenous decline in the long-run rate of return in asset markets through its distributional incidence across households and show quantitatively that this decline mitigates roughly half of the aggregate output loss from markups relative to the case where the return is fixed. Despite the smaller aggregate quantity distortions, the population consumption-equivalent variation welfare loss is nearly five times as large as in the representative household benchmark, with the welfare gains concentrated among the small population of firm owners.

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A Proofs and Derivations

A.1 Aggregation with heterogeneous wedges

This appendix derives the aggregate representation used in Section 2.1. Firm i produces $y_i = z_i v_i$ from a common constant returns input composite $v_i = F(k_i, \ell_i)$. Aggregate output is a homothetic index with $PY = \int p_i y_i \, di$. Define the output-to-index ratio $q_i \equiv y_i/Y$ and the revenue share $s_i \equiv p_i y_i/(PY)$.

Productivity. Aggregate use of the input composite is

$$V = \int \frac{y_i}{z_i} \, di = Y \int \frac{q_i}{z_i} \, di.$$

Hence $Y = ZV$, with

$$Z^{-1} \equiv \int q_i z_i^{-1} \, di.$$

The allocation of output across producers determines Z. For a general homothetic output index, the q_i need not integrate to one, so Z is a productivity index rather than a harmonic mean. In the additive special case, where $\int q_i \, di = 1$, it is the quantity-weighted harmonic mean of firm productivities.

Aggregate markup. Let $c_{v,i}$ denote firm i's shadow unit cost of the input composite, evaluated at its shadow factor prices, and let $\mathcal{C}_i \equiv c_{v,i} v_i$ denote its shadow variable cost. Constant returns and the definition of the markup imply $p_i y_i = \mu_i \mathcal{C}_i$. Letting $\mathcal{C} \equiv \int \mathcal{C}_i \, di$ allows shadow factor prices, and hence $c_{v,i}$, to differ across firms. Aggregation gives

$$PY = \mu \mathcal{C}, \qquad \mu \equiv \int \mu_i \frac{\mathcal{C}_i}{\mathcal{C}} \, di.$$

Moreover,

$$s_i = \frac{\mu_i}{\mu} \frac{\mathcal{C}_i}{\mathcal{C}}, \qquad \mu^{-1} = \int s_i \mu_i^{-1} \, di.$$

Thus the aggregate markup is the cost-weighted arithmetic mean of firm markups, equivalently their sales-weighted harmonic mean. If all firms face a common shadow unit cost c_v for the composite, then $\mathcal{C}_i = c_v v_i$, $\mathcal{C} = c_v V$, and the special-case identities $\mu = \int \mu_i (v_i/V) \, di$ and $P = \mu c_v / Z$ follow. The factor-share derivation below does not impose this common-cost restriction.

Factor shares and aggregate markdown. Now specialize the common composite to Cobb–Douglas, $v_i = k_i^{\alpha} \ell_i^{1-\alpha}$. Capital is rented competitively and labor is marked down. Euler's theorem

and the firm's first-order conditions give $\mathcal{C}_i = r^k k_i + \nu_i w_i \ell_i$. The firm factor shares are

$$\frac{r^k k_i}{p_i y_i} = \frac{\alpha}{\mu_i}, \qquad \frac{w_i \ell_i}{p_i y_i} = \frac{1-\alpha}{\mu_i \nu_i}.$$

Integrating against revenue shares gives

$$s_K = \alpha \int \frac{s_i}{\mu_i}\, di = \frac{\alpha}{\mu}, \qquad s_W = (1-\alpha) \int \frac{s_i}{\mu_i \nu_i}\, di = \frac{1-\alpha}{\mu\nu}.$$

The aggregate markdown separates the labor wedge from the product wedge:

$$\nu^{-1} = \frac{\int s_i (\mu_i \nu_i)^{-1}\, di}{\int s_i \mu_i^{-1}\, di}.$$

Using the firm labor share condition,

$$\nu_i w_i \ell_i = (1-\alpha) \mu_i^{-1} s_i P Y,$$

so the same object can be written as

$$\nu^{-1} = \int c_i \nu_i^{-1}\, di, \qquad c_i \equiv \frac{\nu_i w_i \ell_i}{\int \nu_j w_j \ell_j\, dj}.$$

The weights c_i are shares of the shadow wage bill, $\nu_i w_i \ell_i$, where $\nu_i w_i$ is the marginal cost of labor implied by the firm's first-order condition. This is the labor-market analogue of the cost weights in the aggregate markup; observed wage-bill shares would not recover the aggregate markdown.

The aggregate factor shares are therefore

$$s_K = \frac{\alpha}{\mu}, \qquad s_W = \frac{1-\alpha}{\mu\nu}, \qquad s_\Pi = 1 - s_K - s_W.$$

Under the common Cobb–Douglas technology, heterogeneous markups and markdowns enter factor shares only through μ and ν.

Misallocation. Let $\mathcal{Z}^*$ denote frontier TFP and let $\Omega \leq 1$ denote misallocation, so realized TFP is $A = \Omega \mathcal{Z}^*$. Under Cobb–Douglas, Ω contains two components. The first is the output allocation term $Z/\mathcal{Z}^*$ from the allocation of output across producers. The second is an input mix component. Since

$$V = \int k_i^\alpha \ell_i^{1-\alpha}\, di = K^\alpha L^{1-\alpha} \int \hat{k}_i^\alpha \hat{\ell}_i^{1-\alpha}\, di,$$

where $\hat{k}_i \equiv k_i/K$ and $\hat{\ell}_i \equiv \ell_i/L$, aggregate output can be written as

$$Y = Z\left(\int \hat{k}_i^{\alpha}\hat{\ell}_i^{1-\alpha}\,di\right)K^{\alpha}L^{1-\alpha}, \qquad \Omega = \frac{Z}{\mathcal{Z}^*}\int \hat{k}_i^{\alpha}\hat{\ell}_i^{1-\alpha}\,di.$$

Markup dispersion affects Ω through Z. Dispersion in labor shadow prices, $\nu_i w_i$, also affects Ω through the input mix component because it changes how firms combine capital and labor. Substituting the factor demands gives

$$\hat{k}_i \propto \frac{s_i}{\mu_i}, \qquad \hat{\ell}_i \propto \frac{s_i}{\mu_i \nu_i w_i}, \qquad \frac{\hat{\ell}_i}{\hat{k}_i} \propto (w_i\nu_i)^{-1}.$$

The markup cancels from the relative input choice, while the labor shadow price remains. Hölder's inequality implies that the input mix component is at most one, with equality if and only if $w_i\nu_i$ is constant across active firms. A second-order expansion around a common labor shadow price gives

$$\int \hat{k}_i^{\alpha}\hat{\ell}_i^{1-\alpha}\,di \simeq 1 - \frac{\alpha(1-\alpha)}{2}\mathrm{Var}_{\hat{k}}\big(\log w_i\nu_i\big),$$

where $\mathrm{Var}_{\hat{k}}$ is the variance under the capital-share weights $\{\hat{k}_i\}$. Thus dispersion in labor shadow prices lowers Ω by distorting the input mix across firms. Realized output is

$$Y = AK^{\alpha}L^{1-\alpha}.$$

This is the production side separation used in the main text: under Cobb–Douglas, wedge dispersion affects realized TFP, while factor shares depend on the cross section only through the aggregate wedges.

A.2 Misallocation and the Asset Market Channel

This appendix measures how much the baseline separation changes when production is CES in place of Cobb–Douglas. It is a sensitivity exercise, not a general aggregation result for heterogeneous-firm CES economies: aggregate wedges are held fixed, and dispersion is represented by the Hicks-neutral productivity term $A = \Omega\mathcal{Z}^*$ in a common-wedge CES firm. The question is whether a change in Ω moves the asset market by changing factor shares.

Under Cobb–Douglas, a fall in Ω lowers output but leaves factor shares unchanged. The return equation is therefore unchanged. Under CES, factor shares depend on the user cost of capital relative to productivity, r^k/A. At a fixed return, a fall in Ω raises r^k/A, changes the capital and labor shares, and shifts the asset market equation. The response is proportional to $1-\sigma_{KL}$ and vanishes at $\sigma_{KL} = 1$.

Let production be constant returns to scale and CES,

$$Y = A\,F(K,L), \qquad F = \left[\alpha K^{\rho} + (1-\alpha)L^{\rho}\right]^{1/\rho}, \qquad \rho = \frac{\sigma_{KL}-1}{\sigma_{KL}},$$

with capital elasticity $\varepsilon_K \equiv F_K K/F$, where σ_{KL} is the elasticity of substitution between capital and labor and is distinct from the household's CRRA parameter. The user cost satisfies $r^k = (A/\mu)F_K$, and the factor shares are $s_K = \varepsilon_K/\mu$, $s_W = (1-\varepsilon_K)/(\mu\nu)$, and $s_\Pi = 1 - s_K - s_W$.

The capital cost share responds to the relative price of capital with elasticity $1-\sigma_{KL}$, so, holding the wedges fixed,

$$\begin{aligned} d\log\varepsilon_K &= (1-\sigma_{KL})\,d\log\frac{r^k}{A}, \\ ds_K &= (1-\sigma_{KL})\,s_K\,d\log\frac{r^k}{A}, \\ ds_W &= -\frac{1}{\nu}\,ds_K, \qquad ds_\Pi = -\frac{\nu-1}{\nu}\,ds_K. \end{aligned}$$

A fall in Ω raises r^k/A at a fixed return. Under gross complements ($\sigma_{KL} < 1$) the capital share s_K rises and the labor share s_W falls; under gross substitutes ($\sigma_{KL} > 1$) s_K falls and s_W rises. The business profit share s_Π moves by $-(\nu-1)/\nu$ times the move in s_K and is unchanged at the calibration, which sets $\nu = 1$.

The asset market equation remains

$$\underbrace{\bar{D}(r)\left[s_W(\Omega,r) + \lambda(r)s_\Pi(\Omega,r)\right]}_{\text{asset demand}} = \underbrace{\frac{s_K(\Omega,r)}{r+\delta_K} + Q(r)\omega s_\Pi(\Omega,r)}_{\text{asset supply}}.$$

These share movements therefore shift asset demand and asset supply, and r adjusts while asset market clearing holds. Define the fixed-share semielasticity of aggregate household asset demand by

$$\epsilon_D \equiv \frac{\partial}{\partial r}\log\left\{\bar{D}(r)\left[s_W + \lambda(r)s_\Pi\right]\right\}\Big|_{s_W,s_\Pi} = \frac{\partial\log\bar{D}(r)}{\partial r} + \frac{\lambda'(r)s_\Pi}{s_W + \lambda(r)s_\Pi}.$$

This is the aggregate household-demand semielasticity used in the numerical exercise; if λ is held fixed, its second term is zero. Differentiating at the Cobb–Douglas baseline and using $r^k = r + \delta_K$,

Table A1: Effect of misallocation on shares and the return under CES production ($d\log\Omega = -0.03$, $\epsilon_D = 16.2963$).

σ_{KL}	dr	ds_K	ds_W
0.6	+8 bp	+0.47 pp	−0.47 pp
0.8	+4 bp	+0.21 pp	−0.21 pp
0.9	+2 bp	+0.10 pp	−0.10 pp
1.0	0 bp	0.00 pp	0.00 pp
1.1	−2 bp	−0.09 pp	+0.09 pp
1.2	−3 bp	−0.18 pp	+0.18 pp

Notes: The baseline sets $\nu = 1$, so $ds_\Pi = -\frac{\nu-1}{\nu} ds_K = 0$ in this exercise. Here ϵ_D is the fixed-share semielasticity of aggregate household asset demand defined above, including the response of $\lambda(r)$. Calibration $r = 0.06$, $\delta_K = 0.07$, $\delta_C = 0.05$, $\omega^{\text{inc}} = 0.40$, $\omega = (1-\theta_F)\omega^{\text{inc}}$, $\theta_F = 0.15$, $\lambda = 3.31$, $\mu = 1.0417$, $\nu = 1$, $\varepsilon_K = 1/3$.

the return solves

$$\left[\frac{1}{r} + \epsilon_D - \frac{\delta_K s_K (r+\delta_K)^{-2} + \omega\delta_C s_\Pi (r+\delta_C)^{-2}}{\dfrac{r}{r+\delta_K} s_K + rQ(r)\omega s_\Pi}\right] dr$$

$$= (1-\sigma_{KL})\, s_K \left[\frac{\dfrac{r}{r+\delta_K} - rQ(r)\omega\dfrac{\nu-1}{\nu}}{\dfrac{r}{r+\delta_K} s_K + rQ(r)\omega s_\Pi} + \frac{1+\lambda(\nu-1)}{\nu(s_W+\lambda s_\Pi)}\right]\left(\frac{dr}{r+\delta_K} - d\log\Omega\right).$$

Let $\mathcal{L}$ denote the bracket on the left and let $\mathcal{B}$ denote the bracket multiplying $(1-\sigma_{KL})s_K$ on the right. After collecting the feedback through r^k, the relevant local-equilibrium slope is

$$\mathcal{S}_{CES} \equiv \mathcal{L} - \frac{(1-\sigma_{KL})s_K \mathcal{B}}{r+\delta_K}.$$

The local-stability condition is $\mathcal{S}_{CES} > 0$, not merely $\mathcal{L} > 0$. It holds at the calibration for the values reported in Table A1, where $\mathcal{B} > 0$. Hence a fall in Ω raises r under complements and lowers it under substitutes. A smaller ϵ_D produces a larger response as long as the stability condition continues to hold. The endogenous change in r^k reinforces the share movement under complements and offsets it under substitutes.

The perturbation $d\log\Omega = -0.03$ lies within the 2 to 6% value added misallocation losses measured by Edmond, Midrigan and Xu (2023). Attributing the entire loss to the experiment would roughly double the entries. Over the central range $\sigma_{KL} \in [0.6, 1.2]$, the return moves by at most 8.3 basis points. The corresponding business profit share movement is zero because the baseline sets $\nu = 1$. At $\sigma_{KL} = 0.5$, a stress case below most estimates, the return moves about 11 basis points. These movements are small compared with the main transition, which puts the return -94.0 basis points relative to baseline under the observed aggregate markup path. Thus the separation is exact under Cobb–Douglas, and away from Cobb–Douglas the feedback from misallocation to the return

remains quantitatively small at estimated elasticities.

A.3 Business profit shares from a homothetic single-aggregator (HSA) demand system

This appendix microfounds the business profit share state z and shows it carries no markup dispersion, so the cross-section stays in the common-wedge case of Section 2. Demand comes from a homothetic single-aggregator (HSA) system (Matsuyama and Ushchev, 2017) with a multiplicative appeal shifter. The asset market needs a distribution of business profit shares with the properties the calibration targets, and the construction below is one microfoundation that delivers them.

Demand. A continuum of product lines $m \in \mathcal{M}$ is consumed in quantities q_m. Demand is

$$q_m = z_m \frac{E}{\mathcal{A}} D\Big(\frac{p_m}{\mathcal{A}}\Big),$$

where E is aggregate expenditure and $z_m > 0$ is a multiplicative appeal shifter with $\int_{\mathcal{M}} z_m \, dm = 1$. The function D is positive, twice differentiable, and strictly decreasing, with elasticity $\varepsilon(x) \equiv -xD'(x)/D(x) > 1$ and ε nondecreasing (Marshall's second law), so the pricing problem below has a unique interior solution. The single aggregator $\mathcal{A}$ is defined by the adding-up condition $\int_{\mathcal{M}} z_m \, (p_m/\mathcal{A}) \, D(p_m/\mathcal{A}) \, dm = 1$, which has a unique positive root under these conditions. The appeal z_m is a *pure horizontal* taste shifter, orthogonal to marginal cost; quality interpretations that raise cost are excluded precisely because they would move the markup. The system nests CES, $D(x) = x^{-\varepsilon}$.

Appeal is markup-neutral. Lines are priced atomistically: each takes $\mathcal{A}$ and E as given, even though ownership is organized into positive-measure baskets below. The perceived demand elasticity is

$$-\frac{\partial \log q_m}{\partial \log p_m} = \varepsilon\Big(\frac{p_m}{\mathcal{A}}\Big),$$

which depends on $p_m/\mathcal{A}$ alone and not on z_m: a multiplicative shifter scales the demand curve without rotating it. With common marginal cost c, every line solves the same first-order condition $p_m = \mu(p_m/\mathcal{A}) \, c$, where $\mu(x) = \varepsilon(x)/(\varepsilon(x) - 1)$, so all lines set the same price $p^\star$ and the same markup $\mu^\star$. The cross-section is markup-homogeneous by construction.

Business profit shares. At the symmetric price the adding-up condition gives $(p^\star/\mathcal{A})D(p^\star/\mathcal{A}) = 1$, so the revenue share is $s_m = z_m(p_m/\mathcal{A})D(p_m/\mathcal{A}) = z_m$ and line profit is $\pi_m = (1 - 1/\mu^\star)p_m q_m \propto z_m$. The measure of lines is fixed and there is no free entry, so Π is a pure rent to incumbent ownership with no free-entry dissipation. Aggregate profit income is

$$\Pi = \int_{\mathcal{M}} \pi_m \, dm = \Big(1 - \tfrac{1}{\mu^\star}\Big) E = \Big(1 - \tfrac{1}{\mu^\star}\Big) Y,$$

set by the common markup and independent of the cross-sectional distribution of $\{z_m\}$. With a common labor markdown ν, markdown rents accrue to each line in proportion to its size and distribute in the same shares, so z is the entrepreneur's share of *total* aggregate profit income.

Ownership and dynamics. Entrepreneurs own baskets of lines $\{B_i\}$ that partition $\mathcal{M}$. Entrepreneur i earns

$$\pi_i = \Pi z_i, \qquad z_i \equiv \int_{B_i} z_m \, dm, \qquad \int z_i \, di = \int_{\mathcal{M}} z_m \, dm = 1.$$

The owner-specific shock is a *basket-level common factor* that scales the appeal of every line in B_i together, so it survives aggregation within the basket and makes z_i a non-degenerate, stationary, mean-one process—the AR(1) for z in the calibration. Were appeal shocks instead independent across an entrepreneur's continuum of lines, a law of large numbers would make z_i deterministic and the idiosyncratic profit income risk would vanish. Because the shocks are relative, $\int z_i \, di = 1$ at every date: they reallocate rents across entrepreneurs without moving aggregate profit income Π, the factor shares, or realized TFP. The business profit share distribution is thus orthogonal to the Section 2 accounting, which is what lets the household problem scale exactly with Π.

What the construction requires. Three features are essential, and each is a modeling choice. (i) Lines are infinitesimal and priced atomistically, so each entrepreneur takes substitution within her basket as external; with internalization, markups would rise in basket share (Atkeson and Burstein, 2008) and dispersion would return. (ii) Appeal enters multiplicatively and is orthogonal to cost; cost heterogeneity, or quality entering as a price equivalent, would move $p_m/\mathcal{A}$ across lines and break markup homogeneity. (iii) The appeal shocks are relative, so the cross-sectional reallocation leaves the aggregate markup, Π, and realized TFP untouched.

A.4 Stationary private, incorporated, and liquid masses

Let N_P, N_C, N_L be the masses of private businesses, incorporated businesses, and liquid entrepreneurs. The laws of motion are

$$N_{P,t+1} = (1-\kappa)N_{P,t} + \delta_C N_{L,t}, \quad N_{C,t+1} = (1-\delta_C)N_{C,t} + \kappa N_{P,t}, \quad N_{L,t+1} = (1-\delta_C)N_{L,t} + \kappa N_{P,t}.$$

In the stationary composition with active-business mass $N_P^* + N_C^* = \eta$: $\kappa N_P^* = \delta_C N_C^*$ and $N_L^* = N_C^*$, hence

$$N_C^* = \eta \frac{\kappa}{\kappa + \delta_C}, \qquad N_P^* = \eta \frac{\delta_C}{\kappa + \delta_C}, \qquad N_L^* = \eta \frac{\kappa}{\kappa + \delta_C}.$$

Define the publicly incorporated profit share $\omega^{\text{inc}} = \int \tau_j z_j \, dj$, where $\tau_j = 1$ for publicly incorporated businesses. We assume that a firm's probability of public incorporation is independent of its business profit share z_j. Publicly incorporated businesses then carry the stationary z-distribution,

and $\omega^{\text{inc}} = N_C^*/\eta = \kappa/(\kappa + \delta_C)$. The publicly traded equity share of aggregate profits is $\omega = (1-\theta_F)\omega^{\text{inc}}$, as in (9). The private, retained, and publicly traded profit shares are respectively $1-\omega^{\text{inc}}$, $\theta_F\omega^{\text{inc}}$, and ω. Without independence, ω^{inc} is the profit-weighted share of publicly incorporated businesses. The aggregate market value of tradable equity is $\int_{\tau_j=1} q(z_j;\Pi,r)\,dj = Q(r)(1-\theta_F)\omega^{\text{inc}}\Pi = Q(r)\omega\Pi$.

A.5 Equity valuation

Dividing (7) by Π, the normalized price $\theta_q(z;r) = q(z;\Pi,r)/\Pi$ solves

$$(1+r)\theta_q(z;r) = \mathbb{E}\big[(1-\theta_F)z' + (1-\delta_C)\theta_q(z';r) \mid z\big],$$

a recursion free of Π. Taking expectations under the stationary distribution of z (mean one) and solving the resulting scalar equation gives $\mathbb{E}_\pi\theta_q = (1-\theta_F)/(r+\delta_C)$. Dividing by the publicly traded fraction $1-\theta_F$ gives $Q(r) = 1/(r+\delta_C)$ in (8); when $\theta_F = 1$, this multiple is defined by continuity and tradable equity has zero value. If z is constant the pointwise equity value is $q = (1-\theta_F)\Pi z/(r+\delta_C)$.

A.6 Proof of Proposition 2 (linearity)

Proof. **Workers.** With GHH preferences, the intratemporal labor choice can be solved before the savings problem. Let

$$y^W(w,e) = \max_\ell\{we\ell - v(\ell)\}.$$

For $v(\ell) = \psi_\ell \ell^{1+1/\varphi}/(1+1/\varphi)$,

$$y^W(w,e) = \frac{1}{1+\varphi}\psi_\ell^{-\varphi}(we)^{1+\varphi}.$$

Aggregate labor income is

$$W = \int we\ell^*(w,e)\,d\Lambda_e = \psi_\ell^{-\varphi}w^{1+\varphi}\int e^{1+\varphi}\,d\Lambda_e,$$

so $y^W(w,e) = W\widetilde{e}$ with $\widetilde{e} = e^{1+\varphi}/[(1+\varphi)\int e^{1+\varphi}d\Lambda_e]$, which is independent of the aggregate scale W. Setting $x = c - v(\ell)$, $b = a/W$, $b' = a'/W$, and $\hat{x} = x/W$, the worker problem is equivalent to

$$v^W(b,e) = \max_{\hat{x},\,b'\geq 0}\big\{u(\hat{x}) + \beta_W\mathbb{E}[v^W(b',e') \mid e]\big\}, \qquad \hat{x} + b' = Rb + \widetilde{e}.$$

The normalized problem contains no W; the guess $V^W(a,e) = W^{1-\sigma}v^W(a/W,e)$ verifies by homogeneity of u. Normalized policies g_W and the invariant distribution Λ_W^r are therefore independent

of W, and

$$D_W(r, W) = W \int g_W(b, e)\, d\Lambda_W^r \equiv \bar{D}(r)\, W.$$

Entrepreneurs. Set $b = a/\Pi$, $\widetilde{t} = t/\Pi$, $\tilde{c} = c/\Pi$, and let $\theta_q(z; r)$ be the normalized tradable equity price of Appendix A.5, which is independent of Π. Guess $V^P(a, z) = \Pi^{1-\sigma} v^P(b, z)$ and $V^L(t, a, z) = \Pi^{1-\sigma} v^L(\widetilde{t}, b, z)$. The normalized budgets are

$$\tilde{c} + b' = Rb + z \qquad \text{and} \qquad \tilde{c} + b' = Rb + \widetilde{t} + \theta_F z,$$

incorporation sends the entrepreneur to the normalized state $v^L(\theta_q(z; r), b', z)$, and the transfer state resets to zero while the business profit share state follows its Markov process until the incorporated business exits. Neither normalized problem contains Π, so normalized policies g_O and the invariant distribution Λ_O^r are independent of Π, and

$$D_O(r, \Pi) = \Pi \int g_O(b, z, \widetilde{t})\, d\Lambda_O^r \equiv \lambda(r)\, \bar{D}(r)\, \Pi.$$

□

A.7 Proof of Theorem 1

Proof. The clearing identity follows from Proposition 2 (demand side), the supply construction $S(r) = K + Q(r)\omega\Pi$ of Appendix A.4–A.5, and market clearing. Dividing by output and using $K/Y = s_K/(r + \delta_K)$ gives

$$\bar{D}(r)\left[s_W + \lambda(r) s_\Pi\right] = \frac{1}{r + \delta_K} s_K + Q(r)\omega s_\Pi.$$

This is (11). □

A.8 Derivations for Section 3.3

The derivations of these special cases impose complete sale, $\theta_F = 0$, and no capital depreciation, $\delta_K = 0$, as in Assumption 1. Write $H(r) \equiv r\bar{D}(r)$, with $H'(r) > 0$. This slope converts shifts in the right side of the sufficient statistic into interest rate responses.

Redistribution ($\omega = 0$). Under the transparent benchmark $\delta_K = 0$, the statistic is $r\bar{D} = s_K/(s_W + \lambda^P s_\Pi)$ where λ^P is the relative wealth-to-income ratio of owners to workers when all businesses remain private. With markup-only Cobb–Douglas shares $(s_K, s_W, s_\Pi) = (\alpha/\mu, (1-\alpha)/\mu, 1 - 1/\mu)$:

$$r\bar{D}(r) = \frac{\alpha/\mu}{(1-\alpha)/\mu + \lambda^P(r)\,(1 - 1/\mu)},$$

which at $\lambda^P = 1$ collapses to $\alpha/(\mu - \alpha)$, decreasing in μ. With markdown-only shares $(\alpha, (1 -$

$\alpha)/\nu, (1-\alpha)(1-1/\nu)$):

$$r\bar{D}(r) = \frac{\alpha}{(1-\alpha)/\nu + \lambda^P(r)\,(1-\alpha)(1-1/\nu)},$$

which at $\lambda^P = 1$ collapses to $\alpha/(1-\alpha)$: the denominator $(1-\alpha)/\nu + (1-\alpha)(1-1/\nu) = (1-\alpha)$ exactly, so ν drops out at the equal-asset-demand case. For $\lambda^P > 1$ the direct right-side shift is decreasing in ν; under the local-stability condition stated below, r^* falls.

Asset Creation ($\omega = 1$, $\lambda = 0$). With $\kappa > 0$ and $\delta_C = 0$, every business eventually incorporates and its founder sells the publicly traded equity once. Complete sale gives $Q(r) = 1/r$, and $\beta_O(1+r) < 1$ makes founders draw down their assets (Lemma 1). At this endpoint $\lambda = 0$, so

$$r\bar{D}(r) = \frac{s_K + s_\Pi}{s_W} = \frac{1 - s_W}{s_W}.$$

For markup-only and markdown-only Cobb–Douglas shares, respectively,

$$r\bar{D}(r) = \frac{\mu - 1 + \alpha}{1-\alpha}, \qquad r\bar{D}(r) = \frac{\nu - 1 + \alpha}{1-\alpha}.$$

The first expression rises with μ for every $\alpha \in (0,1)$; the second rises with ν. With both wedges,

$$r\bar{D}(r) = \frac{1 - s_W}{s_W} = \frac{\mu\nu}{1-\alpha} - 1.$$

Hence $\partial r^*/\partial\mu = \nu/[(1-\alpha)H'(r^*)] > 0$ and $\partial r^*/\partial\nu = \mu/[(1-\alpha)H'(r^*)] > 0$. Pure Asset Creation raises the interest rate under either form of market power for every $\alpha \in (0,1)$.

Local sign comparison. Write the equilibrium as

$$F(r; s_W, s_K, s_\Pi) \equiv r\bar{D}(r) - \mathrm{RHS}(r; s_W, s_K, s_\Pi) = 0, \qquad \mathrm{RHS} = \frac{\dfrac{r}{r+\delta_K}s_K + rQ(r)\omega s_\Pi}{s_W + \lambda s_\Pi}.$$

Assume the local-stability condition $F_r > 0$, which includes the r-dependence of the capital and equity valuation terms and of λ. The implicit function theorem then implies that r^* moves in the direction of the direct shift in RHS. A markup raises s_Π by ds_Π and lowers s_K, s_W by $\alpha\, ds_\Pi$ and $(1-\alpha)\, ds_\Pi$; a markdown leaves s_K fixed and moves $ds_W = -ds_\Pi$. Evaluating the direct change in RHS at the undistorted baseline, with the incidence terms evaluated at the initial return, gives, for the markup,

$$\mathrm{sign}\left(dr^*\right) = \mathrm{sign}\left[(1-\alpha)Q(r)\omega - \frac{\alpha\lambda}{r+\delta_K}\right],$$

so r^* falls iff $\lambda > (1-\alpha)Q(r)\omega(r+\delta_K)/\alpha$; for the markdown the same computation gives sign$[(1-\alpha)Q(r)\omega - \alpha(\lambda-1)/(r+\delta_K)]$, i.e. $\lambda > 1 + (1-\alpha)Q(r)\omega(r+\delta_K)/\alpha$. At $\alpha = 1/3$, $r/(r+\delta_K) = 0.50$,

and $rQ(r)\omega = 0.218$, the transparent cutoffs are 0.93 and 1.93; around the calibrated product markup $\mu_0 = 1.0417$, the markdown cutoff is 1.99.

These are direct share comparisons: the incidence objects are held fixed when the shares move, while their dependence on the return enters F_r. Appendix B instead varies the structural hazards, so ω and λ move jointly.

B Redistribution and Asset Creation: The Two Special Cases

In the literature, profits typically enter asset-market clearing in two distinct ways. In models of entrepreneurial saving, profits are income received by business owners. They increase the wealth those owners wish to hold but do not create a traded asset (Quadrini, 2000; Cagetti and De Nardi, 2006; Mian, Straub and Sufi, 2021*b*). In models with capitalized profits, the market value of tradable equity backed by future profit income enters asset supply (Eggertsson, Robbins and Wold, 2021; Auclert et al., 2025).

This appendix studies the complete-sale subcase $\theta_F = 0$ in order to establish the two special cases. The baseline with retained owner equity in the main text instead holds $\theta_F = 0.15$ and defines $\omega = (1 - \theta_F)\omega^{\text{inc}}$, with $Q(r) = 1/(r + \delta_C)$ per dollar of publicly traded profit income. Throughout this appendix, complete sale implies $\omega = \omega^{\text{inc}}$, so the formulas below use ω for both shares.

These are special cases of our model. The pure redistribution case is obtained by ruling out the incorporation of private businesses into tradable equity, or setting the probability of incorporation to $\kappa = 0$. Consequently, the share of aggregate profit income represented by tradable equity is zero: $\omega = 0$. The asset-market clearing condition then becomes

$$\bar{D}(r)\left[W + \lambda(r)\Pi\right] = K. \tag{B1}$$

At the pure asset-creation endpoint, tradable equity backed by future profit income is fully tradable. This case is obtained by setting the exit rate of public firms to zero, $\delta_C = 0$, while maintaining $\kappa > 0$. Every private business then eventually incorporates, and its founder sells the equity once. Because the resulting tradable equity is permanent, $\omega = 1$ and $Q(r) = 1/r$. Profits therefore contribute Π/r to asset supply. The corresponding asset-market-clearing benchmark is

$$\bar{D}(r)W = K + \frac{\Pi}{r}. \tag{B2}$$

The absence of profits from the demand side does not follow mechanically from $\omega = 1$, which determines only asset supply. It also requires the assets founders accumulate from private-profit income and sale proceeds to vanish as the endpoint is approached. To illustrate the mechanism, we assume $\beta_O R < 1$, as in standard stationary incomplete-market environments in which precautionary saving keeps the equilibrium risk-free return below $1/\beta_O$. A liquidated firm owner who receives sale proceeds once then draws down the additional assets generated by that payment.

For every $\delta_C > 0$, however, stationary founder asset demand reflects recurring flows of private-profit income and sale proceeds. The relevant question is whether their contribution to asset demand vanishes as $\delta_C \downarrow 0$. We first record the asset-market accounting and then study the drawdown of a one-time payment. Finally, we show that both flows are of order δ_C and, under a uniform bound on the persistence of founder saving, that the associated stock of founder assets converges to zero. Thus asset-market clearing converges to (B2). Away from the endpoint, both

sources of founder asset demand are encoded in $\lambda(r;\kappa,\delta_C)$.

B.1 Asset-market accounting

For general (κ,δ_C), write asset demand as

$$D(r) = \bar{D}(r)W + D_O(r,\Pi), \qquad \lambda(r;\kappa,\delta_C) \equiv \frac{D_O(r,\Pi)}{\bar{D}(r)\Pi}, \tag{B3}$$

where D_O is aggregate founder asset demand and its dependence on (κ,δ_C) is suppressed in the arguments. Because incorporation is independent of a firm's share of aggregate profit income, the stationary ownership masses also determine the shares of aggregate profit income represented by private and public firms. Asset supply is therefore

$$S(r) = K + Q(r)\omega\Pi, \qquad Q(r) = \frac{1}{r+\delta_C}, \qquad \omega = \frac{\kappa}{\kappa+\delta_C}. \tag{B4}$$

Thus the stationary asset-market equation is

$$\bar{D}(r)\big[W + \lambda(r;\kappa,\delta_C)\Pi\big] = K + Q(r)\omega\Pi. \tag{B5}$$

At $\kappa = 0$, the pure redistribution endpoint has $\omega = 0$ and

$$\bar{D}(r)\big[W + \lambda(r)\Pi\big] = K. \tag{B6}$$

At $\delta_C = 0$, $\omega = 1$, $Q(r) = 1/r$, and the asset-market equation has the form

$$\bar{D}(r)W + D_O(r,\Pi) = K + \frac{\Pi}{r}. \tag{B7}$$

The remaining demand-side result is that, along the interior stationary economies, $D_O(r_{\delta_C},\Pi)$ converges to zero as $\delta_C \downarrow 0$. This result yields the benchmark in (B2).

B.2 A one-time sale and the drawdown mechanism

When there is no exit for firms, $\delta_C = 0$, a founder sells her equity at most once. Normalize the liquidated firm owner's sale date to $j = 0$. Let $a_0 \geq 0$ denote financial assets at the beginning of that date and let $x > 0$ denote the sale proceeds. Total resources at the sale date are $q \equiv Ra_0 + x$, and there is no subsequent nonasset income. Hold the gross return fixed at $R = 1 + r > 1$. Taking (a_0, x), and hence q, as given, the founder solves

$$\max_{\{c_j,a_{j+1}\}_{j\geq 0}} \sum_{j=0}^{\infty} \beta_O^j u(c_j), \qquad u(c) = \begin{cases} c^{1-\sigma}/(1-\sigma), & \sigma \neq 1, \\ \log c, & \sigma = 1, \end{cases} \tag{B8}$$

subject to

$$c_0 + a_1 = Ra_0 + x \equiv q, \qquad c_j + a_{j+1} = Ra_j, \quad j \geq 1, \qquad c_j > 0, \quad a_{j+1} \geq 0, \qquad j \geq 0, \tag{B9}$$

and the present-value exhaustion condition $\lim_{J\to\infty} a_{J+1}/R^J = 0$.

Lemma 1 (Drawdown after a one-time sale). *Suppose $\sigma > 0$, $0 < \beta_O < 1$, and $\beta_O R \leq 1$. Define*

$$\gamma \equiv (\beta_O R)^{1/\sigma}. \tag{B10}$$

The optimal paths are

$$c_j(q) = q\left(1 - \frac{\gamma}{R}\right)\gamma^j, \tag{B11}$$

$$a_{j+1}(q) = \frac{q}{R}\gamma^{j+1}, \qquad j \geq 0. \tag{B12}$$

If $\beta_O R < 1$, asset holdings converge to zero at rate $\gamma < 1$. Moreover, the assets attributable to the sale proceeds satisfy

$$a_{j+1}(Ra_0 + x) - a_{j+1}(Ra_0) = \frac{x}{R}\gamma^{j+1}, \qquad j \geq 0, \tag{B13}$$

and therefore converge to zero at the same rate. If $\beta_O R = 1$, the founder holds q/R after date zero and consumes the interest income in every subsequent period.

Proof. The Euler equation gives $c_{j+1}/c_j = \gamma$. Combining this condition with the intertemporal budget $c_0 + \sum_{j=1}^{\infty} c_j/R^j = q$ gives (B11). Substituting consumption into (B9) and iterating forward gives (B12). Evaluating that expression at $Ra_0 + x$ and at Ra_0, and then taking the difference, gives (B13). □

The resources q determine the size of the initial saving response, while γ determines how quickly that response decays. Under the assumed inequality $\beta_O R < 1$, the liquidated firm owner does not maintain the additional assets generated by the one-time proceeds indefinitely.

This example isolates the saving response to one-time IPO proceeds. Date zero is the individual founder's sale date, not a common incorporation date for all firms. The calculation supplies the drawdown intuition; the convergence result for the full nonlinear model instead uses the uniform policy bound imposed below.

B.3 Convergence of asset-market clearing at the asset-creation endpoint

Fix $\kappa > 0$ and consider stationary economies indexed by $\delta_C > 0$. Public-firm exit returns a liquid founder to private ownership, after which the new private business may incorporate at rate κ. These flows generate founder asset demand in every interior economy. We now ask whether that asset demand remains positive as $\delta_C \downarrow 0$.

The stationary flow of aggregate private profit income received by founders is

$$Y_{P,\delta_C} = (1 - \omega_{\delta_C})\Pi = \frac{\delta_C}{\kappa + \delta_C}\Pi. \tag{B14}$$

Here $\omega_{\delta_C} = \kappa/(\kappa + \delta_C)$. By the maintained assumption that incorporation is independent of a firm's share of aggregate profits, this ownership share is also the share of aggregate profit income represented by tradable equity. In a stationary equilibrium, the value of newly sold equity equals the value of tradable equity removed by incorporated business exit. Aggregate sale proceeds in each period are therefore

$$X_{\text{IPO},\delta_C} = \delta_C \omega_{\delta_C} Q(r_{\delta_C})\Pi = \frac{\kappa\delta_C}{(\kappa + \delta_C)(r_{\delta_C} + \delta_C)}\Pi. \tag{B15}$$

Both Y_{P,δ_C} and X_{IPO,δ_C} are of order δ_C provided that r_{δ_C} is bounded away from zero. The question is whether the founder asset demand generated by these flows also vanishes in the limit.

To translate these flows into the level of the entrepreneur's asset demand, let s collect the founder's current nonfinancial states. Let $y_{\delta_C}(s)$ denote current private-profit income or sale proceeds, excluding returns on financial assets, and let $g_{\delta_C}(a, s)$ denote next-period assets. We impose the following uniform bound on the founder's policy function near the endpoint: there are constants $0 \leq \rho_a < 1$ and $0 \leq b < \infty$, independent of δ_C, such that

$$g_{\delta_C}(a, s) \leq \rho_a a + b\, y_{\delta_C}(s). \tag{B16}$$

This inequality is a restriction on the nonlinear saving policy. Economically, it rules out saving behavior that becomes arbitrarily persistent as δ_C approaches zero[15]. Integrating (B16) across founders and using stationarity gives

$$\begin{aligned} D_{O,\delta_C} &\leq \rho_a D_{O,\delta_C} + b\left[Y_{P,\delta_C} + X_{\text{IPO},\delta_C}\right], \\ D_{O,\delta_C} &\leq \frac{b}{1 - \rho_a}\left[Y_{P,\delta_C} + X_{\text{IPO},\delta_C}\right]. \end{aligned} \tag{B17}$$

The bound is the only stock-flow relationship needed for the endpoint result. It does not assign particular dollars of founder wealth to past private profits or sale proceeds.

[15] We also require the stationary founder distribution to have a finite first moment.

Proposition 3 (Convergence to the asset-creation benchmark). *Fix $\kappa > 0$ and a production-side allocation (W, K, Π) with $\Pi > 0$. For each sufficiently small $\delta_C > 0$, suppose a stationary equilibrium exists with return r_{δ_C} and founder asset demand D_{O,δ_C}. Suppose that (B16) holds with constants independent of δ_C, that the stationary founder distributions have finite first moments, that $r_{\delta_C} \in [\underline{r}, \bar{r}] \subset (0, \infty)$, and that $\bar{D}(r)$ is continuous and strictly positive on this interval. Provided that the equation*

$$\bar{D}(r)W = K + \frac{\Pi}{r} \tag{B18}$$

has a unique solution $r \in [\underline{r}, \bar{r}]$, then, as $\delta_C \downarrow 0$,

$$r_{\delta_C} \longrightarrow r, \qquad \omega_{\delta_C} \longrightarrow 1, \qquad D_{O,\delta_C} = O(\delta_C), \qquad \lambda(r_{\delta_C}; \kappa, \delta_C) = O(\delta_C) \longrightarrow 0, \tag{B19}$$

and

$$Q(r_{\delta_C})\omega_{\delta_C}\Pi \longrightarrow \frac{\Pi}{r}. \tag{B20}$$

Proof. The positive lower bound on r_{δ_C}, together with (B14) and (B15), gives

$$Y_{P,\delta_C} + X_{\mathrm{IPO},\delta_C} = O(\delta_C).$$

Equation (B17) therefore implies $D_{O,\delta_C} = O(\delta_C)$. Because $\bar{D}(r)$ is strictly positive on the compact interval,

$$\lambda(r_{\delta_C}; \kappa, \delta_C) = \frac{D_{O,\delta_C}}{\bar{D}(r_{\delta_C})\Pi} = O(\delta_C).$$

Consider any sequence $\delta_C \downarrow 0$. Compactness gives a subsequence along which $r_{\delta_C} \to \widehat{r}$. Asset-market clearing along this subsequence is

$$\bar{D}(r_{\delta_C})W + D_{O,\delta_C} = K + \frac{\kappa}{(\kappa + \delta_C)(r_{\delta_C} + \delta_C)}\Pi.$$

Taking the limit and using continuity gives

$$\bar{D}(\widehat{r})W = K + \frac{\Pi}{\widehat{r}}.$$

Uniqueness of the solution to (B18) implies $\widehat{r} = r$. Every convergent subsequence has the same limit, so the entire sequence converges to r. The remaining claims follow directly. □

The qualifications in the proposition identify when convergence can fail. A vanishing exit rate need not eliminate founder asset demand if returns approach zero, if saving becomes arbitrarily persistent (e.g., "status-seeking"), or if the limiting asset-market equation has multiple solutions. Subject to these qualifications, profits enter demand as private-owner income at one endpoint and supply as tradable equity at the other. Along this limit, ω and λ move jointly.

C Microfoundations of Linear Asset Demand

The sufficient-statistic representation uses the homogeneity of group asset demand in the relevant income scale. An upward-sloping asset-demand schedule is a separate condition used for equilibrium comparative statics; homogeneity by itself does not establish that condition. This appendix gives three examples that deliver exact linearity and then states the general argument.

C.1 Standard incomplete markets (Aiyagari)

A type-g household has CRRA utility, faces the borrowing constraint $a' \geq 0$, and receives idiosyncratic income $X_g e_t$, where $X_g > 0$ is a group income scale and e_t is a stationary Markov process. This is the standard Bewley–Huggett–Aiyagari income-fluctuation problem (Bewley, 1977; Huggett, 1993; Aiyagari, 1994). At a fixed gross return $R = 1 + r$, assume that the normalized household problem has a unique invariant distribution with a finite first asset moment. The usual impatience condition $\beta_g R < 1$, together with appropriate ergodicity and moment conditions on income, is a standard sufficient set of restrictions.

For $\sigma \neq 1$, scaling assets and consumption by X_g gives

$$V_g(a, e; X_g) = X_g^{1-\sigma} v_g(a/X_g, e), \qquad D_g(r, X_g) = \bar{D}_g(r) X_g. \tag{C1}$$

For log utility, the first equality is replaced by $V_g(a, e; X_g) = v_g(a/X_g, e) + (1 - \beta_g)^{-1} \log X_g$; the policy and demand scaling are unchanged. Homogeneity therefore establishes linearity in X_g. It does not by itself establish that $\bar{D}_g(r)$ is increasing in r. In the canonical incomplete-markets model desired wealth often rises sharply as $\beta_g R$ approaches one from below, but monotonicity and finiteness must be verified under the specified income process and parameterization.

C.2 Perpetual-youth OLG

Consider a Blanchard–Yaari perpetual-youth economy (Blanchard, 1985). A household survives to the next period with probability $v \in (0, 1)$, receives the constant endowment X_g while alive, is born with zero assets, cannot borrow, and leaves no bequest. Deaths are replaced by newborns, so population mass is constant. Expected utility is $\sum_{j \geq 0} (\beta_g v)^j u(c_{g,j})$. Actuarially fair annuities pay the gross return R/v conditional on survival, where $R = 1 + r > 1$. The budget of an age-j household is therefore

$$c_{g,j} + a_{g,j+1} = \frac{R}{v} a_{g,j} + X_g, \qquad a_{g,0} = 0. \tag{C2}$$

With CRRA utility, the Euler equation gives $c_{g,j+1}/c_{g,j} = \gamma_g$, where

$$\gamma_g(r) \equiv \left(\beta_g R\right)^{1/\sigma}. \tag{C3}$$

The budget equations and the no-Ponzi condition then imply

$$a_{g,j} = \frac{v}{R - v}\left(\gamma_g^j - 1\right) X_g. \tag{C4}$$

The stationary share of households of age j is $(1 - v)v^j$. Hence aggregate group asset demand is

$$\begin{aligned} D_g(r, X_g) &= \sum_{j=0}^{\infty} (1 - v) v^j a_{g,j} \\ &= \underbrace{\frac{v^2\left(\gamma_g(r) - 1\right)}{(1 + r - v)\left(1 - v\gamma_g(r)\right)}}_{\bar{D}_g(r)} X_g. \end{aligned} \tag{C5}$$

The admissible domain $1 < \gamma_g < 1/v$ makes lifecycle asset holdings nonnegative and the cross-sectional first moment finite. On this domain, $\bar{D}_g$ is increasing in γ_g. Thus, holding survival probabilities and other parameters fixed, $\beta_O > \beta_W$ implies $\lambda > 1$. Whether $\bar{D}_g(r)$ is increasing in the return is a separate comparative-static restriction because r affects both consumption growth and the capitalization term in (C5).

C.3 Wealth in utility

As a static representation of preferences for wealth, let $0 < \theta_g < 1$ and $\rho(r) > 0$, and consider

$$\max_{c,a \geq 0} \frac{\left(c^{1-\theta_g} a^{\theta_g}\right)^{1-\sigma}}{1 - \sigma} \quad \text{subject to} \quad c + \rho(r)a = X_g. \tag{C6}$$

For $\sigma = 1$, use the logarithm of the Cobb–Douglas composite. Because the outer CRRA transformation is increasing, the expenditure shares satisfy $c = (1 - \theta_g)X_g$ and $\rho(r)a = \theta_g X_g$. Thus

$$D_g(r, X_g) = \frac{\theta_g}{\rho(r)} X_g. \tag{C7}$$

Homotheticity gives exact linearity. Demand is increasing in r if and only if $\rho'(r) < 0$. If groups face the same $\rho(r)$, then $\lambda = \theta_O/\theta_W > 1$ requires $\theta_O > \theta_W$, not heterogeneity alone.

C.4 A general homogeneity principle

The same argument applies to a broader class of household problems. Let $X_{g,t} > 0$ denote the group income scale and let normalized wealth be $b_t = a_t/X_{g,t}$. Suppose that: (i) period utility is CRRA; (ii) all nonasset income, transfers, borrowing limits, and other constraints measured in goods scale with $X_{g,t}$; (iii) feasible portfolios and the conditional return distribution depend on $(a_t, X_{g,t})$ only through b_t and other normalized states; and (iv) $X_{g,t+1} = \Gamma_{t+1} X_{g,t}$, where the

law of Γ_{t+1} does not depend on the level of $X_{g,t}$. Finally, assume that the normalized Markov decision problem is well defined and has an invariant distribution with the moments needed to aggregate asset holdings. With constant X_g this is an ordinary stationary distribution; with stochastic growth it is a stationary distribution in normalized variables, so the level allocation follows a balanced-growth path.

For $\sigma \neq 1$, the value function and policy functions take the form

$$V_g(a, X_g, s) = X_g^{1-\sigma} v_g(a/X_g, s), \qquad a' = X_g g_g(a/X_g, s), \tag{C8}$$

where s collects the remaining normalized states. To see the scaling, write $b' = a'/X_g$. The normalized continuation value is

$$\frac{\beta_g \mathbb{E}[V_g(a', X_g', s')]}{X_g^{1-\sigma}} = \beta_g \mathbb{E}\left[(\Gamma')^{1-\sigma} v_g\left(\frac{b'}{\Gamma'}, s'\right)\right], \tag{C9}$$

which contains no level X_g. The required integrability conditions include the relevant moments of Γ' and of normalized wealth; the single moment $\mathbb{E}[(\Gamma')^{1-\sigma}] < \infty$ need not be sufficient on its own. For log utility, the multiplicative value-function relation becomes $V_g = v_g + (1-\beta_g)^{-1} \log X_g$, while policy homogeneity is unchanged.

With several assets, the same argument gives $a_j' = X_g g_{g,j}(a/X_g, s)$ for every portfolio position. Integrating these policies over the normalized invariant distribution gives asset-specific demands

$$D_{g,j}(r, X_g) = \bar{D}_{g,j}(r) X_g. \tag{C10}$$

Borrowing need not be prohibited; any borrowing limit must simply scale with X_g. The result accommodates uninsurable idiosyncratic investment risk (Angeletos, 2007), portfolio choice over multiple assets, and return heterogeneity across groups or normalized states, as motivated empirically by Fagereng et al. (2020). It does not accommodate fixed dollar participation costs, fixed nominal debt limits, or lump-sum taxes that remain unchanged when X_g changes. In those cases a local approximation generally introduces an intercept and additional derivatives; it does not preserve the sufficient statistic automatically. In the one-asset model, exact homogeneity is what permits the asset block to be summarized by $\bar{D}(r)$ and the relative demand coefficient $\lambda(r)$.

D Measurement Appendix

D.1 Measuring the path of market power

This section describes the estimation of the aggregate markup path.

The Census matched sample. The series is estimated by Ren and Zhang (2025) on publicly traded firms linked to the Longitudinal Business Database (LBD) of the U.S. Census Bureau. The LBD is a census of private nonfarm establishments in the United States. It is built from the Business Register, which the Census Bureau maintains from its own surveys and from administrative records such as those of the IRS, and it reports annual payroll and employment for each establishment together with an identifier of the firm that owns it (Chow et al., 2021). Establishments are aggregated to the firm, and firms are linked to the CRSP/Compustat merged file through a crosswalk developed by Lawrence Schmidt. In the resulting firm year panel, the income statement of a public firm sits beside the administrative record of its U.S. payroll and employment, where the compustat data has limited information on: its coverage of labor expenses is highly incomplete, the reason why the markup literature has folded labor into a composite variable input.

The panel covers 1977 to 2019. It excludes utilities, finance, real estate, public administration, and unclassified firms (NAICS 22, 52, 53, 92, and 99), drops firms without a four digit industry code, and requires positive sales, COGS, SG&A, total assets, and capital. Observations are trimmed by year at the 1st and 99th percentiles of the ratios of COGS, SG&A, and materials to sales and at the 99th percentile of R&D to sales. About 69,500 firm year observations remain: roughly 1,400 firms in 1980, about 2,000 during the 1990s, and 1,000 in 2018.[16] The matched firms employ between 11 and 16 million workers in the United States in every year of the sample. The series therefore describes market power among publicly traded firms, measured on firm years for which U.S. payroll is observed. This second property is what allows the measured wedge to be taken apart in Section D.2.

Output, the flexible input, and capital. Output is sales and the flexible input is cost of goods sold, both deflated with the GDP deflator. COGS is the conventional choice since De Loecker, Eeckhout and Unger (2020): it is the one broadly reported expense that varies with production.

Capital plays two roles in the estimation, and both shape how it is built. It is an argument of the production function, so the elasticity of the flexible input cannot be estimated apart from it. And it provides the timing restriction that identifies that elasticity: capital in place at the start of a year was chosen before the year's productivity innovation, whereas the flexible input responds to it. The estimation therefore needs a stock that is measured consistently over the life of a firm and dated before production takes place. Ren and Zhang (2025) construct it by accumulating investment on a book value benchmark. At a firm's first observation the stock equals gross property, plant, and equipment (Compustat `PPEGT`, bounded below by zero). In each later year it grows by net investment, measured as the change in net property, plant, and equipment (`PPENT`). The stock is deflated with the price index for nonresidential fixed investment (NIPA Table 1.1.9, line 9), because the price of capital goods has not followed the GDP deflator. Balance sheets record capital at the

[16] Counts are rounded under Census disclosure rules. The COGS based measure used in this paper and the labor separated estimates of Section D.2 are computed on the same firm years.

end of the fiscal year, so the capital input for year t is the stock at the end of year $t-1$.

Production function and estimation. Within each two digit NAICS industry, log output is

$$y_{it} = \sum_a \beta_a x_{it}^a + \sum_a \beta_{aa} \left(x_{it}^a\right)^2 + \sum_{a<b} \beta_{ab}\, x_{it}^a x_{it}^b + \omega_{it} + \varepsilon_{it}, \tag{D1}$$

where x_{it} collects the logs of the flexible input v_{it} and of capital k_{it}, ω_{it} is Hicks neutral productivity, and ε_{it} captures measurement error and shocks realized after inputs are chosen. The translog is a second order approximation to an arbitrary technology. Its coefficients are constant within an industry over the entire sample, but the output elasticities they imply are specific to the firm and the year,

$$\theta_{it}^v = \beta_v + 2\beta_{vv} v_{it} + \sum_{b \neq v} \beta_{vb}\, x_{it}^b, \tag{D2}$$

so the measured wedge reflects each firm's input mix as well as its accounting ratio of sales to COGS.

Estimation follows the two stage procedure of De Loecker and Warzynski (2012), which builds on Olley and Pakes (1996), Levinsohn and Petrin (2003), and Ackerberg, Caves and Frazer (2015). The first stage regresses log sales on a second order polynomial in the inputs and year effects. Its fitted value $\widehat{\phi}_{it}$ is output net of ε_{it}, and its residual $\widehat{\varepsilon}_{it}$ is retained. In the second stage, a candidate coefficient vector β implies productivity $\omega_{it}(\beta) = \widehat{\phi}_{it} - f(x_{it}; \beta)$, where f is the deterministic part of (D1). Productivity follows a first order autoregression with innovation $\xi_{it}(\beta)$. Under the timing assumption the innovation is orthogonal to capital in place and to last year's flexible input, which gives the moment conditions

$$\mathbb{E}\left[\xi_{it}(\beta)\, z_{it}\right] = 0, \tag{D3}$$

with z_{it} containing current capital, the lagged flexible input, and their squares and interactions.

The elasticity of the flexible input is the coefficient on which the wedge depends, and it is the one these moments identify least well. Without variation in the relative price of the flexible input, the proxy variable approach leaves it underdetermined (Gandhi, Navarro and Rivers, 2020). Following Flynn, Traina and Gandhi (2019), the estimation adds a restriction on returns to scale: constant returns enters as the additional moment $\mathbb{E}[\sum_a \theta_{it}^a(\beta) - 1] = 0$. This is the restriction our model places on the technology F in Section 2.1, so the estimated elasticities and the model's factor shares rest on a common assumption. The size of the rise is not sensitive to it. Without the restriction the index rises 10.3 percent between 1980 and 2018, against 9.6 percent with it.[17]

The firm level wedge. A cost minimizing firm hires a flexible input until its price equals its marginal product valued at marginal cost. When the firm takes the input price as given, this

[17] The moments are weighted equally. In each industry the objective is minimized from 1,000 random starting values, with first order coefficients restricted to $[0, 1]$ and second order coefficients to $[-0.05, 0.05]$, and the estimates are interior.

condition says that the ratio of price to marginal cost equals the output elasticity of the input divided by its share of revenue (Hall, 1988; De Loecker and Warzynski, 2012). Applied to COGS, the firm level measure in (16) is

$$\widehat{\mu}_{it}^{\text{meas}} = \frac{\widehat{\theta}_{it}^{v}}{\widehat{\alpha}_{it}^{v}}, \qquad \widehat{\alpha}_{it}^{v} = \frac{\text{COGS}_{it}}{\text{Sales}_{it} / \exp(\widehat{\varepsilon}_{it})}. \tag{D4}$$

Dividing sales by $\exp(\widehat{\varepsilon}_{it})$ removes the variation in revenue that is unrelated to input choices, as in De Loecker and Warzynski (2012). Because a ratio magnifies errors in either term, the elasticities and the revenue shares are winsorized at the 5th and 95th percentiles before the ratio is formed. The requirement that the firm be a price taker in the market for the flexible input is what licenses reading the ratio as a product markup. Section D.2 examines it.

Aggregation within and across industries. The firm level wedges are aggregated as the theory prescribes. Within each two digit industry j,

$$\widehat{\mu}_{jt}^{\text{meas}} = \left(\sum_{i \in j} s_{it}^{j} \left(\widehat{\mu}_{it}^{\text{meas}} \right)^{-1} \right)^{-1}, \tag{D5}$$

where s_{it}^{j} is firm i's share of industry sales. This is (15) applied to the industry.

Across industries, levels cannot be combined in the same way, for a reason of identification. The data record revenue, and without firm level prices the estimated output elasticity carries a multiplicative bias of unknown size (Bond et al., 2021; De Ridder, Grassi and Morzenti, 2025). The bias is common to the firms that share a production function, here an industry. It therefore leaves the growth of each industry series informative and its level uninformative. An aggregate of levels would treat the industry constants as if they were data: a shift of sales toward industries with a high estimated level would register as rising market power whether or not the true wedge in those industries is high. The index removes the constants before aggregating. Each industry series is divided by its 1977 value, and the normalized series are averaged with industry sales weights w_{jt}:

$$I_t = \sum_{j} w_{jt} \frac{\widehat{\mu}_{jt}^{\text{meas}}}{\widehat{\mu}_{j,1977}^{\text{meas}}}. \tag{D6}$$

I_t is a sales weighted average of the growth of the wedge within industries.

Panel A of Table D1 shows what each choice does. The harmonic mean across all firms, taken in levels, rises 16.7 percent between 1980 and 2018, against 9.6 percent for the index. The difference of 7 points reflects sales moving toward industries with higher estimated levels, the component that cannot be told apart from the industry constants. The arithmetic sales weighted counterpart of the index, the statistic most often reported in the literature, rises 15.4 percent. It exceeds the harmonic index because reallocation toward high markup firms raises an arithmetic mean by more than a

Table D1: The measured wedge under alternative constructions

	Index, 1980 = 1				Share of rise
	1990	2000	2010	2018	by 1990 (%)
Panel A. Census matched sample					
Harmonic index within industries (baseline)	1.027	1.059	1.096	1.096	28
without the returns to scale restriction	1.042	1.084	1.104	1.103	41
with arithmetic sales weights	1.044	1.091	1.162	1.154	29
Harmonic mean across all firms, in levels	1.049	1.147	1.175	1.167	29
Aggregate sales over COGS	1.079	1.120	1.161	1.180	44
Labor separated price markup	1.006	1.025	1.033	1.022	
Labor separated wage markdown	1.131	1.393	1.590	1.760	
Panel B. Full Compustat sample					
Translog	1.069	1.103	1.107	1.100	69
Cobb–Douglas	1.081	1.094	1.103	1.105	77
Rolling Cobb–Douglas elasticities	1.095	1.130	1.098	1.112	85
Aggregate sales over COGS	1.101	1.126	1.134	1.128	79

Notes: Each series is divided by its 1980 value. The last column is the share of the increase between 1980 and 2018 that has occurred by 1990. Panel A uses the Census matched sample of Ren and Zhang (2025). The first three rows are indexes built as in (D6); the fourth is the harmonic mean (15) of firm level estimates across all firms without the industry normalization; the fifth requires no estimation. The labor separated price markup is aggregated as in the first row, and the wage markdown is weighted by the wage bill. Panel B uses Compustat firms reporting sales, COGS, SG&A, and capital, excluding finance and real estate, with COGS as the flexible input and gross property, plant, and equipment as capital. The translog and Cobb–Douglas rows estimate time invariant production functions by two digit industry; the third row applies the rolling window elasticities of De Loecker, Eeckhout and Unger (2020). Firm level estimates in these three rows are trimmed at the 1st and 99th percentiles by year and aggregated with (15) across all firms.

harmonic one (Edmond, Midrigan and Xu, 2023). Factor shares depend on inverse markups, so the harmonic index is the one the model requires.

From the index to the model path. The same identification problem governs how the series enters the model. The estimated level is uninformative even in the aggregate: the harmonic mean across all firms is 0.82 in 1980, and a wedge below one would have large public firms selling below marginal cost. The growth of the series is informative, and growth is all the model takes from it. We rebase the index to 1980 and attach it to a level taken from the national accounts, as in (17). By (5) the level of the wedge is pinned down by the profit share, $\mu_{1980} = 1/(1 - s_{\Pi,1980})$. The 1980 profit share of 4 percent in Table 2 gives $\mu_{1980} = 1.0417$ and, with $\alpha = 1/3$, a labor share of 64 percent. Firm level data thus discipline the proportional change in the wedge, and national accounts discipline the level to which that change applies. The model path rises from 1.042 in 1980 to 1.142 in 2018, which lowers the labor share to 58.4 percent and raises the profit share to 12.4 percent.

D.2 What the measured wedge captures

The ratio estimator recovers a product markup from an input only if the firm is a price taker in the market for that input. Cost of goods sold fails the condition on accounting grounds: it includes the wages of production workers, and a firm with labor market power pays a wage below the marginal revenue product of labor. Let $\nu_{it} \geq 1$ denote this wage markdown, the ratio of the marginal revenue product of labor to the wage. Ren and Zhang (2025) show that the ratio estimator applied to labor returns the product of the two wedges, $\theta^{\ell}_{it}/\alpha^{\ell}_{it} = \mu_{it}\nu_{it}$, while applied to a competitively supplied input such as materials it returns the markup alone, $\theta^{m}_{it}/\alpha^{m}_{it} = \mu_{it}$. COGS bundles the two. For a proportional change in the bundle, its output elasticity is $\theta^{m}_{it} + \theta^{\ell}_{it}$ and its revenue share is $\alpha^{m}_{it} + \alpha^{\ell}_{it}$, so the conventional measure is

$$\widehat{\mu}^{\text{meas}}_{it} = \frac{\theta^{m}_{it} + \theta^{\ell}_{it}}{\alpha^{m}_{it} + \alpha^{\ell}_{it}} = \mu_{it}\left[1 + (\nu_{it} - 1)\,\frac{\alpha^{\ell}_{it}}{\alpha^{m}_{it} + \alpha^{\ell}_{it}}\right]. \tag{D7}$$

The measured wedge is the product markup times a weighted average of one and the wage markdown, with weight equal to labor's share of the bundle. When the flexible input is labor alone it equals $\mu_{it}\nu_{it}$, the case in (20). A rise in the conventional measure therefore establishes that total market power has risen. It is silent on the market in which it rose.

The matched data can separate the two components because payroll is observed. On the same firm years, Ren and Zhang (2025) estimate a production function in which labor, taken from the LBD, and materials, measured as COGS plus SG&A net of payroll, rents, and the nonlabor part of intangible investment, enter as separate inputs alongside physical and intangible capital. Materials are the price taking flexible input and identify the product markup, $\mu_{it} = \theta^{m}_{it}/\alpha^{m}_{it}$. The wage markdown follows as $\nu_{it} = (\theta^{\ell}_{it}/\alpha^{\ell}_{it})/\mu_{it}$. The two series part ways, as Figure 9 and the last rows of Panel A show. Between 1980 and 2018 the price markup index rises 2 percent and the wage markdown index rises 76.0 percent.

The reason is visible in the cost shares, before any estimation. From 1977 to 2019 payroll falls from 21 to 10 percent of sales in the matched sample, while the estimated output elasticity of labor declines only from about 0.25 to 0.22. The ratio of the two roughly doubles, and with it the markdown. The materials share and the materials elasticity instead rise together, from 0.66 to 0.70 and from 0.65 to 0.69, which leaves the price markup flat. Over the same years COGS falls from 73 to 62 percent of sales. The eleven points of sales that leave COGS are matched by eleven points that leave payroll. In this sample the conventional measure rises because labor is paid a declining share of what firms sell, a fact read directly from the accounts that does not depend on the production function.

Two implications follow for this paper. The first concerns the reading of the baseline. Section 4 attributes the measured wedge to the product markup, under which labor and capital payments fall in proportion as the wedge rises. The labor separated estimates point to the wage markdown,

Table D2: Implied and observed changes in the labor share

	Change since 1980 (points)				RMS gap to observed		
	1990	2000	2010	2018	Business	Corporate	Aggregate
Panel A. Implied by the measured wedge							
Census matched index (baseline)	−1.7	−3.5	−5.6	−5.6	1.6	2.0	1.0
Compustat, translog	−4.1	−6.0	−6.2	−5.8	3.1	3.4	2.3
Compustat, Cobb–Douglas	−4.8	−5.5	−6.0	−6.1	2.6	2.8	1.9
Compustat, rolling elasticities	−5.5	−7.4	−5.7	−6.4	3.4	3.5	2.9
Compustat, sales over COGS	−5.9	−7.2	−7.6	−7.3	3.5	3.6	3.1
Panel B. Observed							
Nonfarm business sector	−0.8	−0.3	−7.2	−7.0			
Corporate sector	−0.3	−0.2	−7.7	−7.3			
Aggregate	−1.4	−2.9	−7.2				

Notes: Changes are in percentage points of the labor share relative to 1980. The implied labor share is $(1-\alpha)/\mu_t$ with $\alpha = 1/3$ and μ_t from (17) for each series in Table D1; it equals 64 percent in 1980. The nonfarm business series is the BLS labor share used by Elsby, Hobijn and Şahin (2013), the corporate series is compensation over gross value added of the corporate sector as in Karabarbounis and Neiman (2014), both updated through 2018, and the aggregate series is the EU KLEMS labor share used by Autor et al. (2020), which ends in 2010. The last three columns report the root mean squared difference between the implied and the observed change across the years 1980 to 2018, and 1980 to 2010 for the aggregate series.

under which the capital share is unchanged and the whole increase in profits comes out of labor income, as in (21). Section 6.2 runs the model under this second reading. Both readings deliver the shift on which the asset market channel operates, from labor income to the income of those who own the rents. For this reason we refer to $\widehat{\mu}^{\text{meas}}$ as a measure of total market power.

The second concerns factor shares. A wedge that mirrors a falling payroll share at the firm level should agree with the labor share in the national accounts, and that agreement is where the Census matched path differs most from conventional Compustat estimates. Panel B of Table D1 reports four such estimates. All use the sample definition of De Loecker, Eeckhout and Unger (2020), COGS as the flexible input, and the harmonic aggregate (15) across all firms. They differ in the output elasticity: a translog and a Cobb–Douglas production function with time invariant coefficients by two digit industry, the rolling Cobb–Douglas elasticities of De Loecker, Eeckhout and Unger (2020), and an elasticity of one, which is the ratio of aggregate sales to aggregate COGS. By 2018 the four series stand 10 to 13 percent above 1980, close to the 9.6 percent of the baseline. The disagreement is about timing. Each Compustat series completes at least two thirds of its rise by 1990. The Census matched index completes 28 percent.

The national accounts favor the later timing. Table D2 passes each path through (5) at the baseline calibration and compares the implied labor share with three published series. In the data the labor share is at most 1.4 points below its 1980 value in 1990 and about 7 points below it after 2010. The baseline path lowers the labor share by 1.7 points by 1990 and by 5.6 points by 2018. The Compustat paths lower it by 4 to 6 points by 1990, a decade in which the measured labor share barely moved. Over 1980 to 2018 the root mean squared gap between the implied and the observed

Table D3: Initial private and incorporated profit shares

Calibration	Private share $(1-\omega^{\text{inc}})$	Publicly incorporated share ω^{inc}
Initial model target	0.60	0.40

Notes: Entries are the initial model allocation of aggregate profit income. They are calibration targets. The owner retained share is $\theta_F = 0.15$; the publicly traded fraction of incorporated profit income is $1-\theta_F$. The retained and publicly traded shares split incorporated profit income; they do not change the private/publicly incorporated partition. The publicly traded equity share of aggregate profits is $\omega = (1-\theta_F)\omega^{\text{inc}} = 0.34$ in the baseline, while $\omega^{\text{inc}} = 0.40$.

change is 1.6 points for the baseline against the nonfarm business labor share and 2.6 to 3.5 points for the Compustat paths, and the ranking is the same against the other two series.

Panel A of Table D1 traces the later timing to two features of the baseline. The cost ratio of the matched sample keeps rising after 1990: sales over COGS stands 7.9 percent above 1980 in 1990 and 18.0 percent above it in 2018, whereas in the full Compustat sample it has already risen 10.1 percent by 1990 and ends at 12.8 percent. And the index within industries removes the part of the early rise that comes from sales moving across industries.

The long run results of Section 4.3 depend on the 2018 level of the wedge, on which all series roughly agree. The path between 1980 and 2018 depends on timing, and for the path we use the series whose timing is consistent with the factor shares through which market power enters the model.

D.3 The private, publicly incorporated, and publicly traded profit shares

TODO: Update this content due to new calibration

D.4 The relative wealth-to-income ratio of owners to workers in the SCF λ

TODO: Update this content due to new calibration Table D4 reports the moments behind $\lambda(r_0) = 3.31$. The 1989 asset ratio $D_O/D_W = 0.252$ proxies the 1980 stationary ratio and is scaled by the calibrated income ratio $s_W/s_\Pi = 7.54$. The average across the 11 waves from 1989 to 2019 is 0.256, and every wave lies between 0.20 and 0.31, a relative wealth-to-income ratio of owners to workers between 3.3 and 5.0 with 1989 near the center of the range. These period comparisons describe asset holdings; the baseline relative wealth-to-income ratio uses the 1989 proxy throughout. The active income robustness measure scales each group's assets by its reported wages and business income. Including private business equity and averaging across all waves instead gives $\lambda = 7.52$. We evaluate that broader asset concept in a separate calibration.

Table D4: SCF moments for the relative wealth-to-income ratio of owners to workers

1983 SCF moment	Value
Owner / worker nonbusiness assets	0.3617
Owner / worker total household income	0.1506

Notes: The calibration sample is the 1983 SCF with reference-person ages 25 to 64. Owners are the exact survey-weighted top 25 percent of active business owners, ranked by total household net worth within active owners, with proportional allocation at boundary ties. Active ownership uses the historical management-business indicator. Empirical workers report zero business/farm/professional income and receive no selected-owner weight; losses, small-loss codes, and missing business income are excluded. Assets are signed net worth less all private-business equity, including employer accounts. The income denominator is total household income, with the existing survey-year price adjustment. Each group ratio divides its survey-weighted asset total by its income total. The ratio of these group ratios supplies λ_0; model owner mass equals the selected owners' population share, and model workers are its complement. The empirical samples are not complementary. The 1983 estimate anchors the model's 1980 stationary economy.

E Quantitative Appendix

E.1 Computation

We solve the nonlinear economy in stationary equilibrium and along an annual perfect-foresight transition. We calibrate the group discount factors, productivity normalization, and baseline wedges to match the targets in Table 2; the equilibrium then determines the return, household policies, invariant distributions, and asset demand. Along the transition, the return clears the asset market each year, with market-clearing errors below 9×10^{-9}. The sufficient-statistic calculation uses the same sales-harmonic markup series and baseline calibration as the nonlinear transition.

E.2 Representative-household welfare benchmark

Consider a simple welfare benchmark implied by a representative household with GHH preferences, $U(C, L) = u(C - v(L))$. Proposition **??** connects the output response to the distorted user cost $\tilde{r}^k$. We use this result to connect changes in output Y to representative household welfare, accounting for replacement investment and labor disutility.

In an initial steady state with aggregate production wedge μ^0, the household consumes C^0 and supplies labor L^0. For a change to μ^1, with allocation (C^1, L^1), we compute the consumption equivalent variation (CEV) by solving

$$U\Big((1 + \text{CEV}^{\text{RA}})C^0, L^0\Big) = U(C^1, L^1).$$

A positive CEV denotes a welfare gain: it is the proportional increase in baseline consumption that delivers the counterfactual utility. With investment share $s_I = I/Y$ and resource constraint

$C = Y - I$, GHH preferences give

$$1 + \text{CEV}^{\text{RA}} = \frac{C^1}{C^0} - \frac{v(L^1) - v(L^0)}{C^0} = \frac{Y^1(1 - s_I^1)}{Y^0(1 - s_I^0)} - \frac{v(L^1) - v(L^0)}{Y^0(1 - s_I^0)}.$$

The same distorted user cost introduced in Section 2,

$$\tilde{r}^k = \mu(r + \delta_K) = AF_K(K, L), \tag{E1}$$

governs the local output response. In a steady state, replacement investment gives $s_I = \delta_K K/Y$; the other income shares retain their earlier definitions.

For $r > 0$, Proposition **??** gives

$$\frac{d \log \tilde{r}^k}{d \log \mu} = 1 + \frac{r}{r + \delta_K} \frac{d \log r}{d \log \mu}, \tag{E2}$$

and

$$\frac{d \log Y}{d \log \mu} = -\varphi - (\sigma_{KL} + \varphi) \frac{s_K}{s_W} \frac{d \log \tilde{r}^k}{d \log \mu}. \tag{E3}$$

The following proposition connects this output response to consumption C and representative household welfare.

Proposition 4 (Local consumption and representative household welfare responses)**.** *Under the production and GHH labor supply assumptions of Proposition* **??**, *consider a local change in the aggregate production wedge μ, holding effective productivity A, labor supply parameters, and depreciation δ_K fixed. The consumption response is*

$$\frac{d \log C}{d \log \mu} = \frac{d \log Y}{d \log \mu} + \sigma_{KL} \frac{s_I}{1 - s_I} \frac{d \log \tilde{r}^k}{d \log \mu}. \tag{E4}$$

Evaluating these allocations with representative household GHH preferences gives

$$\begin{aligned} \left. \frac{d \, \text{CEV}^{\text{RA}}(\mu^1; \mu^0)}{d \log \mu^1} \right|_{\mu^1 = \mu^0 = \mu} &= \frac{d \log C}{d \log \mu} - \frac{s_W}{1 - s_I} \frac{d \log L}{d \log \mu} \\ &= \frac{d \log C}{d \log \mu} + \frac{\varphi s_W}{1 - s_I} \left(1 + \frac{s_K}{s_W} \frac{d \log \tilde{r}^k}{d \log \mu} \right). \end{aligned} \tag{E5}$$

The consumption response accounts for the change in replacement investment, while the CEV also values the change in labor. In particular, a decline in labor reduces disutility and offsets part of the consumption loss. These adjustments extend the output result without requiring another asset market calculation: the equilibrium interest rate response $d \log r / d \log \mu$ remains the link to the household block.

In the inelastic labor limit, $\varphi \to 0$, the output, consumption, and CEV responses become exactly proportional to the elasticity of the distorted user cost $\tilde{r}^k$:

$$\frac{d \log Y}{d \log \mu} = -\sigma_{KL} \frac{s_K}{s_W} \frac{d \log \tilde{r}^k}{d \log \mu},$$
$$\left. \frac{d \, \mathrm{CEV}^{\mathrm{RA}}(\mu^1; \mu^0)}{d \log \mu^1} \right|_{\mu^1 = \mu^0 = \mu} = \frac{d \log C}{d \log \mu} = -\sigma_{KL} \left(\frac{s_K}{s_W} - \frac{s_I}{1 - s_I} \right) \frac{d \log \tilde{r}^k}{d \log \mu}.$$

To illustrate the role of the asset market channel, compare a local increase in the aggregate production wedge μ in the representative household economy (RA) and the incorporation economy (HA). Both economies share the production and labor supply assumptions above and begin at a common reference allocation; their output responses differ through the equilibrium interest rate response in the HA economy.

With inelastic labor, the ratio of their output elasticities is

$$\left. \frac{d \log Y^{\mathrm{HA}} / d \log \mu}{d \log Y^{\mathrm{RA}} / d \log \mu} \right|_{\varphi = 0} = 1 + \frac{r}{r + \delta_K} \frac{d \log r^{\mathrm{HA}}}{d \log \mu} \approx \underbrace{1 + \frac{0.06}{0.06 + 0.0685} (-2.15) \simeq 0}_{\text{1980 calibration}}.$$

Here we use the reference values $\alpha = 1/3$, $r = 0.06$, $\delta_K = 0.0685$, $s_K = 0.32$, and $s_W = 0.64$. The two-group sufficient statistic gives the local interest rate elasticity $d \log r^{\mathrm{HA}} / d \log \mu = -2.15$. Holding this interest rate response fixed in the illustration, the decline in r almost exactly offsets the local output loss when labor supply is inelastic. With elastic labor supply, the aggregate production wedge μ also reduces the wage w directly. At $\varphi = 0.5$, the same local interest rate response mitigates the output loss:

$$\left. \frac{d \log Y^{\mathrm{HA}} / d \log \mu}{d \log Y^{\mathrm{RA}} / d \log \mu} \right|_{\varphi = 0.5} \approx \underbrace{0.40}_{\text{1980 calibration}}.$$

As a simple heuristic, consider how a representative household would value the welfare effects of this change in markups when the interest rate r is determined by the asset market channel. We compare representative household CEV under RA allocations with the same welfare evaluation under HA aggregate allocations. Denote these by RA|RA and RA|HA, respectively. At the same reference allocation with $\varphi = 0.5$, Proposition 4 gives

$$\frac{d \, \mathrm{CEV}^{\mathrm{RA|RA}}}{d \log \mu} = \frac{d \log C^{\mathrm{RA}}}{d \log \mu} - \frac{s_W}{1 - s_I} \frac{d \log L^{\mathrm{RA}}}{d \log \mu} \approx \underbrace{-0.466}_{\text{1980 calibration}},$$
$$\frac{d \, \mathrm{CEV}^{\mathrm{RA|HA}}}{d \log \mu} = \frac{d \log C^{\mathrm{HA}}}{d \log \mu} - \frac{s_W}{1 - s_I} \frac{d \log L^{\mathrm{HA}}}{d \log \mu} \approx \underbrace{-0.114}_{\text{1980 calibration}}.$$

Viewed through aggregate consumption and labor, this local calculation implies a smaller welfare loss with the asset market channel. While RA|HA is not a measure of average welfare in the actual

HA economy, it is a useful benchmark for the household welfare calculations below: changes in aggregate allocations may conceal significant welfare effects across households. These numerical comparisons use the local sufficient statistic and the representative household benchmark at the stated reference allocation. They are not the full transition welfare results; the next subsections evaluate both economies along fully solved markup transitions.

E.3 Welfare calculation and source decomposition

Individual, type, and population CEVs. The preceding representative household calculation only values changes in aggregate consumption and labor. We now turn to welfare in the heterogeneous household economy, where these aggregate changes can conceal large differences across households. We define individual-, type-, and population-level CEVs that account for household heterogeneity and the full transition induced by the markup counterfactual in Section 4.3. We first describe the perfect foresight transition; the same CEV definitions apply to the realized myopic transition in Table 4. The economy begins in its calibrated 1980 stationary equilibrium. At date 0, households unexpectedly learn the future aggregate markup path, while continuing to face idiosyncratic risk. The reference economy retains the initial markup indefinitely. With $t = 0$ corresponding to 1980, $T = 2018 - 1980$, and estimated markups $\widehat{\mu}^{y}$, the paths are

$$\mu_t^0 = \mu^{1980}, \qquad \mu_t^1 = \begin{cases} \mu^{1980} \dfrac{\widehat{\mu}^{1980+t}}{\widehat{\mu}^{1980}}, & t = 0, \ldots, T, \\ \mu^{2018}, & t > T. \end{cases}$$

Here μ^{2018} is the normalized counterfactual markup at the end of the sample. Because the change is unanticipated, households enter date 0 with the assets, productivity states, and ownership statuses inherited from the initial stationary distribution. Households have perfect foresight over the aggregate transition but continue to face idiosyncratic uncertainty. Prices, asset valuations, labor, consumption, and saving can therefore respond from the announcement date onward.

Let $g \in \{W, O\}$ denote workers and business owners, and let x collect a household's initial state. Write the announced path as $\boldsymbol{\mu}^1 = \{\mu_t^1\}_{t=0}^{\infty}$. Expected lifetime utility along the counterfactual is

$$V_g^1(x) = V_g(x; \boldsymbol{\mu}^1) = \mathbb{E}_1 \left[\sum_{t=0}^{\infty} \beta_g^t U_g(c_{g,t}^1, \ell_{g,t}^1) \,\middle|\, x_{g,0} = x \right],$$

where $U_W(c, \ell) = u(c - v(\ell))$ and $U_O(c, \ell) = u(c)$ are the preferences specified in Section 3. Expectations integrate over future idiosyncratic histories. For a permanent proportional change in baseline consumption, define

$$V_g^0(x; \mathrm{CEV}) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta_g^t U_g\Big((1 + \mathrm{CEV}) c_{g,t}^0, \ell_{g,t}^0 \Big) \,\middle|\, x_{g,0} = x \right].$$

Compensation scales actual consumption, holding baseline labor and policies fixed. For workers, it therefore changes c inside the GHH composite, rather than scaling consumption net of labor disutility.

The individual CEV equates these values for each initial state:

$$V_g^0(x; \text{CEV}_g^{\text{ind}}(x)) = V_g^1(x).$$

Its value can differ across initial states, but the compensation for a given household is constant across dates and future histories. To evaluate welfare within a type, integrate over its initial-state distribution Λ_g^0:

$$\overline{V}_g^0(\text{CEV}) = \int V_g^0(x; \text{CEV})\, d\Lambda_g^0(x), \qquad \overline{V}_g^1 = \int V_g^1(x)\, d\Lambda_g^0(x).$$

The type-level CEV applies a common compensation within that type,

$$\overline{V}_g^0(\text{CEV}_g^{\text{type}}) = \overline{V}_g^1.$$

It generally differs from the mean individual CEV because utility is nonlinear. The population-level CEV additionally aggregates across types:

$$N_W \overline{V}_W^0(\text{CEV}^{\text{pop}}) + (1 - N_W)\overline{V}_O^0(\text{CEV}^{\text{pop}}) = N_W \overline{V}_W^1 + (1 - N_W)\overline{V}_O^1.$$

The type- and population-level measures use a utilitarian criterion, with initial-state probabilities and population shares as weights. The population measure also depends on the cardinal normalization of utility across types; it is one social welfare benchmark rather than an average of individual CEVs.

For the myopic calculation in Table 4, each observed annual markup is believed permanent. Only the first year of each subjective forecast is realized until the next observation arrives. The final forecast supplies the constant-markup continuation for lifetime welfare. The compensation definitions above are applied to the resulting realized allocation. Both calculations use the same initial household distribution and the same markup observations.

Lifetime values cover the 39 observed annual allocations from 1980 through 2018, a 120-year convergence tail, and the exact terminal stationary continuation. We invert the baseline value function for every positive-mass initial worker and business owner cell, including private, IPO-period, and ongoing incorporated founder states. Baseline labor and policies are held fixed on the compensation side. In particular, worker compensation scales actual consumption inside the GHH composite rather than scaling consumption net of labor disutility.

For either aggregate allocation path $M \in \{\text{RA}, \text{HA}\}$, the representative-household benchmark

solves

$$\frac{1}{1-\beta}U\Big((1+\text{CEV}^{\text{RA}|M})C^0, L^0\Big) = \sum_{t=0}^{\infty}\beta^t U(C_t^M, L_t^M).$$

The RA|HA measure evaluates aggregate allocations from the incorporation economy with representative-household preferences; it is not the population-level CEV in that economy.

Source decomposition under perfect foresight. Figure E1 shows how changes in household income and asset returns contribute to welfare. For workers, changes in individual labor income $we\ell$ account for the principal loss. Initial asset revaluation partly offsets this loss, while lower subsequent returns reduce the offset. Business owners gain from changes in individual business profit income π, retained individual profit income $\theta_F \Pi z$, equity sale proceeds, and initial asset revaluation, partly offset by lower subsequent returns. These are the household consequences of the income and valuation changes that determine asset demand D and asset supply S in the earlier sections.

The individual CEV combines every change in a household's equilibrium consumption and labor path. To attribute this total welfare effect to its economic sources, we gradually increase the markup change, measure each source's marginal contribution, and add these contributions. We connect the reference and counterfactual markup paths through

$$\log \mu_t^\tau = \log \mu_t^0 + \tau(\log \mu_t^1 - \log \mu_t^0), \qquad \tau \in [0,1]. \tag{E6}$$

Here τ indexes the size of the markup change, while t indexes calendar time. At each of 129 homotopy nodes, we solve the complete nonlinear equilibrium transition and define

$$V_g^0(x; \text{CEV}_g^{\text{ind}}(x;\tau)) = V_g^\tau(x).$$

The individual CEV is zero at $\tau = 0$ and equals the reported counterfactual CEV at $\tau = 1$.

Let $j \in \mathcal{J}_g$ index the income and return sources for household type g: labor or business profit income, equity sale proceeds, initial asset revaluation, and subsequent asset returns, as applicable. Initial asset revaluation is the change in the value of assets held when the counterfactual is announced. Let $\mathcal{R}_{g,j,t}^\tau$ denote the resources from each source. The envelope condition gives

$$\begin{aligned} \frac{dV_g^\tau(x)}{d\tau} &= \sum_{j\in\mathcal{J}_g} G_{g,j}^\tau(x), \\ G_{g,j}^\tau(x) &:= \mathbb{E}_\tau\left[\sum_{t=0}^{\infty}\beta_g^t U_{g,c,t}^\tau \frac{\partial \mathcal{R}_{g,j,t}^\tau}{\partial \tau}\,\middle|\, x_{g,0} = x\right]. \end{aligned} \tag{E7}$$

Here $G_{g,j}^\tau(x)$ is source j's contribution to the marginal change in lifetime utility, and $U_{g,c,t}^\tau$ is marginal utility of actual consumption. Because households choose consumption, labor, and saving optimally,

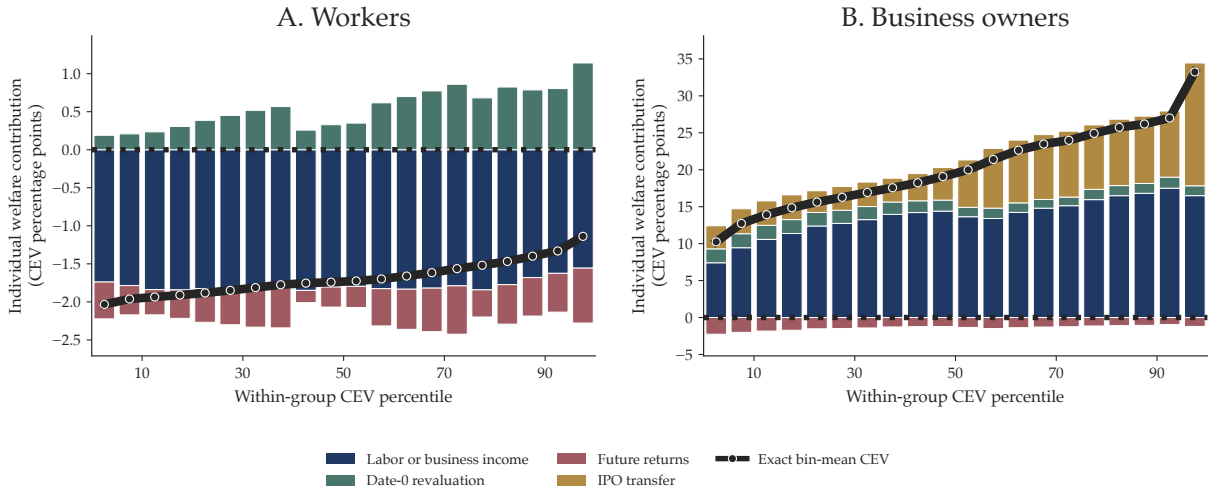


Figure E1: Sources of individual welfare effects under perfect foresight. Within each household type, households are sorted by individual CEV into 20 equal mass bins. Colored bars show income and asset return contributions in CEV percentage points; the black line shows the exact mean CEV in each bin. Finite grid reconciliation is retained separately and omitted from the colored bars.

their marginal welfare changes can be measured from changes in income and asset returns, holding their choices fixed at each intermediate equilibrium. Converting these utility changes into CEV units and integrating gives

$$\mathrm{CEV}^{\mathrm{ind}}_{g,j}(x) := \int_0^1 \frac{G^{\tau}_{g,j}(x)}{\partial_{\mathrm{CEV}} V^0_g(x; \mathrm{CEV}^{\mathrm{ind}}_g(x;\tau))}\, d\tau, \tag{E8}$$

$$\mathrm{CEV}^{\mathrm{ind}}_g(x) = \sum_{j \in \mathcal{J}_g} \mathrm{CEV}^{\mathrm{ind}}_{g,j}(x). \tag{E9}$$

The denominator is the marginal lifetime utility value of a permanent proportional increase in baseline actual consumption. Dividing by it expresses each source's contribution in the same consumption units as CEV. The contributions sum to each household's total CEV. They attribute that welfare change to income and asset returns along the specified sequence of equilibria; they do not measure the effects of changing each source separately.

Figure E1 reports these contributions for households grouped by their total CEV. For business owners, the income component combines private business profit income and retained individual profit income. Taxes are zero in this counterfactual.

E.4 Sufficient statistic implementation

The sufficient-statistic calculation takes the normalized sales-harmonic markup path as given. For each date t, the markup determines the factor shares $(s_{K,t}, s_{W,t}, s_{\Pi,t})$. The calibration supplies

$\lambda_0 = \lambda(r_0)$, ω, θ_F, r_0, δ_K, δ_C, and the fixed-share semi-elasticity of total household asset demand,

$$\epsilon_D = \frac{\partial}{\partial r} \log\{\bar{D}(r)\left[s_{W,0} + \lambda(r) s_{\Pi,0}\right]\}\Big|_{r_0} = \frac{\partial \log D(r)}{\partial r}\Big|_{r_0},$$

where $D(r)$ is the income share weighted average defined in footnote 9. This is the aggregate slope measured in the heterogeneous-agent model. It includes the direct saving responses of both groups and the local movement in $\lambda(r)$. **TODO**: Update this content due to new calibration Let $\widehat{D}(r)$ denote the reduced-form demand coefficient used in the scalar approximation. Define

$$M(r, s_t) = \frac{\frac{1}{r+\delta_K} s_{K,t} + Q(r)\omega s_{\Pi,t}}{s_{W,t} + \lambda_0 s_{\Pi,t}}, \qquad Q(r) = \frac{1}{r+\delta_C}.$$

Normalize $\widehat{D}(r_0) = M(r_0, s_0)$ so that the baseline asset market clears, and close the demand side locally by

$$\widehat{D}(r_t) = \widehat{D}(r_0) \exp\{\epsilon_D (r_t - r_0)\}.$$

The return plotted in the sufficient-statistic figures is the root of

$$\epsilon_D (r_t - r_0) = \log M(r_t, s_t) - \log M(r_0, s_0).$$

This scalar equation evaluates the observed share movements, the capital-depreciation adjustment, and tradable equity valuation at the solved return. Holding λ_0 fixed in M does not discard its local return response: that response is already part of the aggregate slope ϵ_D. The resulting equation is the first-order representation of the exact asset market around the baseline income shares. It holds the other incidence objects and the demand slope fixed, so the statistic isolates the asset-market effect of the markup path before transition distributions and wealth accumulation are added.

E.5 Robustness over entrepreneur definitions

TODO: Update this content due to new calibration The baseline uses active owners in the top five percent ranked by total working-age net worth, but measures their assets as net worth less private-business equity. Two kinds of variants clarify the mapping from the SCF to the model object. Changing the owner cutoff changes both the entrepreneur mass and the measured asset-to-income ratio. Changing which assets or income components are included changes the object held fixed within each group. In particular, using total net worth gives $\lambda = 7.52$, but includes private-business value that the model excludes before incorporation. Each empirical variant is therefore a separate calibration: we re-solve both discount factors and recompute its asset-demand semi-elasticity rather than combining an alternative λ with the baseline slope.

E.6 The asset-demand semi-elasticity in the data

The elasticity that governs how much of the output cost the asset market absorbs is the asset-demand semi-elasticity $\epsilon_D = \partial \log D / \partial r$. In our calibration it is an equilibrium object, recovered by perturbing the return in the solved model, and it equals 16.2963 at the baseline (Table E1). The SCF ratio identifies the level $\lambda(r_0)$, not this slope. Its empirical counterpart is the semi-elasticity of household capital supply, equivalently asset demand, to the return. In a stationary equilibrium the capital stock and household assets move together, so $d \log K / dr$ and $\partial \log D / \partial r$ coincide up to the asset-to-capital ratio.

Moll, Rachel and Restrepo (2022) survey this semi-elasticity in their study of automation and wealth inequality, where it likewise sets how far the return moves as desired capital shifts. They observe that the available quasi-experimental estimates all lie below 35, far more inelastic than the value near 50 their own model implies. The estimates come from wealth and capital-income tax reforms: Zoutman (2015) and Jakobsen et al. (2020) from Dutch and Danish reforms, Brülhart et al. (2022) from the Swiss wealth tax, and Kleven and Schultz (2014) from Danish income-tax reforms, with implied semi-elasticities $d \log K / dr$ from roughly 1 to 35. Our equilibrium $\epsilon_D = 16.2963$ lies within this range. The mapping from ϵ_D to the mitigation share is transparent through the sufficient statistic (11): a more elastic asset market requires a smaller fall in the return to clear and mitigates less, while a more inelastic one mitigates more. The absorption we report rests on an elasticity within the measured range.

E.7 Asset demand sensitivity across steady states

The sufficient statistic holds the worker and business owner semi elasticities at their baseline values defined in Section 4.2, while their asset demand weights and the aggregate semi elasticity vary. A smaller aggregate semi elasticity requires a larger interest rate adjustment for a given shift in asset demand D or asset supply S.

Table E1 reports the local asset demand slopes at the calibrated steady state and at the long run steady state that holds the 2018 markup fixed. The final column reports the semi elasticity of equity values, $-\partial \log Q(r) / \partial r$, because changes in equity values alter IPO proceeds and thereby enter business owner asset demand.

At the long run steady state associated with the 2018 markup, the aggregate semi elasticity is 13.97, and the decline reflects the valuation of tradable equity rather than the direct saving response. These changes explain why fixed baseline group slopes need not reproduce the full nonlinear response.

E.8 The asset-demand semi-elasticity in the model: saving and valuation

At the long run steady state associated with the 2018 markup, the aggregate semi elasticity is 13.97. The business owner slope moves from 15.49 to 12.24, while the equity value semi elasticity

Table E1: Asset demand and equity value semi elasticities

State	r	All households	Workers	Business owners	Equity value
Calibrated baseline	6.00%	**16.2963**	16.46	15.49	9.09
Long run steady state at 2018 markup	4.98%	13.97	15.22	12.24	10.02

Notes: All households, Workers, and Business owners report local semi elasticities of desired asset holdings with respect to the annual interest rate. Equity value reports $-\partial \log Q(r)/\partial r$. Semi elasticities are expressed per unit change in the annual interest rate. Thus an entry of 16.2963 implies that a 100 basis point increase in the interest rate raises desired assets by approximately 16.2963 percent. The second row evaluates the local slopes in the steady state that holds the 2018 markup fixed.

moves from 9.09 to 10.02. Figure E2 shows the source of this movement. Business owner asset demand responds to the interest rate through two channels. A higher interest rate directly raises desired saving, holding incorporation proceeds fixed. At the same time, it lowers the value of future tradable equity and therefore the IPO proceeds received upon incorporation, partially offsetting the direct saving response. This valuation response becomes stronger at lower interest rates as the effective duration of the tradable equity increases.

At the calibrated baseline, the direct saving response is 17.04 and the valuation response is -1.55, yielding a total business owner semi elasticity of 15.49. At the long run steady state associated with the 2018 markup, the corresponding components are 14.37 and -2.13, for a total of 12.24. **TODO**: Update this content due to new calibration The sufficient statistic uses the baseline aggregate slope, while the nonlinear perfect foresight transition allows these slopes and the wealth distribution to evolve and incorporates anticipation of the future markup path. Despite these additional margins, the 2018 endpoints remain close.

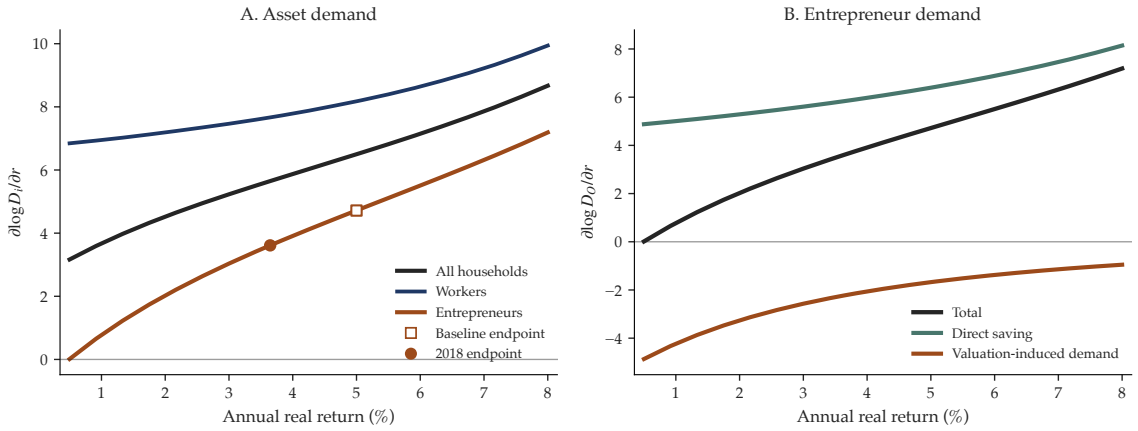


Figure E2: Asset demand semi elasticities and valuation. Panel A shows household semi elasticities in the cash flow environment of the long run equilibrium at the 2018 markup. The square and circle mark the baseline and long run business owner slopes. Panel B separates the business owner saving response, varying the saving return with claim prices fixed, from the valuation response, varying claim prices and incorporation transfers with the saving return fixed.

E.9 Sensitivity to profit incidence

Figure E3 shows how the balance between Redistribution and Asset Creation varies with the two incidence parameters, λ and ω^{inc}. It reports the 2018 sufficient statistic response across $(\lambda_0, \omega^{\text{inc}})$ pairs while holding the worker and business owner semi elasticities fixed at 16.46 and 15.49, respectively. Each pair is normalized to clear the baseline asset market at the common interest rate r_0. The exercise therefore varies baseline profit incidence without re-solving household problems; $\lambda(r)$ and the aggregate asset demand semi elasticity vary endogenously.

A higher λ strengthens Redistribution by increasing the asset demand generated by the shift of income toward business owners, pushing the equilibrium interest rate further down. A higher ω^{inc} strengthens Asset Creation because more profit income is capitalized into tradable equity, expanding asset supply and pushing the equilibrium interest rate up. The baseline lies in the region where Redistribution dominates Asset Creation.

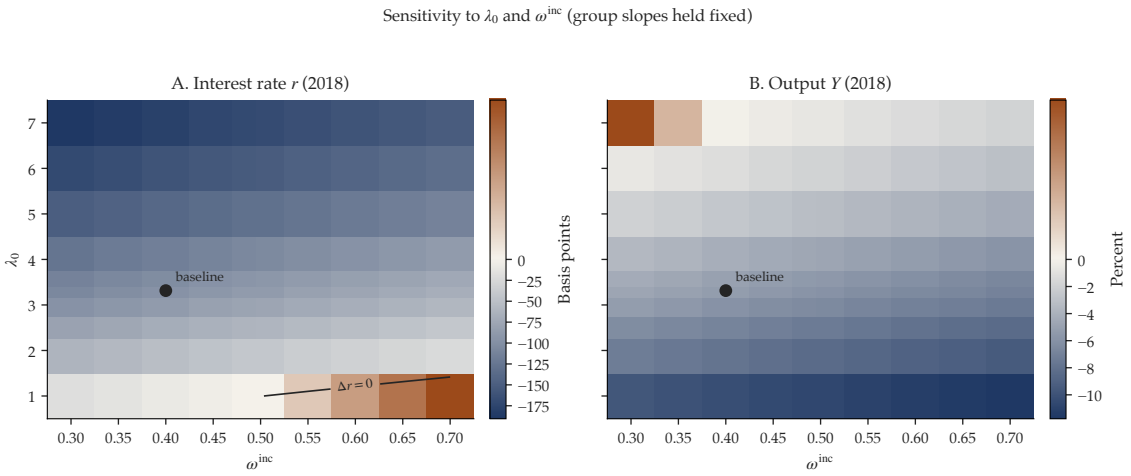


Figure E3: Incidence sensitivity of the 2018 markup counterfactual. Panels A and B show interest rate r and output Y changes from 1980, in basis points and percent, across baseline relative wealth-to-income ratios of owners to workers λ_0 and publicly incorporated profit shares ω^{inc}. Black points mark the baseline calibration. The publicly traded share is $\omega = (1 - \theta_F)\omega^{\text{inc}}$. Worker and business owner semi elasticities are held fixed; $\lambda(r)$ adjusts endogenously.

E.10 Dynamic sensitivity to selected $(\lambda, \omega^{\text{inc}})$ values

Figure E4 reports dated sufficient-statistic paths for 5 selected $(\lambda, \omega^{\text{inc}})$ combinations. **TODO**: Update this content due to new calibration Each path uses the cost-weighted markup series and the same fixed baseline asset-demand semi-elasticity as Figure E3. Panel A plots the return that clears the asset market. Panel B plots output and adds the representative household benchmark generated by the same markup path. The representative household line is common across points because λ and ω^{inc} enter asset-market pricing, while benchmark output depends only on the production-side markup movement. This diagnostic is distinct from the empirically recalibrated robustness exercises described above.

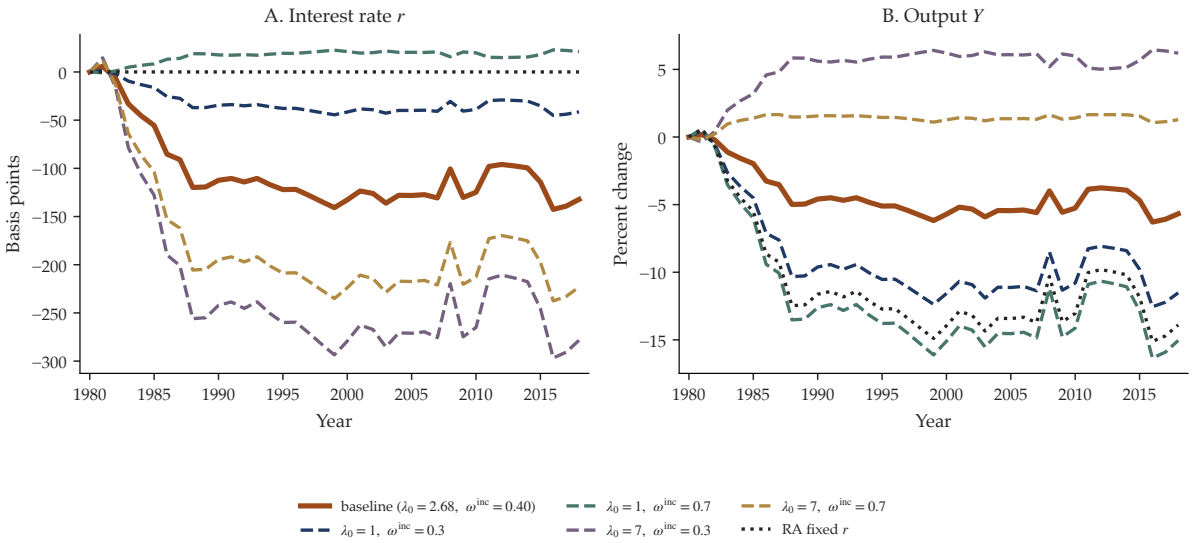


Figure E4: Incidence sensitivity along the markup path. Panels A and B show interest rate r and output Y changes relative to 1980, in basis points and percent, under selected $(\lambda_0, \omega^{\text{inc}})$ pairs and the representative household benchmark. Worker and business owner semi elasticities are held fixed, while $\lambda(r)$ adjusts endogenously. The publicly traded share is $\omega = (1 - \theta_F)\omega^{\text{inc}}$. The baseline uses $\lambda_0 = 3.31$ and $\omega^{\text{inc}} = 0.40$.

F Wealth Inequality

TODO: Update this content due to new calibration The markup transition generates increases in the Gini and top decile wealth share close to those observed in the Survey of Consumer Finances (SCF). Over 1983–2019, the model generates 53 percent of the observed increase in the Gini and 59 percent of the increase in the top 10 percent share. The top 1 percent share also rises substantially and reaches a level moderately above the SCF estimate. These untargeted outcomes show that the asset market channel generates a quantitatively large rise in wealth concentration alongside the decline in the equilibrium interest rate.

The SCF comparison measures household wealth after subtracting closely held business equity, the closest available counterpart to the model's household asset holdings. The World Inequality Database (WID) supplies a broader historical benchmark in Table F2 over the full model window. We examine the Gini and ranked wealth shares together, then connect their changes to asset accumulation by business owners and workers. **TODO**: Update this content due to new calibration The initial business owner asset share is partly disciplined by the calibrated owner to worker asset ratio. The Gini, ranked shares, and their changes provide additional implications of the equilibrium transition.

F.1 The asset measure and its empirical counterparts

Household asset holdings are the wealth households have saved in the model's asset market. We measure that stock at the beginning of each period, before the current saving decision. These holdings include amounts saved out of business income and proceeds from selling tradable equity. The value of the owner's private business and retained equity is excluded. That ownership instead affects income and the resources available for consumption and saving. This distinction connects the wealth measure directly to household asset demand.

We approximate this asset stock in the SCF by subtracting closely held business equity, `BUS`, from household net worth, `NETWORTH`. This leaves housing, retirement accounts, financial assets, vehicles, and other nonbusiness assets, net of household liabilities. The measure retains publicly traded founder holdings as well as negative observations. We rank households by this signed nonbusiness net worth when computing concentration, separately from the total net worth ranking used to identify business owners in the calibration.

The age restriction follows the calibration sample. We also report both SCF wealth concepts for all households, so the effect of including older households can be read separately from the effect of including business equity. WID uses individuals as its population unit and divides wealth equally within couples. Its net personal wealth concept includes funded pension wealth and excludes consumer durables, unfunded pension entitlements, human capital, and capitalized Social Security benefits. The SCF summary extract includes retirement accounts but does not capitalize defined benefit pension promises or Social Security. The total wealth concepts consequently differ as well

Table F1: Wealth concepts and comparison populations

Source	Wealth measure	Population and role
Model	Household asset holdings at the start of the period; excludes private business value and retained equity.	Workers and business owners in the calibrated economy; equilibrium distribution along the markup transition.
SCF	Signed nonbusiness net worth, `NETWORTH−BUS`.	Households with reference person ages 18 to 64; primary empirical comparison.
SCF	Total net worth, including private business equity.	The same age restriction and all households; wealth concept and population comparisons.
WID	Total household financial and nonfinancial assets less debt, including private business and funded pension wealth.	Equal-split adults ages 20 and older; broader historical benchmark in Table F2.

as their population units.

F.2 Wealth dynamics along the markup transition

Concentration rises across the household asset distribution. Between 1980 and 2018, the model's Gini increases from 0.620 to 0.677. The top 1 percent share rises from 12.51 to 22.77 percent, and the top 10 percent share rises from 45.13 to 53.66 percent. The implied changes are +0.058 Gini units, +10.26 percentage points for the top 1 percent, and +8.54 percentage points for the top 10 percent. These distributional changes emerge from the household policies and the equilibrium transition; they are untargeted outcomes of the markup experiment.

F.3 Common dates and alternative wealth concepts

TODO: Update this content due to new calibration The model closely reproduces the cumulative increase in broad wealth concentration over the common SCF window, 1983–2019. Its Gini increases by +0.066, compared with +0.123 in the SCF. The top 10 percent share increases by +9.52 percentage points, compared with +16.01 in the data. These moments summarize the rise in inequality across the distribution and the shift of wealth toward the top decile. Table F2 reports their starting levels, ending levels, and changes together.

The model also generates a substantial concentration of wealth at the very top. The top 1 percent share reaches 23.17 percent in 2019, compared with 31.88 percent in the data, **TODO**: Update this content due to new calibration a moderate endpoint overshoot of -8.72 percentage points. The model starts with a slightly lower share and generates an increase of +11.09 percentage points, compared with +5.73 in the SCF. Its rise is smoother and begins earlier than the pronounced increase in the SCF after 2007.

Panel B of Table F2 uses WID to place these results alongside the longer history of total household wealth. WID's annual coverage spans the full markup experiment, including the years

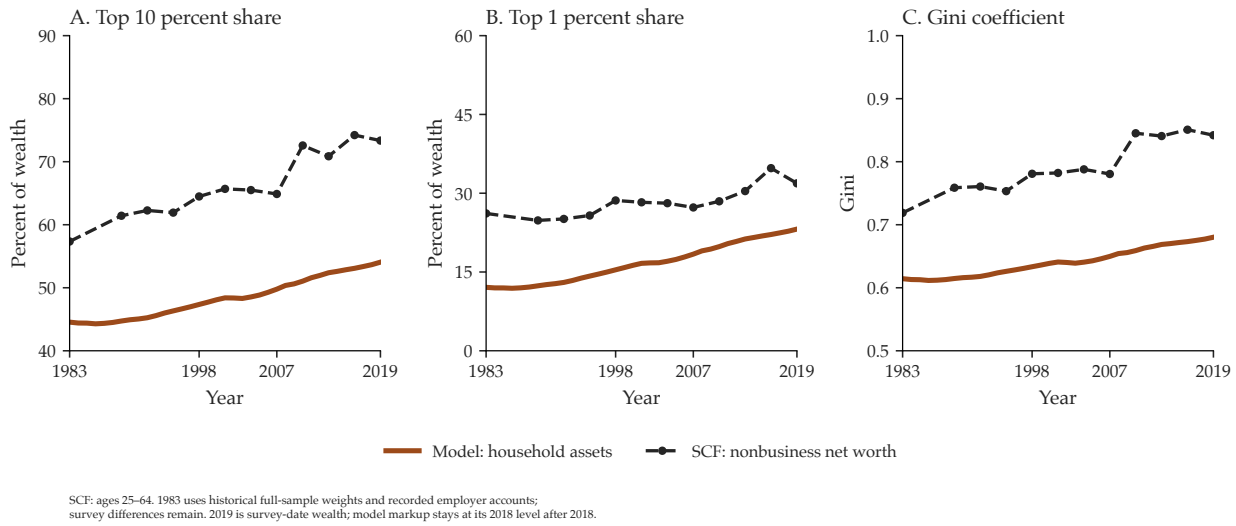


Figure F1: Wealth concentration in the model and SCF over 1983–2019. Panels A–C show the top 10 percent share, top 1 percent share, and Gini. Orange lines use model household assets at the beginning of each period; black markers use survey-weighted SCF nonbusiness net worth for households ages 18 to 64. Dashed segments connect survey waves. Top shares are percentages; the Gini is unscaled.

before the first SCF comparison. Over that window, WID records a Gini change of +0.043, a top 1 percent share change of +11.97 percentage points, and a top 10 percent share change of +6.66 percentage points. The endpoint levels are 0.851, 35.20 percent, and 71.65 percent, respectively. These total wealth series document the historical distribution over the same dates as the markup experiment. The experiment holds the other determinants of saving and asset demand fixed and traces the equilibrium consequences of market power.

The model begins with lower concentration and ends with a Gini of 0.680 and a top 10 percent share of 54.06 percent in 2019. Both are below the SCF values of 0.842 and 73.35 percent and the WID values of 0.863 and 72.78 percent. Within the top decile, a larger fraction of model wealth belongs to the top 1 percent. The Lorenz curves and disjoint ranked groups below show how this pattern relates to the overall rise in concentration.

The two SCF measurement comparisons answer different questions. Adding private business equity broadens the wealth concept. Including all households broadens the population to cover the full life cycle. Each change affects both the wealth held by the ranked group and the aggregate wealth against which its share is measured. Households are ranked again under every definition. Panel C therefore measures concentration within each distribution directly.

WID and the SCF also differ in their construction. The SCF records households' reported assets and liabilities and uses a survey design that oversamples wealthy families. WID combines tax information with wealth aggregates and other evidence to estimate the distribution of total household wealth. For the United States, its series builds on Saez and Zucman's distributional macroeconomic accounts. These sources bring different information to the upper tail, where

Table F2: Wealth concentration: levels, changes, and SCF measurement comparisons

A. Common SCF window

	Model: household assets			SCF: nonbusiness		
Statistic	1989	2016	Change	1989	2016	Change
Gini	0.614	0.680	+0.066	0.719	0.842	+0.123
Top 1 percent	12.08	23.17	+11.09	26.15	31.88	+5.73
Top 10 percent	44.54	54.06	+9.52	57.34	73.35	+16.01

B. Full transition and WID

	Model: household assets			WID: total wealth		
Statistic	1980	2018	Change	1980	2018	Change
Gini	0.620	0.677	+0.058	0.808	0.851	+0.043
Top 1 percent	12.51	22.77	+10.26	23.23	35.20	+11.97
Top 10 percent	45.13	53.66	+8.54	64.99	71.65	+6.66

C. SCF sensitivity: changes from 1989 to 2016

Wealth concept	Population	Gini	Top 1 percent	Top 10 percent
Nonbusiness	Ages 18 to 64	+0.123	+5.73	+16.01
Total net worth	Ages 18 to 64	+0.103	+7.15	+13.56
Nonbusiness	All households	+0.082	+3.16	+10.98
Total net worth	All households	+0.073	+5.69	+9.80

Notes: Top wealth shares are in percent and their changes in percentage points. Gini coefficients and their changes are unscaled. Changes are computed before rounding. Panel A holds calendar dates fixed between the model and the SCF. Panel B uses the full model window and WID observations at the same dates. Panel C recomputes SCF statistics under each wealth concept and population, including new ranks within each sample. All SCF statistics average the five implicate estimates. Model statistics use beginning of period distributions. The calibration targets the owner asset demand ratio; the ranked shares and Gini changes reported here are untargeted.

valuations of businesses and the mapping from capital income to asset stocks are especially consequential.

F.4 The distribution and the ownership of assets

The Lorenz curves show how concentration changes across the ranked distribution. Figure F2 compares Lorenz curves at the initial and final SCF dates. Each curve reports the wealth held by the bottom fraction of households, including negative nonbusiness wealth in the SCF. In 2019, the model assigns more wealth to the bottom 90 percent than the SCF and less to the bottom 99 percent, so the curves cross near the top. This pattern connects the lower model Gini and top 10 percent share to its higher top 1 percent share.

The top decile gains wealth share in both sources, with a larger reallocation toward the top 1 percent in the model. The next 9 percent's model share falls from 32.47 to 30.89 percent between 1983 and 2019, while its SCF share rises from 31.19 to 41.47 percent. The top 1 percent holds 42.86 percent of the top decile's asset holdings in the model in 2019, compared with 43.47 percent in the SCF and 48.78 percent in WID. Panel (c) separates the bottom 90 percent, next 9 percent, and top 1

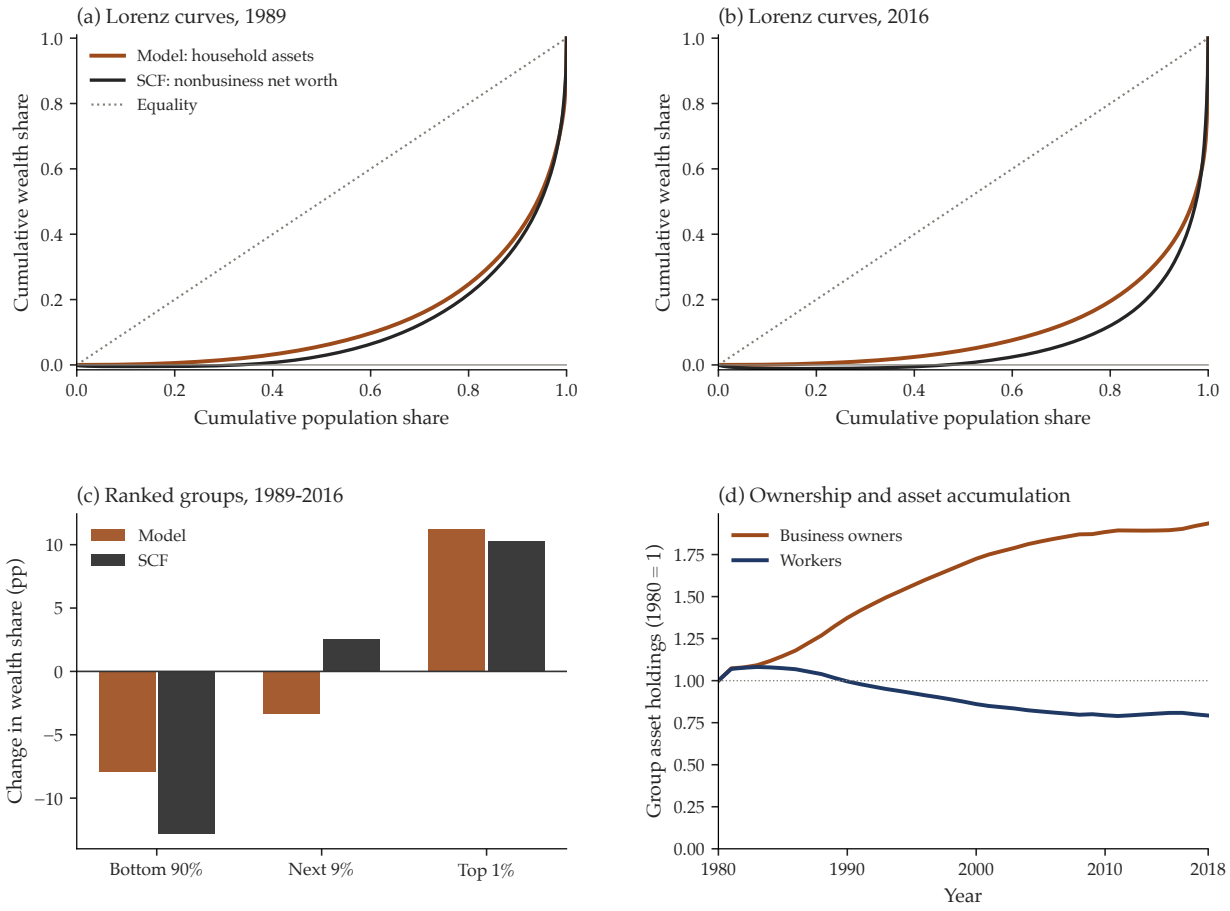


Figure F2: Wealth distribution and asset accumulation. Panels (a) and (b) compare model and SCF Lorenz curves at matched dates; the dotted diagonal denotes equality. Panel (c) shows wealth share changes for the bottom 90 percent, next 9 percent, and top 1 percent over 1983–2019, with ranks specific to each source and date. Panel (d) shows model worker and business owner assets, each normalized to its initial stock. Household types are fixed; ranked groups can change members.

percent. Their share changes sum to zero and show how much of the movement occurs within the top decile.

Panel (d) follows ownership rather than ranks. Workers and business owners remain the same permanent household types throughout the experiment, while the households in a top percentile can change. Over 1980–2018, the business owner share of household assets rises from 17.16 to 30.40 percent. Across the same transition, the total assets held by business owners rise by 83.75 percent, while those held by workers fall by 12.85 percent. These paths reveal how the asset demand response is distributed across the two household groups.

Redistribution links these changes to the equilibrium interest rate. Business owners hold more assets relative to their aggregate income base than workers. As market power shifts income toward profits, their asset demand grows. With inelastic asset demand, the resulting decline in the equilibrium interest rate mitigates the contraction in productive capital. At the household level,

that adjustment goes together with a change in who receives business income and who accumulates assets.

Asset Creation connects incorporation to this accumulation. Selling the publicly traded share of a business adds tradable equity at market value to asset supply and gives its founder sale proceeds. Those proceeds become part of the founder's resources for consumption and saving. Subsequent asset holdings reflect that saving decision as well as the returns earned on assets already held. The path of wealth concentration thus combines the distribution of profit income, the timing of incorporation, and household saving along the equilibrium transition.

An accumulation identity makes the ownership adjustment transparent. Let $A_{g,t}$ be the assets held by group g at the beginning of period t, R_t the common gross return on those assets, and $B_{g,t}$ net additions after that return. These additions collect the other resources allocated to asset holdings, including labor and business income and IPO proceeds, less consumption. For owners O and workers W,

$$A_{g,t+1} = R_t A_{g,t} + B_{g,t}, \qquad \frac{A_{O,t+1}}{A_{O,t}} - \frac{A_{W,t+1}}{A_{W,t}} = \frac{B_{O,t}}{A_{O,t}} - \frac{B_{W,t}}{A_{W,t}}. \tag{F1}$$

The common proportional return cancels from the difference in asset growth rates. That return alone leaves the groups' relative wealth shares unchanged. Owners gain wealth share when their net additions are larger relative to their initial stock. Interest rates still affect those additions through household income and saving decisions. Redistribution and IPO receipts give owners more resources to accumulate, and the equilibrium return changes the decisions of both groups.

This identity accounts for the realized group paths. A causal decomposition of their changes requires separately solved counterfactual transitions, since each force changes prices and saving jointly. The welfare decomposition answers a different question again: it values income and consumption throughout the transition. Private rents, retained income, and IPO proceeds affect both asset accumulation and welfare through their incidence across households; the ownership paths show the associated changes in stocks.

F.5 Construction of the distributional statistics

Let a_i denote the wealth measure and w_i its population weight. We normalize the weights to sum to one and order observations by a_i. The cumulative weight and cumulative wealth define the Lorenz curve $L(p)$, where p is a population fraction. Within a weighted observation, both quantities are allocated proportionally. The top share at a population fraction q and the Gini are

$$S_q = 1 - L(1-q), \qquad G = 1 - 2\int_0^1 L(p)\,dp. \tag{F2}$$

Fractional allocation at the cutoff gives exactly the requested population share even when one survey observation spans it. The Gini integrates the full weighted empirical Lorenz curve. The

finer grid displayed in the figure is used to draw the curve; numerical integration uses its original population and wealth knots. Negative wealth is retained in both the ranking and the aggregate. With signed wealth and a positive aggregate, a Lorenz curve can lie below zero and its Gini can exceed one. We neither truncate observations nor bound the calculated Gini; the figure axes accommodate the observed values.

SCF statistics are constructed separately within each of the five implicates. Within each implicate, we apply the population restriction, construct the wealth measure, and rank households using survey weights. We then average the resulting statistics across implicates. Lorenz ordinates are averaged at common population fractions. This treatment preserves the SCF's multiple imputation structure and avoids treating alternative imputations of one family as distinct households. The primary time series contains 12 survey waves from 1983 through 2019.

Model moments use the distribution at the beginning of each dated period. We sum over idiosyncratic states while retaining each asset grid point and its population mass. The distribution's total asset stock is reconciled with the beginning of period aggregate in the saved equilibrium path. The owner and worker asset stocks in Figure F2 use the same timing and weights. Consequently, the ranked statistics and the ownership shares refer to one common distribution.

The numerical checks cover the full 99 periods of the solved path, including the adjustment after the historical window. Workers place zero mass on their upper five asset grid points. For business owners, the largest conditional mass on those points is 1.06×10^{-59}, and the largest fraction of owner assets held there is 1.88×10^{-56}. Reconstructed asset totals and concentration moments at the exported dates reproduce the saved distributional statistics to a maximum absolute error of 0.00×10^{0}. These checks assess the represented distribution directly, including its upper tail.

WID supplies the Gini and top shares directly. All three series use net personal wealth, `hweal`, equal-split individuals, `j`, and adults ages 20 and older, `992`. The current codes are `ghwealj992` for the Gini at `p0p100` and `shwealj992` for shares at `p99p100` and `p90p100`. The frozen inputs retain the published annual observations through the model's endpoint, from the common September 1, 2026 data and metadata vintage, retrieved September 7, 2026 (UTC). Source files and checksums preserve these observations and definitions across exhibit rebuilds.

The distributional results give the asset market channel an additional empirical implication. Under the existing calibration, the markup transition generates increases in the Gini and top decile share close to the SCF increases, together with a substantial rise in the top 1 percent share. Business owners accumulate more assets as income shifts toward profits. The same rent incidence that lowers the equilibrium interest rate therefore also generates a large and economically relevant redistribution of household wealth.

Sources. The empirical inputs are the Federal Reserve Board's Survey of Consumer Finances summary extracts and the World Inequality Database United States data, together with its series metadata. The WID codes dictionary defines household net wealth and equal-split adults. The

U.S. wealth series draws on Saez and Zucman (2020); the WID wealth update documents its subsequent coverage. The Federal Reserve's discussion of SCF wealth concepts and Smith, Zidar, and Zwick (2023) describe the role of pension coverage, private business valuation, and upper tail measurement in comparisons across sources.

G Extensions

This appendix asks whether three changes to the one-asset environment alter the asset market channel. Fiscal policy adds government debt and after-tax income; balanced growth changes both household normalization and the duration of tradable equity; and CES production makes factor shares respond endogenously to the return. In every case, the central question remains whether the redistribution of profit income toward entrepreneurs raises asset demand by more or less than the market value of tradable equity raises asset supply.

G.1 Common quantitative protocol

The quantitative comparisons use a matched-baseline design. Each extension is recentered, when necessary, to reproduce the common 1980 return and its extension-appropriate asset-market targets, and its household block is re-solved whenever the extension changes the 1980 household problem. We then feed in the same observed market-power path used in the main text through each extension's own sufficient statistic and report differences from the baseline computed the same way. For the growth and CES exercises, re-solving the exact stationary household problems at every date provides a check on that closure. None of these paths is a perfect-foresight transition.

Section G.2 studies fiscal policy. Section G.3 introduces balanced growth, and Section G.4 replaces Cobb–Douglas production with CES. Each extension states its exact asset-market equation, identifies the new calibration input, and verifies that shutting down the new margin recovers the corresponding one-asset benchmark.

G.2 Fiscal Policy in the One-Asset Economy

The main analysis abstracts from government debt and the taxes used to service it. Government debt is a quantitatively important component of asset supply and therefore potentially important for the equilibrium return. This appendix extends the one-asset sufficient statistic to include government bonds, after-tax factor income, and the tax-adjusted user cost of capital.

The fiscal closure creates a fiscal externality that amplifies the response of the equilibrium return. We hold the debt-to-GDP ratio fixed and let the labor income tax adjust to balance the government budget. When the return falls, interest payments on government debt decline relative to GDP. The government therefore requires less labor-tax revenue. The resulting increase in workers' after-tax income raises household asset demand, requiring a further decline in the return to clear the asset market. Thus the fiscal externality strengthens the decline in the equilibrium return, even as the lower labor tax cushions the decline in output.

We first show that proportional labor and profit taxes preserve the income-scaling property underlying the sufficient statistic and derive asset-market clearing with government debt. We then use the government budget to isolate the fiscal externality and derive its implications for output.

Finally, we calibrate the fiscal closure to 1980 and quantify the externality along the observed markup path.

G.2.1 After-tax income and capital taxation

Let τ_W denote the proportional tax on labor income. The GHH labor choice satisfies $v'(\ell) = (1-\tau_W)we$. The maximized labor surplus and aggregate after-tax labor income scale with the same object,

$$\widetilde{W} \equiv (1-\tau_W)W.$$

The normalization in the proof of Proposition 2 can therefore be repeated with $\widetilde{W}$ in place of W. Worker asset demand is

$$D_W(r, \widetilde{W}) = \bar{D}(r)\widetilde{W}. \tag{G1}$$

Let τ_Π denote the proportional tax on pure profits and define aggregate after-tax profit income $\widetilde{\Pi} = (1-\tau_\Pi)\Pi$. Because private profit income, retained profit income, and the equity sold at incorporation are all scaled by $1-\tau_\Pi$, the after-tax individual equity value is $q^\tau = (1-\tau_\Pi)q$. The entrepreneur problem remains homogeneous in $\widetilde{\Pi}$, including the sale proceeds, and hence

$$D_O(r, \widetilde{\Pi}) = \lambda(r)\bar{D}(r)\widetilde{\Pi}. \tag{G2}$$

Here $\lambda(r)$ is the relative wealth-to-income ratio of owners to workers, measured using after-tax profit income for owners and after-tax labor income for workers. The main text reports the corresponding ratio per dollar of gross income. At the calibration date the two are related by

$$\lambda^{\text{after}} = \lambda^{\text{gross}} \frac{1-\tau_{W,0}}{1-\tau_\Pi}. \tag{G3}$$

We use λ for the relative wealth-to-income ratio of owners to workers, measured using after-tax income, below.

Capital taxation affects both the marginal cost of installing capital and the average revenue collected from the corporate tax base. These are different objects. Let τ_K be the statutory corporate income-tax rate and let z^τ be the present value of tax depreciation deductions generated by one dollar of investment. With no inflation or investment tax credit, the one-asset user cost is

$$u_K(r) = (r+\delta_K)\frac{1-\tau_K z^\tau}{1-\tau_K}. \tag{G4}$$

For asset types j, the one-asset specification uses capital-stock weights ϑ_j and $z^\tau = \sum_j \vartheta_j z_j^\tau$. The tax term in (G4) is therefore the capital-stock-weighted average of the asset-specific user-cost wedges used by Barkai (2020).

The firm's capital first-order condition implies $s_K = u_K K/Y$. The part paid as a net return to

savers is therefore

$$\widetilde{s}_K \equiv \frac{rK}{Y} = \frac{r}{u_K(r)} s_K. \tag{G5}$$

Let T_K denote corporate income-tax receipts assigned to capital taxation and let $t_K = T_K/Y$. We do not infer t_K from the marginal wedge in (G4). Depreciation rules, the tax base, losses, and other provisions break that equivalence. The calibration uses the user-cost wedge for the firm's marginal condition and measured receipts for the government budget.

G.2.2 Government debt and the fiscal sufficient statistic

Let B be government debt held by the public and G government consumption and gross investment. Write $b = B/Y$ and $g = G/Y$. Let ξ denote residual net government revenue relative to GDP: unmodeled receipts net of transfers and other primary outlays, including differences in the coverage of the fiscal sources. Because the exercise compares stationary economies without trend growth, maintaining a constant debt-to-GDP ratio requires interest payments equal to rb. The government budget is therefore

$$\tau_W s_W + t_K + \tau_\Pi s_\Pi + \xi = g + rb. \tag{G6}$$

The fiscal closure fixes b, g, ξ, τ_Π, and the capital tax system. The labor tax is the rate that balances the budget:

$$\tau_W = \frac{g + rb - t_K - \tau_\Pi s_\Pi - \xi}{s_W}. \tag{G7}$$

The model has no structural measure of taxable corporate income. The benchmark holds capital-tax receipts fixed relative to GDP, $t_{K,t} = t_{K,0}$, while retaining the separately measured marginal user-cost wedge in (G4). This is a fiscal closure, not a claim that a fixed statutory rate delivers constant receipts.

Define the after-tax income shares

$$\widetilde{s}_W = (1 - \tau_W) s_W, \qquad \widetilde{s}_\Pi = (1 - \tau_\Pi) s_\Pi, \tag{G8}$$

and let $Q(r) = 1/(r + \delta_C)$ as in the main text.

Proposition 5 (Fiscal sufficient statistic)**.** *Under the conditions of Proposition 2, proportional labor and profit taxes preserve the linearity of household asset demand. In a stationary one-asset equilibrium with government debt, asset-market clearing is*

$$\bar{D}(r) \left[\widetilde{s}_W + \lambda(r) \widetilde{s}_\Pi\right] = \frac{\widetilde{s}_K}{r} + Q(r) \omega \widetilde{s}_\Pi + b. \tag{G9}$$

Together, (G6) *and* (G9) *determine the labor tax and the return from after-tax factor shares, the bond stock, and the same incidence objects as in Theorem 1.*

Proof. Proportional labor taxation preserves the worker normalization in (G1), while proportional profit taxation scales private and retained profit income and sale proceeds by $1 - \tau_\Pi$. Equation (G2) follows by dividing entrepreneur assets and consumption by $\widetilde{\Pi}$. Aggregate household asset demand is therefore

$$\bar{D}(r)\left[\widetilde{W} + \lambda(r)\widetilde{\Pi}\right].$$

Asset supply consists of physical capital, after-tax tradable equity, and government bonds:

$$K + Q(r)\omega\widetilde{\Pi} + B.$$

Equating demand and supply and dividing by Y gives (G9), since $K/Y = \widetilde{s}_K/r$. Equation (G6) supplies the labor tax that enters $\widetilde{s}_W$. □

The left side of (G9) is the asset demand generated by after-tax labor and profit income. The right side is the supply of physical capital, tradable equity, and government bonds. The government budget links these two sides by determining the labor tax.

G.2.3 Fiscal externality

Substituting (G6) into after-tax worker income gives

$$\widetilde{s}_W = s_W + t_K + \tau_\Pi s_\Pi + \xi - g - rb. \tag{G10}$$

The income term that generates household asset demand can therefore be written

$$\widetilde{s}_W + \lambda\widetilde{s}_\Pi = s_W + t_K + \xi - g - rb + [\tau_\Pi + \lambda(1 - \tau_\Pi)]\, s_\Pi. \tag{G11}$$

Holding b, g, ξ, τ_Π, and λ locally fixed,

$$d(\widetilde{s}_W + \lambda\widetilde{s}_\Pi) = ds_W + dt_K + [\tau_\Pi + \lambda(1 - \tau_\Pi)]\, ds_\Pi - b\, dr. \tag{G12}$$

The term $-b\, dr$ is the fiscal externality. When market power lowers the return, $dr < 0$, interest payments on government debt decline. Because the labor tax balances the budget, lower debt service permits a lower labor tax and leaves workers with more after-tax labor income. Household asset demand rises, so a further decline in the return is required to clear the asset market. Holding the debt-to-GDP ratio fixed therefore amplifies the return response through the government budget.

The profit tax changes the incidence of rents. Its coefficient in (G12) is

$$\tau_\Pi + \lambda(1 - \tau_\Pi) = \lambda - \tau_\Pi(\lambda - 1).$$

For $\lambda > 1$, this coefficient lies between one and λ. Taxing profits removes income from entrepreneurs with greater asset demand per dollar of income, while the associated revenue lowers

the balancing labor tax. The profit tax therefore attenuates the redistribution force rather than eliminating it. A closure in which capital-tax receipts track a modeled tax base would enter through dt_K; the benchmark instead holds t_K fixed relative to GDP.

Combining the firm's Cobb–Douglas first-order conditions with GHH labor supply gives the output response

$$\Delta \log Y = \varphi \Delta \log(1 - \tau_W) - \frac{\alpha + \varphi}{1 - \alpha} \Delta \log \mu - \frac{\alpha(1 + \varphi)}{1 - \alpha} \Delta \log u_K. \tag{G13}$$

The lower labor tax raises labor supply, while the return response changes capital demand through the tax-adjusted user cost. The fiscal externality can therefore amplify the decline in the return even as the lower labor tax cushions the decline in output.

Setting $b = 0$ removes government bonds from asset supply and the fiscal externality $-b\,dr$ from (G12); taxes and government purchases may still remain. If all fiscal objects are shut down—$b = g = \xi = \tau_W = \tau_\Pi = 0$, $t_K = 0$, and $u_K = r + \delta_K$—Proposition 5 reduces to Theorem 1.

G.2.4 Calibration and quantitative illustration

The quantitative exercise is a comparative-static fiscal closure, not a reconstruction of complete government accounts. Federal debt, government purchases, capital-tax receipts, and residual net revenue are held fixed relative to GDP at their 1980 values; the profit-tax scenario and marginal capital-tax system are also held fixed. The labor tax adjusts to balance the budget as the observed cost-weighted markup path changes. The residual ξ reconciles fiscal flows that the model does not represent separately. Table G1 summarizes the inputs.

Government debt and the tax wedges change the 1980 asset supply per unit of after-tax income, so we re-solve the household discount factors so that the full fiscal economy clears at $r_0 = 6.00$ percent in 1980 with the same wealth ratio, and we measure the asset-demand semi-elasticity at that calibration: $\epsilon_D = 17.9162$, against 16.2963 in the main calibration. For the quantitative comparison, the *full fiscal closure* contains all of the objects in Table G1. The *matched no-debt benchmark* sets $b = 0$, retains the other fiscal wedges and the same household block, adjusts ξ to preserve the initial labor tax, and recenters $\bar{D}(r)$ to the same initial return, so it isolates the debt term in asset supply and the debt service in the budget. The *matched no-fiscal benchmark* removes taxes, debt, purchases, residual revenue, and the corporate user-cost wedge, uses the gross-income value of λ, and recovers the main calibration's discount factors and semi-elasticity, so it reproduces the main-text sufficient statistic. These are response comparisons around a common 1980 equilibrium, not level effects of adding or removing government debt.

Table G2 reports the response to the observed markup path. By 2018, the full fiscal closure lowers the return by 96.1 basis points, compared with 90.4 basis points in the matched no-debt benchmark. Their difference, -5.7 basis points, is the fiscal externality. The balancing labor tax declines from 26.4 to 25.6 percent. Relative to the matched no-fiscal benchmark, the labor tax and

Table G1: Fiscal inputs for the 1980 closure

Object	Symbol	Value	Source or role
Federal debt held by the public / GDP	b	26.2%	Congressional Budget Office (2020)
Average effective labor tax	$\tau_{W,0}$	26.4%	McDaniel (2007)
Government purchases / GDP	g	20.63%	BEA NIPA Table 1.1.5
Corporate income-tax receipts / GDP		2.39%	BEA NIPA Table 1.14
Pure-profit tax rate	τ_{Π}	20%	Scenario
Capital-tax receipts / GDP	t_K	1.59%	Corporate receipts net of the pure-profit component
Residual net revenue / GDP	ξ	2.92%	Balances omitted receipts and primary outlays
Statutory corporate income-tax rate	τ_K	49.91%	Barkai (2020) replication files
Marginal user-cost tax factor	$(1-\tau_K z^{\tau})/(1-\tau_K)$	1.331	Capital-stock-weighted tax wedge
Relative wealth-to-income ratio of owners to workers (after-tax income)	λ	3.05	Equation (G3)
Asset-demand semi-elasticity	ϵ_D	17.9162	Re-solved household block; 16.2963 in the main calibration

Notes: The pure-profit tax assigns 0.80% of GDP to profit-tax receipts; the remaining 1.59% is assigned to capital taxation. The residual ξ has no separate structural interpretation. The present values of depreciation allowances are 0.753 for equipment, 0.438 for structures, and one for intellectual property and inventories. The tax rate and depreciation allowances determine the marginal user-cost wedge, whereas t_K is the measured average receipt share used in the government budget.

user-cost responses raise output by 0.41 percentage points in 2018.

G.3 Balanced Growth

The baseline is stationary in levels. This subsection instead places the same one-asset economy on a balanced-growth path. Growth does not remove the asset market channel: the markup still reallocates income toward entrepreneurs and creates tradable equity. It does, however, change both the household normalization and the duration of that equity, so it cannot be introduced by changing a denominator in the baseline sufficient statistic while leaving household demand fixed.

The matched balanced-growth calculation below re-solves the household calibration at growth rate $g = 2\%$ and then applies the same market-power path as in the main text. The stationary 2018 return change is -87.2 basis points, compared with -97.6 basis points in the matched no-growth economy. Growth therefore changes the magnitude of the markup effect without eliminating it: in the calibrated comparison it moderately attenuates the decline in the return.

We first detrend the household problems and identify the two places where growth enters asset demand. We then derive the exact discrete-time valuation of a growing tradable equity and the corresponding sufficient statistic. Finally, we recalibrate the initial stationary economy and compare the stationary paths with and without trend growth.

Table G2: Matched fiscal benchmark responses in 2018

Economy	Return (basis points)	Output (percent)
Matched no-fiscal benchmark	−97.6	−5.40
Matched no-debt benchmark	−90.4	−5.61
Full fiscal closure	−96.1	−5.00

G.3.1 Environment and detrended household demand

Let $G \equiv 1 + g$ and suppose aggregate output, labor income, and pure-profit income grow at the common deterministic rate g:

$$Y_{t+1} = GY_t, \qquad W_{t+1} = GW_t, \qquad \Pi_{t+1} = G\Pi_t.$$

The factor shares are therefore stationary conditional on the markup. To display the household normalization, write the worker budget as

$$c_t + a_{t+1} = (1 + r)a_t + W_t\widetilde{e}_t, \qquad a_{t+1} \geq 0. \tag{G14}$$

Define contemporaneous-income consumption $\widehat{c}_t = c_t/W_t$ and beginning-of-period assets in lagged-income units $b_t = a_t/W_{t-1}$. Dividing (G14) by W_t yields

$$\widehat{c}_t + b_{t+1} = \check{R}b_t + \widetilde{e}_t, \qquad \check{R} \equiv \frac{1+r}{1+g}. \tag{G15}$$

With CRRA curvature σ, the detrended Bellman problem has effective discount factor

$$\beta_i^g = \beta_i(1+g)^{1-\sigma}, \qquad i \in \{W, O\}. \tag{G16}$$

Thus growth affects household policies through both the detrended gross return $\check{R}$ and β_i^g. It also affects the reported asset–income ratio through timing: $a_t/W_t = b_t/(1+g)$.

Let $\mathcal{D}_W(\check{R}, \beta_W^g)$ denote mean beginning-of-period worker assets in lagged-income units, with the remaining normalized household states suppressed. The worker asset-demand coefficient in current-income units is

$$\bar{D}(r,g) = \frac{1}{1+g}\,\mathcal{D}_W\left(\frac{1+r}{1+g}, \beta_W(1+g)^{1-\sigma}\right). \tag{G17}$$

The owner problem is detrended in the same way. The relative wealth-to-income ratio of owners to workers is $\lambda(r,g)$, so aggregate household demand remains linear in the two income flows:

$$D_t = \bar{D}(r,g)\left[W_t + \lambda(r,g)\Pi_t\right]. \tag{G18}$$

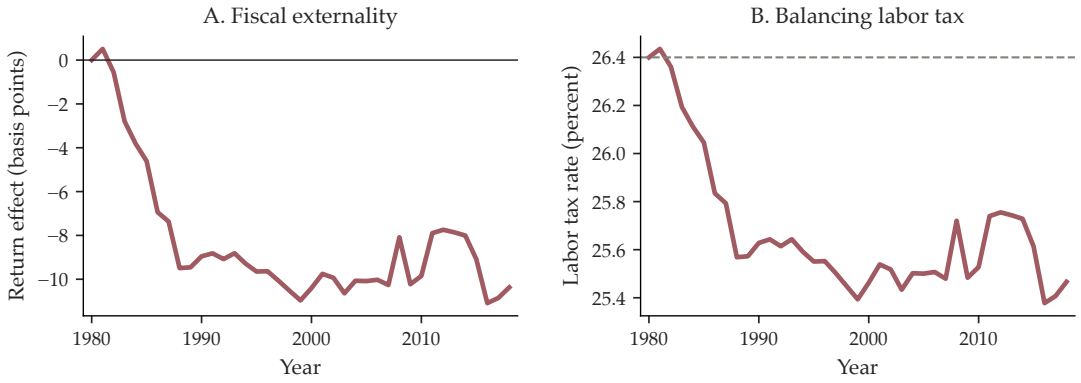


Figure G1: Fiscal externality under the balancing labor tax closure. Panel A shows the interest rate r in the full fiscal economy minus its matched no-debt benchmark, in basis points. Panel B shows the labor tax rate required to balance the budget. The benchmark retains the other fiscal wedges, preserves the initial labor tax, and shares the initial interest rate.

G.3.2 Balanced-growth sufficient statistic

Growth treats reproducible capital and tradable equity differently. The relative price of installed capital remains one because output can be transformed into capital. Its no-arbitrage condition therefore continues to imply

$$\frac{K}{Y} = \frac{s_K}{r + \delta_K}. \tag{G19}$$

Tradable equity has no reproduction technology. It pays next period's publicly traded profit income and survives with probability $1 - \delta_C$. When its price is proportional to the contemporaneous publicly traded profit income flow, its exact discrete-time valuation multiple under balanced growth is

$$Q^g(r, g) = \frac{(1+g)}{r - g + \delta_C(1+g)}, \qquad r - g + \delta_C(1+g) > 0. \tag{G20}$$

Proposition 6 (Balanced-growth sufficient statistic)**.** *Under the conditions of Proposition 2, a stationary balanced-growth equilibrium satisfies*

$$\bar{D}(r, g)\,[s_W + \lambda(r, g)s_\Pi] = \frac{1}{r + \delta_K} s_K + Q^g(r, g)\omega s_\Pi. \tag{G21}$$

Proof. The detrended household problems imply (G18). Asset supply consists of physical capital and tradable equity backed by growing publicly traded profit income, $K + \omega Q^g \Pi$. Equating demand and supply, dividing by Y_t, and using (G19) gives (G21). □

Table G3: Matched baseline calibration with and without growth

Economy	g (%)	β_W	β_O	$\partial \log D / \partial r$	Q_0	A_0/Y_0
Matched no-growth	0.0	0.8872	0.9032	16.296	9.09	2.61
Balanced growth	2.0	0.9233	0.9389	16.680	11.21	2.64

Notes: Both economies match the common initial return and direct relative wealth-to-income ratio of owners to workers measured in the SCF and clear their initial asset markets. Because growth changes the detrended household problem, the discount factors and baseline local asset-demand semi-elasticity are re-solved separately.

G.3.3 Mechanism

The supply-side effect of growth is transparent. With $\Delta_C(r,g) \equiv r - g + \delta_C(1+g)$,

$$\frac{\partial Q^g}{\partial g} = \frac{(1+r)}{\Delta_C(r,g)^2} > 0, \qquad \frac{\partial Q^g}{\partial r} = -\frac{(1+g)}{\Delta_C(r,g)^2} < 0. \tag{G22}$$

Growth raises the value and duration of tradable equity, expanding the asset supply created by a given profit share. This tends to attenuate the decline in the clearing return. The demand-side effect has no general sign: growth lowers the detrended return, changes the effective discount factor when $\sigma \neq 1$, and changes the timing conversion in (G17). This is why a balanced-growth exercise must re-solve the household block rather than combine Q^g with the zero-growth demand slope.

G.3.4 Calibration and quantitative illustration

We set trend growth to 2% and retain the main calibration's markup path, factor-share mapping, depreciation rates, incorporated profit share, founder retention, household risk, and entrepreneur definition. For the growth economy, we jointly re-solve (β_W, β_O) so that the 1980 stationary equilibrium clears at $r_0 = 6.00\%$ and matches the direct SCF target $\lambda(r_0, g) = 3.31$. The entrepreneur asset share and the aggregate asset–output ratio remain untargeted implications. The matched no-growth comparison is solved under the same protocol with $g = 0$.

For each date, the observed markup determines the factor shares and we solve (G21) with the local closure recovered from the corresponding matched baseline; re-solving the nonlinear stationary household-demand schedules at each date provides the check reported alongside. Specifically, the local closure uses

$$\bar{D}^{\text{loc}}(r,g) = \bar{D}_0^g \exp\left\{\epsilon_D^g (r - r_0)\right\}, \qquad \lambda^{\text{loc}}(r,g) = \lambda_0, \tag{G23}$$

where ϵ_D^g is the reduced-form slope of aggregate household asset demand, computed from the exact household blocks at the matched baseline income shares. Thus the local calculation follows the main sufficient-statistic convention: it lets one aggregate demand index absorb the baseline response and holds the relative wealth-to-income ratio of owners to workers fixed. It should not be read as a separate approximation to the worker and entrepreneur demand schedules. The local statistic is the reported result, as in the main text; the exact stationary calculation checks how

Table G4: Balanced growth and the stationary 2018 endpoint

Economy	r_{2018} (%)	Δr (bp)	Exact HA Δr (bp)	A_{2018}/Y_{2018}	Q^g_{2018}
Matched no-growth	5.02	-97.6	-102.1	2.88	9.98
Balanced growth	5.13	-87.2	-91.2	2.96	12.40

Notes: r_{2018}, Δr, A/Y, and Q^g come from the sufficient statistic evaluated with the slope from the corresponding matched baseline. "Exact HA" re-solves the nonlinear stationary household problems at every date and is reported as a check. The endpoint is associated with the 2018 value of the common market-power path, not the dated perfect-foresight outcome.

much of it survives re-solving both household problems. Neither path incorporates anticipation or transitional wealth distributions.

Nesting check. Setting $g = 0$ gives $\check{R} = 1 + r$, $\beta_i^g = \beta_i$, and $Q^g(r, 0) = 1/(r + \delta_C) = Q(r)$. Equations (G17) and (G21) then reduce to the stationary household demand and one-asset clearing equation used in the main text. This nesting check separates the effect of trend growth from differences in initial calibration.

G.4 CES Production and the Markup Experiment

The baseline uses Cobb–Douglas production, under which a movement in the return does not change the technological split between capital and labor. With CES production, factor shares depend on the capital–labor ratio and hence on the user cost. The observed markup path therefore changes the asset market through a third force: in addition to redistributing income and creating tradable equity, it changes capital intensity and the division of non-profit income between workers and capital.

The empirically preferred specification uses $\sigma_{KL} = 0.7$. Its stationary 2018 return and output changes are -95.6 basis points and -5.32 percent, respectively, compared with -97.6 basis points and -5.40 percent under Cobb–Douglas. The representative household CES benchmark instead holds $r_t = r_0$ and gives an output change of -9.71 percent in the preferred CES specification. The quantitative issue is therefore whether endogenous factor shares materially amplify or dampen the asset-market channel, not whether CES production removes that channel.

We first parameterize the CES share functions using the observed 1980 shares, avoiding the need to separately identify a distribution parameter and productivity normalization. We then derive the implicit scalar sufficient statistic, isolate the capital-share mechanism, and map the stationary return into output. Finally, we solve matched stationary economies over an empirically disciplined range of substitution elasticities.

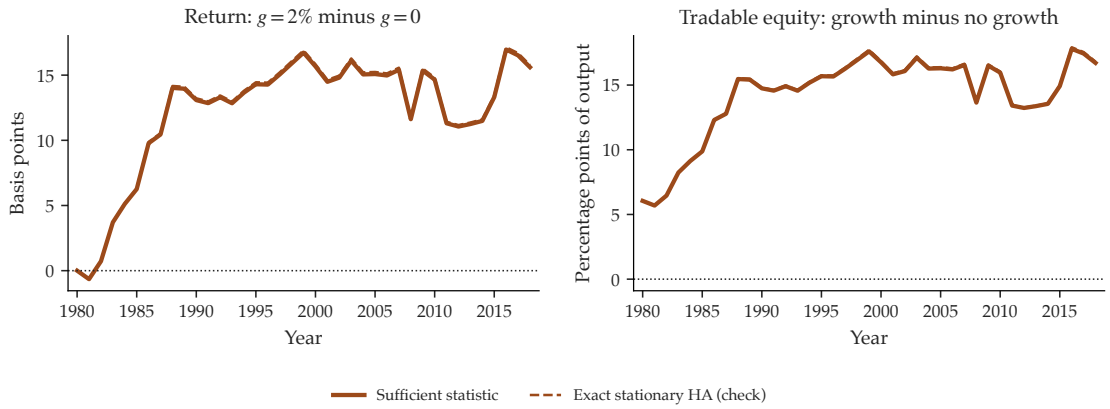


Figure G2: Incremental effects of balanced growth. Panels A and B show growth-minus-no-growth differences in the interest rate r (basis points) and tradable equity relative to output (percentage points). Both economies use the same markup path and initial interest rate. Solid lines show the sufficient statistic; dashed lines re-solve the exact stationary household problems. These are stationary comparisons, not perfect foresight transitions.

G.4.1 Environment and factor-share mapping

Let production be constant returns to scale and CES:

$$Y = Z\left[\alpha K^{\rho} + (1-\alpha)L^{\rho}\right]^{1/\rho}, \qquad \rho = \frac{\sigma_{KL}-1}{\sigma_{KL}}, \tag{G24}$$

where σ_{KL} is the capital–labor elasticity of substitution and is distinct from household CRRA curvature. A firm with gross markup μ_t pays each factor $1/\mu_t$ times its value marginal product. Constant returns then imply

$$s_{\Pi,t} = 1 - \frac{1}{\mu_t}, \qquad s_{W,t} + s_{K,t} = \frac{1}{\mu_t}. \tag{G25}$$

Normalize the CES economy to reproduce the observed capital share $s_{K,0}$ at the initial pair (r_0, μ_0). The capital first-order condition then gives the exact share mapping

$$\boxed{s_{K,t}(r) = s_{K,0}\left(\frac{\mu_t}{\mu_0}\right)^{-\sigma_{KL}}\left(\frac{r+\delta_K}{r_0+\delta_K}\right)^{1-\sigma_{KL}}} \tag{G26}$$

and

$$s_{W,t}(r) = \frac{1}{\mu_t} - s_{K,t}(r), \qquad \frac{K_t}{Y_t} = \frac{s_{K,t}(r)}{r+\delta_K}. \tag{G27}$$

This representation absorbs (α, Z) into one observed reference share. The factor $(\mu_t/\mu_0)^{-\sigma_{KL}}$ is essential: a formula that conditions on a fixed markup is not valid for the historical markup experiment.

Conditional on the return, CES production does not alter either *normalized* household saving

problem. Consequently $\bar{D}(r)$ and $\lambda(r)$ are the same nonlinear stationary demand schedules, measured per unit of contemporaneous group income, as in the calibrated one-asset model. What changes is that the income-share weights and capital intensity in the clearing equation are functions of the return.

G.4.2 CES sufficient statistic

Proposition 7 (CES sufficient statistic)**.** *Under the conditions of Proposition 2 and production technology* (G24)*, the stationary return at date t solves*

$$\bar{D}(r)\left[s_{W,t}(r)+\lambda(r)s_{\Pi,t}\right]=\frac{1}{r+\delta_K}s_{K,t}(r)+Q(r)\omega s_{\Pi,t}, \tag{G28}$$

where the shares are given by (G25)*–* (G27)*. Thus CES production makes the sufficient statistic implicit through the share functions but leaves it a scalar equation in r.*

Proof. Household homogeneity gives aggregate demand $\bar{D}(r)[s_{W,t}(r)+\lambda(r)s_{\Pi,t}]Y_t$. Physical capital is $s_{K,t}(r)Y_t/(r+\delta_K)$, and tradable equity is $Q(r)\omega s_{\Pi,t}Y_t$. Equating demand and supply and dividing by Y_t gives (G28). □

G.4.3 Mechanism and output mapping

Two derivatives summarize the new force. Holding the return fixed while the markup moves,

$$\frac{\partial s_K}{\partial \mu}=-\sigma_{KL}\frac{s_K}{\mu},\qquad \frac{\partial s_\Pi}{\partial \mu}=\frac{1}{\mu^2},\qquad \frac{\partial s_W}{\partial \mu}=-\frac{1}{\mu^2}+\sigma_{KL}\frac{s_K}{\mu}. \tag{G29}$$

A higher markup compresses the capital share and capital intensity at a fixed return. The worker share also moves because $s_W=1/\mu-s_K$; its response depends on the substitution elasticity and the initial shares. These movements jointly change the income-demand weights and asset supply per unit of output at the initial return. Along the equilibrium response,

$$\frac{\partial s_K}{\partial r}=(1-\sigma_{KL})\frac{s_K}{r+\delta_K},\qquad \frac{\partial \log(K/Y)}{\partial r}=-\frac{\sigma_{KL}}{r+\delta_K}. \tag{G30}$$

When $\sigma_{KL}<1$, capital and labor are complements: capital intensity responds less to the decline in r, and the induced share movement tends to require a somewhat larger decline to clear the asset market. When $\sigma_{KL}>1$, capital intensity is more return-elastic and tends to stabilize the equilibrium. Equation (G28) measures the net of the initial capital-intensity compression and this equilibrium feedback.

The return solution maps into an exact stationary output level using the same GHH labor supply as in the main model. Let $M_t\equiv\mu_t/\mu_0$, $U_t\equiv(r_t+\delta_K)/(r_0+\delta_K)$, and define the baseline technological shares $\eta_{K,0}\equiv\mu_0 s_{K,0}$ and $\eta_{L,0}\equiv\mu_0 s_{W,0}$. Normalizing the common 1980 economy to

$Y_0 = 1$, the CES unit-cost equation and GHH labor supply give

$$M_t^{\sigma_{KL}-1} = \eta_{K,0} U_t^{1-\sigma_{KL}} + \eta_{L,0} \left(\frac{w_t}{w_0}\right)^{1-\sigma_{KL}}, \qquad \frac{L_t}{L_0} = \left(\frac{w_t}{w_0}\right)^{\varphi},$$
$$\frac{Y_t}{Y_0} = M_t^{\sigma_{KL}} \left(\frac{w_t}{w_0}\right)^{\sigma_{KL}+\varphi}, \qquad \frac{K_t}{K_0} = \frac{K_t/Y_t}{K_0/Y_0}\frac{Y_t}{Y_0}. \tag{G31}$$

At $\sigma_{KL} = 1$, the continuous limit is $\log(w_t/w_0) = [-\log M_t - \eta_{K,0} \log U_t]/\eta_{L,0}$. This closure holds aggregate productivity, the labor wedge, the GHH labor-supply shifter, and the composition of worker efficiency units fixed. The last equality reconciles the output normalization with the capital intensity already used in asset-market clearing; it does not introduce a second capital quantity. Equation (G31) is exact for finite changes between matched stationary economies. Its local differential around the baseline is

$$d \log Y = -\varphi \, d \log \mu - (\sigma_{KL} + \varphi) \frac{s_K}{s_W} d \log[\mu(r + \delta_K)] \, . \tag{G32}$$

The first term is the wage-induced GHH labor-supply contraction. The second is the CES capital-intensity channel: the equilibrium decline in r lowers the user cost and offsets part of the markup loss. Equation (G32) is used to interpret the level mapping (G31), not to construct the reported output column.

The representative household CES benchmark sets $r_t = r_0$ and recomputes the CES shares and K_t/Y_t at that return; it need not clear the asset market in the heterogeneous household economy. In that stationary equilibrium, summing household budget constraints and using firm accounting, capital accumulation, and asset-market clearing also delivers goods-market clearing; security trades and equity transfers cancel.

This exercise is distinct from Appendix A.2. That appendix holds the aggregate wedge fixed and asks whether a Hicks-neutral misallocation shock affects the asset market through CES factor shares. Here productivity is fixed and the observed markup itself moves. The term $(\mu_t/\mu_0)^{-\sigma_{KL}}$ in (G26) is therefore specific to the experiment studied in this subsection.

G.4.4 Calibration and quantitative illustration

We use $\sigma_{KL} = 0.7$ as the preferred CES calibration and report a grid spanning complementary, Cobb–Douglas, and high-substitution specifications. This choice gives greatest weight to estimates below one based on plant-level and aggregate manufacturing variation (Oberfield and Raval, 2021), while retaining the higher elasticity used in the aggregate factor-share literature as a robustness case (Karabarbounis and Neiman, 2014). The generated table reports the precise grid used in the calculation.

Every CES economy is normalized to reproduce the same 1980 return, markup, factor shares, and output level. Since the normalized household problems and initial asset supplies per unit of

Table G5: CES calibration and reference normalization

Specification	σ_{KL}	r_0 (%)	$s_{K,0}$ (%)	$s_{W,0}$ (%)	$s_{\Pi,0}$ (%)
Low elasticity	0.50	6.00	32.00	64.00	4.00
Preferred	0.70	6.00	32.00	64.00	4.00
Cobb–Douglas	1.00	6.00	32.00	64.00	4.00
High elasticity	1.25	6.00	32.00	64.00	4.00

Notes: Each specification is normalized at the same initial return, markup, factor shares, and output level $Y_0 = 1$. The preferred elasticity and robustness grid are read from the extension configuration; no CES result is calibrated to the endpoint return.

output are unchanged, the main calibration's (β_W, β_O) and direct SCF target $\lambda(r_0) = 3.31$ carry over without recalibration. At every subsequent date, the common market-power path supplies μ_t, equations (G26)–(G27) update the shares at the candidate return, and (G28) determines the stationary equilibrium.

We solve (G28) with the local sufficient statistic, using the baseline reduced-form aggregate asset-demand slope and the CES share mapping in (G26). As in (G23), this closure applies the aggregate slope to a common demand index and fixes λ at λ_0; it does not approximate the two household schedules separately. Re-solving both nonlinear stationary household problems at every date provides the check reported in Table G6; the two return columns differ only in their treatment of household demand away from the common baseline. The output column applies the exact CES–GHH mapping (G31) to the statistic's return. It is not a local output approximation.

Nesting check. At $\sigma_{KL} = 1$, equation (G26) becomes $s_{K,t} = s_{K,0}(\mu_t/\mu_0)^{-1}$, the Cobb–Douglas capital-share path, while $s_{W,t} = 1/\mu_t - s_{K,t}$ and $s_{\Pi,t} = 1 - 1/\mu_t$. Hence (G28) reduces exactly to the one-asset stationary equilibrium used as the comparison row. This nesting restriction is imposed directly in the generated results rather than checked only after rounding.

Table G6: CES production: stationary 2018 responses and outcomes

	2018 responses				2018 outcomes	
σ_{KL}	Δr (bp)	ΔY (%)	Exact HA Δr (bp)	$\Delta r-$CD (bp)	s_K (%)	K/Y
0.50	-93.9	-5.25	-99.2	3.7	29.43	2.47
0.70	-95.6	-5.32	-100.5	2.0	29.32	2.46
1.00	-97.6	-5.40	-102.1	0	29.19	2.46
1.25	-99.0	-5.46	-103.2	-1.4	29.11	2.45

Notes: All rows match the same 1980 return, factor shares, output normalization, and GHH labor-supply schedule and use the same household calibration and market-power path. Δr is the sufficient statistic and $\Delta Y = 100(Y_{2018}/Y_0 - 1)$ applies (G31) at that return, where $Y_0 = 1$ in the common 1980 economy. "Exact HA" re-solves the nonlinear stationary household problems at every date and is reported as a check. "$\Delta r-$CD" subtracts the $\sigma_{KL} = 1$ response of the same method. The last two columns are the capital share and capital–output ratio at the statistic's 2018 return. These are matched stationary comparisons, not a perfect-foresight transition.

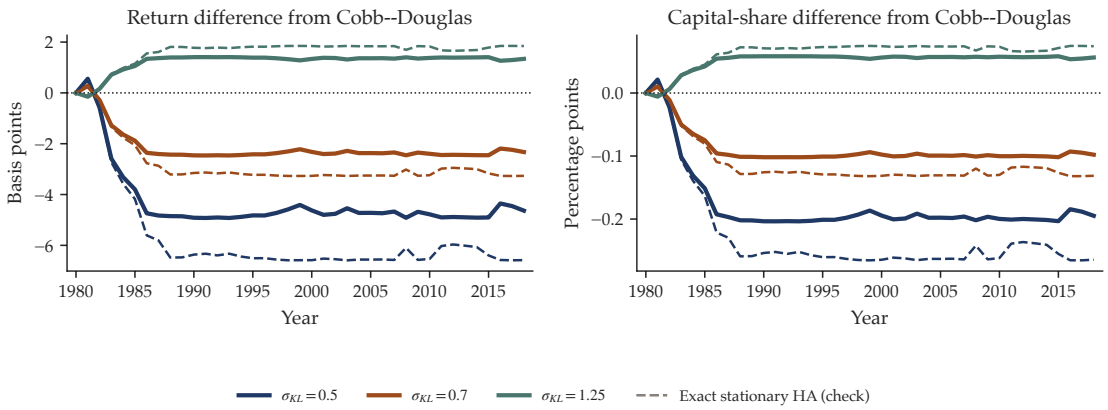


Figure G3: Incremental effects of CES production. Panels A and B show CES-minus-Cobb–Douglas differences in the interest rate r (basis points) and capital share (percentage points). Specifications share the markup path and initial calibration. Solid lines show the sufficient statistic; dashed lines re-solve the exact stationary household problems. These are stationary comparisons, not perfect foresight transitions.

H Multiple Assets

The main text compresses all household wealth and all firm claims into one asset, so it cannot separate the safe return from the required return on equity. This appendix splits the asset market in two and carries the fiscal closure of Appendix G.2 with it, so that government debt sits explicitly inside safe-asset supply. The scalar clearing condition becomes two market-clearing equations in $\mathbf{r} = (r^b, r^e)'$.

Adding the two equations preserves the aggregate accounting of the one-asset model: household demand generated by after-tax income on one side, installed capital, tradable equity backed by aggregate profit income, and government debt on the other. It does not reproduce the one-asset condition term by term, because physical capital is now discounted at the cost of capital while tradable equity backed by aggregate profit income, which is residual equity, is discounted at r^e. With a positive equity premium the same profit income flow supports a smaller stock of tradable equity, so asset creation offsets less of the redistribution that market power generates. This valuation margin, rather than portfolio substitution, accounts for most of the difference between the one-asset and two-asset cost of capital. The second equation is what determines composition, and with it the premium at which households absorb the extra equity; the statistic it requires is the long-run response of the equity share to the equity premium, which we take from Fagereng, Guiso and Ring (2026).

H.1 Asset demand with multiple assets

There are J traded assets. Let s denote a household's idiosyncratic state and Z the aggregate state, and write $\mathbf{s} = (s, Z)$ for the full state. A household in group g enters the period with financial wealth a. Its non-financial income is

$$X_g y_g(\mathbf{s}; \mathbf{r}),$$

where X_g is the group-specific income scale, and y_g may incorporate static intratemporal choices, such as GHH labor supply.

$$\mathbf{r} = (r^1, \ldots, r^J)$$

is the vector of required returns common to all households. Let $\mathbf{R}' = (R^{1\prime}, \ldots, R^{J\prime})$ denote next period's realized gross returns. Conditional on the current state and $\mathbf{r}$, the household faces a joint law for future states, returns, and income growth,

$$\left(\mathbf{s}', \mathbf{R}', \frac{X_g'}{X_g}\right) \sim P_g(\cdot \mid \mathbf{s}; \mathbf{r}).$$

All households trade the same securities and therefore face the same market returns. Their idiosyncratic income risks and their exposure to market risk may differ across groups. The household chooses consumption c, total financial investment $\widetilde{a}$, and portfolio shares $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_J)$, with

$\sum_{j=1}^{J} \phi_j = 1$. Given realized returns, these choices induce next-period financial wealth

$$a' = \widetilde{a}\,\boldsymbol{\phi} \cdot \mathbf{R}'. \tag{H1}$$

The recursive problem is

$$\begin{aligned} V_g(a, \mathbf{s}; X_g, \mathbf{r}) = \max_{c, \widetilde{a}, \boldsymbol{\phi}} \quad & \mathcal{U}_g\Big(c, \Big\{V_g(a', \mathbf{s}'; X_g', \mathbf{r})\Big\}; \boldsymbol{\phi}, \mathbf{s}\Big) \\ \text{s.t.} \quad & c + \widetilde{a} = a + X_g y_g(\mathbf{s}; \mathbf{r}), \\ & \widetilde{a} \geq -\underline{\ell}_g(\mathbf{s}) X_g, \qquad \boldsymbol{\phi} \in \mathcal{P}_g(\mathbf{s}), \end{aligned} \tag{H2}$$

where $\underline{\ell}_g(\mathbf{s}) \geq 0$ is the borrowing limit per unit of group income and $\mathcal{P}_g(\mathbf{s})$ is the set of admissible portfolio shares. The portfolio set may incorporate short-sale, leverage, participation, or other state-dependent portfolio restrictions.

Assumption 2 (Multi-asset homogeneity)**.** *For each group g, the recursive aggregator is homogeneous of degree one in current consumption and continuation utility,*

$$\mathcal{U}_g\left(tc, \{tV'\}; \boldsymbol{\phi}, \mathbf{s}\right) = t\,\mathcal{U}_g\left(c, \{V'\}; \boldsymbol{\phi}, \mathbf{s}\right), \qquad t > 0; \tag{H3}$$

Assumption 2 includes consumption-equivalent representations of time-additive CRRA and Epstein–Zin preferences. Importantly. dependence of $\mathcal{U}_g$ on $\boldsymbol{\phi}$ also permits nonpecuniary portfolio frictions that is necessary to match empirically observed pattern of portfolio holding.

Proposition 8 (Multi-asset homogeneity)**.** *Under Assumption 2, desired holdings of each traded asset are linear in the group income scale:*

$$D_{g,j}(\mathbf{r}, X_g) = X_g\, \overline{D}_{g,j}(\mathbf{r}), \qquad j = 1, \ldots, J, \tag{H4}$$

whenever the relevant first moments exist. Hence, writing $\overline{\mathbf{D}}_g(\mathbf{r})$ for asset demand per dollar of group income, aggregate asset-market clearing is

$$\mathbf{D}(\mathbf{r}) \equiv \sum_g X_g\, \overline{\mathbf{D}}_g(\mathbf{r}) = \mathbf{S}(\mathbf{r}). \tag{H5}$$

Proof. Let $b = \frac{a}{X_g}, \widetilde{b} = \frac{\widetilde{a}}{X_g}, \widehat{c} = \frac{c}{X_g}, G_g' = \frac{X_g'}{X_g}$. After dividing by X_g, the budget and wealth transition become

$$\widehat{c} + \widetilde{b} = b + y_g(\mathbf{s}; \mathbf{r}), \qquad \frac{a'}{X_g'} = \frac{\widetilde{b}\,\boldsymbol{\phi} \cdot \mathbf{R}'}{G_g'}.$$

The normalized problem therefore contains no X_g. Homogeneity of $\mathcal{U}_g$ implies

$$V_g(a, \mathbf{s}; X_g, \mathbf{r}) = X_g v_g(b, \mathbf{s}; \mathbf{r}), \qquad \widetilde{a} = X_g g_g(b, \mathbf{s}; \mathbf{r}),$$

while the optimal portfolio shares depend only on normalized states. Thus the dollar position in every asset scales one-for-one with X_g. Integrating over the normalized household distribution gives (H4); summing across groups gives (H5). □

Two-asset specialization. For the empirical application, we specialize to two traded assets: a class of safe asset (e.g., treasury, safe corporate debt) and equity, with common required-return vector $\mathbf{r} = (r^b, r^e)$. The two household groups are workers and entrepreneurs as before. With proportional taxes, their income scales are

$$X_W = \widetilde{W} = (1 - \tau_W)W, \qquad X_O = \widetilde{\Pi} = (1 - \tau_\Pi)\Pi.$$

The GHH intratemporal problem makes worker resources scale with $\widetilde{W}$, while private and retained profit income and incorporation proceeds scale with $\widetilde{\Pi}$. Proposition 8 therefore applies to both groups. For $g \in \{W, O\}$, define total asset demand per dollar of group income and the equity share by

$$\overline{D}_g(\mathbf{r}) \equiv \overline{D}_{g,b}(\mathbf{r}) + \overline{D}_{g,e}(\mathbf{r}), \qquad \phi_g(\mathbf{r}) \equiv \frac{\overline{D}_{g,e}(\mathbf{r})}{\overline{D}_g(\mathbf{r})}.$$

Taking workers as the reference group, let

$$\bar{D}(\mathbf{r}) \equiv \overline{D}_W(\mathbf{r}), \qquad \lambda(\mathbf{r}) \equiv \frac{\overline{D}_O(\mathbf{r})}{\overline{D}_W(\mathbf{r})}.$$

Here $\lambda(\mathbf{r})$ is the relative wealth-to-income ratio of owners to workers, measured using after-tax income. Total group asset demand is therefore

$$D_W(\mathbf{r}, \widetilde{W}) = \bar{D}(\mathbf{r})\widetilde{W}, \qquad D_O(\mathbf{r}, \widetilde{\Pi}) = \lambda(\mathbf{r})\bar{D}(\mathbf{r})\widetilde{\Pi}, \tag{H6}$$

with safe and equity positions

$$\begin{aligned} D_W^b &= [1 - \phi_W(\mathbf{r})]\bar{D}(\mathbf{r})\widetilde{W}, & D_W^e &= \phi_W(\mathbf{r})\bar{D}(\mathbf{r})\widetilde{W}, \\ D_O^b &= \lambda(\mathbf{r})[1 - \phi_O(\mathbf{r})]\bar{D}(\mathbf{r})\widetilde{\Pi}, & D_O^e &= \lambda(\mathbf{r})\phi_O(\mathbf{r})\bar{D}(\mathbf{r})\widetilde{\Pi}. \end{aligned} \tag{H7}$$

Fiscal closure and asset supply. Government debt is a part of the safe asset and pays r^b. As in Appendix G.2, let $b = B/Y$ denote government debt relative to output. The government budget is

$$\tau_W s_W + t_K + \tau_\Pi s_\Pi + \xi = g + r^b b, \qquad \tau_W(\mathbf{r}) = \frac{g + r^b b - t_K - \tau_\Pi s_\Pi - \xi}{s_W}. \tag{H8}$$

Define after-tax income shares

$$\widetilde{s}_W = (1 - \tau_W)s_W, \qquad \widetilde{s}_\Pi = (1 - \tau_\Pi)s_\Pi.$$

Firms finance installed capital with a fraction θ of bond-like claims (corporate bonds) and $1-\theta$ of equity. Their net cost of capital is

$$r^c(\mathbf{r}) = \theta r^b + (1-\theta) r^e. \tag{H9}$$

Retaining the capital-tax treatment of Appendix G.2,

$$u_K(\mathbf{r}) = [r^c(\mathbf{r}) + \delta_K]\frac{1-\tau_K z^\tau}{1-\tau_K}, \qquad \frac{K}{Y} = \frac{s_K}{u_K(\mathbf{r})}. \tag{H10}$$

Tradable equity backed by aggregate profit income has valuation multiple

$$Q^\Pi(r^e) = \frac{1}{r^e + \delta_C} \tag{H11}$$

per dollar of aggregate after-tax publicly traded profit income. Asset supplies per unit of output are therefore

$$\frac{S^b}{Y} = \theta\frac{s_K}{u_K(\mathbf{r})} + b, \qquad \frac{S^e}{Y} = (1-\theta)\frac{s_K}{u_K(\mathbf{r})} + Q^\Pi(r^e)\omega\widetilde{s}_\Pi. \tag{H12}$$

Proposition 9 (Two-asset sufficient statistic)**.** *Two-asset market clearing is*

$$\bar{D}(\mathbf{r})\begin{pmatrix}[1-\phi_W(\mathbf{r})]\widetilde{s}_W + \lambda(\mathbf{r})[1-\phi_O(\mathbf{r})]\widetilde{s}_\Pi \\ \phi_W(\mathbf{r})\widetilde{s}_W + \lambda(\mathbf{r})\phi_O(\mathbf{r})\widetilde{s}_\Pi\end{pmatrix} = \begin{pmatrix}\theta\frac{s_K}{u_K(\mathbf{r})} + b \\ (1-\theta)\frac{s_K}{u_K(\mathbf{r})} + Q^\Pi(r^e)\omega\widetilde{s}_\Pi\end{pmatrix}. \tag{H13}$$

The first row clears the safe-asset market and the second the equity market.

Equation (H13) follows by summing the group positions in (H7), dividing by output, and equating them to (H12). Adding the two rows eliminates portfolio composition:

$$\bar{D}(\mathbf{r})[\widetilde{s}_W + \lambda(\mathbf{r})\widetilde{s}_\Pi] = \frac{s_K}{u_K(\mathbf{r})} + Q^\Pi(r^e)\omega\widetilde{s}_\Pi + b. \tag{H14}$$

This preserves the aggregate accounting of the one-asset model, but with some differences: physical capital is discounted through $r^c(\mathbf{r})$, while tradable equity is discounted at r^e. Under the common-return limit $r^b = r^e = r$, however, (H14) reduces exactly to the one-asset benchmark, as in (G9).

Risky steady state. To study the effect of market power, we compare risky steady states as market power changes. A risky steady state is the allocation reached after a long sequence of mean aggregate shock realizations, while households continue to expect future aggregate and idiosyncratic risk. Hence $\overline{D}_g(\mathbf{r})$ and $\phi_g(\mathbf{r})$ incorporate precautionary saving, portfolio risk, and hedging motives, and the common required-return vector $\mathbf{r} = (r^b, r^e)$ clears the asset markets at the mean aggregate state.

Across these comparisons, we hold fixed the stochastic environment, including the process for future aggregate and idiosyncratic risk. The sufficient-statistic implementation then approximates household demand locally around the baseline risky steady state, with return vector $\mathbf{r}_0$, allowing household policies and the stationary cross-sectional distribution to adjust with $\mathbf{r}$. For each group,

$$\begin{pmatrix} d\log \overline{D}_g \\ d\phi_g \end{pmatrix} = \begin{pmatrix} \epsilon_g(1-\phi_{g,0}) & \epsilon_g\phi_{g,0} \\ -\eta_g & \eta_g \end{pmatrix} \begin{pmatrix} dr^b \\ dr^e \end{pmatrix}. \tag{H15}$$

Thus ϵ_g governs total asset demand with respect to the return on the baseline reference portfolio, while η_g governs portfolio substitution with respect to the equity premium. The first row imposes that a spread change that leaves the baseline portfolio return unchanged has no additional first-order effect on total asset demand.

This separation is consistent with and inspired by Ring (2025). In his empirical setting, a marginal wealth tax shifts after-tax returns on different assets simultaneously. The paper finds that financial saving responds substantially, while portfolio shares move little. This supports treating common return movements as an amount response and changes in relative returns as a portfolio response. The calibration below therefore disciplines the two margins separately.

H.2 Calibration and results

Fiscal block. We carry over the fiscal calibration of Appendix G.2: federal debt held by the public is 26.2 percent of GDP, the baseline effective labor tax is 26.4 percent, the marginal user-cost tax factor is 1.331, and the relative wealth-to-income ratio of owners to workers, measured using after-tax income, is $\lambda = 3.05$. The only adjustment is the residual ξ. Debt service is now $r_0^b b$ rather than $r_0 b$, so leaving the residual unchanged would lower the initial labor tax to 25.51 percent. We therefore recenter ξ from 2.92 to 2.36 percent of GDP, preserving the common 26.4-percent labor tax in 1980.

Household demand. For the amount margin, we use the group-specific asset demand semi-elasticities measured at the re-solved fiscal household calibration of Appendix G.2: 18.13 for workers and 16.89 for entrepreneurs. Their baseline reference portfolios are measured from gross financial wealth in the SCF. Averaging the 12 waves from 1989 to 2022 with equal weights gives

$$\phi_{W,0} = 48.88 \text{ percent}, \qquad \phi_{O,0} = 52.23 \text{ percent},$$

using the same age and wealth selection as for λ in Appendix D. Together, these objects determine each group's local response of total desired assets to the return on its baseline reference portfolio.

Portfolio composition. For the composition margin, we use the long-run response of equity holdings to the equity premium estimated by Fagereng, Guiso and Ring (2026). Their setting

exploits the 1998 abolition of preferential wealth-tax treatment of equity in Norway, which changed the after-tax equity premium. Because we compare risky steady states, we use their long-run estimate,

$$\eta_g = \frac{\partial \phi_g}{\partial (r^e - r^b)} = 6.9.$$

Thus a 100-basis-point increase in the equity premium raises the equity share by 6.9 percentage points. We use the same slope for workers and entrepreneurs.

Baseline returns and θ. We set the baseline cost of capital to the 6.00 percent 1980 target used in the main text. For the equity premium, we use 4.0 percentage points, a conventional long-run calibration. Given the premium, (H9) implies

$$r_0^b = r_0 - (1-\theta)(r_0^e - r_0^b),$$

while safe-asset market clearing pins down θ. Together, the two market-clearing equations imply $\theta = 45.91\%, r_0^b = 3.84\%, r_0^e = 7.84\%$. At the baseline it implies $r_0^e - r_0^c = \theta(r_0^e - r_0^b) = 1.84\%$. Because firm values are discounted at r^e rather than at the cost of capital, this gap makes a dollar of profit income support less asset value than in the one-asset benchmark. This feature of the two asset economy weakens the asset creation force of market power.

The historical exercise solves (H13) in levels along the observed markup path. For a candidate pair of required returns, the government budget determines the balancing labor tax, while household demand, portfolio composition, the cost of capital, and equity values adjust accordingly. Away from the baseline, we extend the amount response using the calibrated asset demand semi-elasticities and the composition response using a logit schedule that matches the baseline portfolio substitution elasticity.

Table 5 reports the 2018 endpoint. The safe return changes by -128.0 basis points and the equity required return by -80.2 basis points, so the equity premium widens by 47.8 basis points and the aggregate desired equity share rises to 53.56 percent. The cost of capital changes by -102.2 basis points, compared with -96.1 basis points in the one-asset economy. The corresponding output response is -4.54 percent, compared with -5.00 percent. This difference in the cost of capital translates, through the user cost, into roughly 0.46 percentage points of output.

Three changes separate the two columns. We take them in sequence; because the model is nonlinear, the individual contributions are order-dependent. Replacing the one-asset aggregate demand schedule by the two group schedules, still with a single return, moves the cost-of-capital response to -96.1 basis points. The 0-basis-point difference reflects demand aggregation, not the introduction of a second asset.

Next, introduce a separate equity return but hold the equity premium at its baseline value. Tradable equity is now valued at r^e, while portfolio composition does not adjust. This moves the cost-of-capital response to -102.2 basis points. The resulting -6.1-basis-point difference is the

valuation margin and the largest of the three. At $r_0^e = 7.84$ percent rather than 6.00 percent, a dollar of underlying incorporated profit income supports about 14 percent less equity market value, so the asset-creation force offsets less of the redistribution generated by market power. Finally, allowing the equity premium to adjust adds the remaining 0.1 basis points.

In this sequential decomposition, portfolio substitution contributes little to the cost-of-capital response. Its main role is to determine the premium at which households are willing to hold the additional equity. Rising market power adds tradable equity backed by aggregate profit income to equity supply but not to safe supply. Given the estimated portfolio slope, households readily rebalance toward equity, so the premium widens by only 47.8 basis points and the safe rate carries most of the adjustment. This matters for the fiscal closure because debt service is $r^b b$: the balancing labor tax falls to 25.48 percent by 2018, compared with 25.6 percent in the one-asset economy.

Figure 8 in the main text plots the required return paths. The two required returns straddle the one asset benchmark, and the gap between them widens along the market power path.